ACADEMIC REGULATIONS & SYLLABUS

Faculty of Pharmacy
Bachelor of Pharmacy Programme

1st to 8th semester of B. Pharm

(AS PER PCI SYLLABUS)

Ramanbhai Patel College of Pharmacy

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Vision of RPCP

To Become a Premier Pharma Institute by Creating World Class Pharmacists and Researchers.

Mission of RPCP

To Strive for the Excellence in Pharmaceutical Sciences through Quality Education and Research.

Programme Educational Objective

1. To produce competent pharmacy graduates who shall be able to secure their career in pharmaceutical industry and/or institution of higher studies by dissemination of knowledge and training in pharmacy.
2. To acquire the knowledge about advances in pharmacy and to orient towards socially relevant research and development.
3. To encourage students for self-learning through modern information resources.
4. To enhance communication and critical thinking abilities for effective delivery of societal responsibilities.
5. To nurture and facilitate entrepreneurial capabilities and opportunities.

PROGRAM OUTCOMES

1. Pharmacy Knowledge: Possess knowledge and comprehension of the core and basic knowledge associated with the profession of pharmacy, including biomedical sciences; pharmaceutical sciences; behavioral, social, and administrative pharmacy sciences; and manufacturing practices.

2. Planning Abilities: Demonstrate effective planning abilities including time management, resource management, delegation skills and organizational skills. Develop and implement plans and organize work to meet deadlines.

3. Problem analysis: Utilize the principles of scientific enquiry, thinking analytically, clearly and critically, while solving problems and making decisions during daily practice. Find, analyze, evaluate and apply information systematically and shall make defensible decisions.

4. Modern tool usage: Learn, select, and apply appropriate methods and procedures, resources, and modern pharmacy-related computing tools with an understanding of the limitations.

5. Leadership skills: Understand and consider the human reaction to change, motivation issues, leadership and team-building when planning changes required for fulfillment of practice, professional and societal responsibilities. Assume participatory roles as responsible citizens or leadership roles when appropriate to facilitate improvement in health and wellbeing.

6. Professional Identity: Understand, analyze and communicate the value of their professional roles in society (e.g. health care professionals, promoters of health, educators, managers, employers, employees).

7. Pharmaceutical Ethics: Honour personal values and apply ethical principles in professional and social contexts. Demonstrate behavior that recognizes cultural and personal variability in values, communication and lifestyles. Use ethical frameworks; apply ethical principles while making decisions and take responsibility for the outcomes associated with the decisions.

8. Communication: Communicate effectively with u the pharmacy community and with society at large, such as, being able to comprehend and write effective reports, make effective presentations and documentation, and give and receive clear instructions.

9. The Pharmacist and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety and legal issues and the consequent responsibilities relevant to the professional pharmacy practice.

10. Environment and sustainability: Understand the impact of the professional pharmacy solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

11. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. Self-assess and use feedback effectively from others to identify learning needs and to satisfy these needs on an ongoing basis.

REGULATIONS

1. Short Title and Commencement

These regulations shall be called as "The Revised Regulations for the B. Pharm. Degree Program (CBCS) of the Pharmacy Council of India, New Delhi". They shall come into effect from the Academic Year 2016-17. The regulations framed are subject to modifications from time to time by Pharmacy Council of India.

2. Minimum qualification for admission

2.1 First year B. Pharm:

Candidate shall have passed 10+2 examination conducted by the respective state/central government authorities recognized as equivalent to 10+2 examination by the Association of Indian Universities (AIU) with English as one of the subjects and Physics, Chemistry, Mathematics (P.C.M) and or Biology (P.C.B / P.C.M.B.) as optional subjects individually. Any other qualification approved by the Pharmacy Council of India as equivalent to any of the above examinations.

2.2. B. Pharm lateral entry (to third semester):

A pass in D. Pharm. course from an institution approved by the Pharmacy Council of India under section 12 of the Pharmacy Act.

3. Duration of the program

The course of study for B. Pharm. shall extend over a period of eight semesters (four academic years) and six semesters (three academic years) for lateral entry students. The curricula and syllabi for the program shall be prescribed from time to time by Pharmacy Council of India, New Delhi.

4. Medium of instruction and examinations

Medium of instruction and examination shall be in English.

5. Working days in each semester

Each semestershall consist of not less than 100 working days. The odd semesters shall be conducted from the month of June/July to November/December and the even semesters shall be conducted from December/January to May/June in every calendar year.

6. Attendance and progress

A candidate is required to put in at least 80% attendance in individual courses considering theory and practical separately. The candidate shall complete the prescribed course satisfactorily to be eligible to appear for the respective examinations.

7. Program/Course credit structure

As per the philosophy of Credit Based Semester System, certain quantum of academic work viz. theory classes, tutorial hours, practical classes, etc. are measured in terms of credits. On satisfactory completion of the courses, a candidate earns credits. The amount of credit associated with a course is dependent upon the number of hours of instruction per week in that course. Similarly, the credit associated with any of the other academic, co/extra-curricular activities is dependent upon the quantum of work expected to be put in for each of these activities per week.

7.1. Credit assignment

7.1.1. Theory and Laboratory courses

Courses are broadly classified as Theory and Practical. Theory courses consist of lecture (L) and /or tutorial (T) hours, and Practical (P) courses consist of hours spent in the laboratory. Credits (C) for a course is dependent on the number of hours of instruction per week in that course, and is obtained by using a multiplier of one (1) for lecture and tutorial hours, and a multiplier of half (1/2) for practical (laboratory) hours. Thus, for example, a theory course having three lectures and one tutorial per week throughout the semester carries a credit of 4. Similarly, a practical having four laboratory hours per week throughout semester carries a credit of 2.

7.2. Minimum credit requirements

The minimum credit points required for award of a B. Pharm. degree is 208. These credits are divided into Theory courses, Tutorials, Practical, Practice School and Projectover the duration of eight semesters. The credits are distributed semester-wise as shown in Table IX. Courses generally progress in sequences, building competencies and their positioning indicates certain academic maturity on the part of the learners. Learners are expected to follow the semester-wise schedule of courses given in the syllabus.

The lateral entry students shall get 52 credit points transferred from their D. Pharm program. Such students shall take up additional remedial courses of 'Communication Skills' (Theory and Practical) and 'Computer Applications in Pharmacy' (Theory and Practical) equivalent to 3 and 4 credit points respectively, a total of 7 credit points to attain 59 credit points, the maximum of I and II semesters.

8. Academic work

A regular record of attendance both in Theory and Practical shall be maintained by the teaching staff of respective courses.

9. Program Committee

1. The B. Pharm. program shall have a Program Committee constituted by the Head of the institution in consultation with all the Heads of the departments.
2. The composition of the Program Committee shall be as follows:
   A senior teacher shall be the Chairperson; One Teacher from each department handling B.Pharm courses; and four student representatives of the program (one from each academic year), nominated by the Head of the institution.
3. Duties of the Program Committee:
   i. Periodically reviewing the progress of the classes.
   ii. Discussing the problems concerning curriculum, syllabus and the conduct of classes.
   iii. Discussing with the course teachers on the nature and scope of assessment for the course and the same shall be announced to the students at the beginning of respective semesters.
   iv. Communicating its recommendation to the Head of the institution on academic matters.
   v. The Program Committee shall meet at least thrice in a semester preferably at the end of each Sessionalexam (Internal Assessment) and before the end semester exam.

10. Examinations/Assessments

The scheme for internal assessment and end semester examinations is given in Table – X.

10.1 End semester examinations

The End Semester Examinations for each theory and practical coursethrough semesters I to VIII shall beconducted by the university except for the subjects with asterix symbol (*) in table I and II for which examinations shall be conducted by the subject experts at college level and the marks/grades shall be submitted to the university.

10.2 Internal assessment: Continuous mode

The marks allocated for Continuous mode of Internal Assessment shall be awarded as per the scheme given below.

Table-I: Scheme for awarding internal assessment: Continuous mode

Theory		
Criteria	**Maximum Marks**	
Attendance (Refer Table – XII)	4	2
Academic activities (Average of any 3 activities e.g. quiz, assignment, open book test, field work, group discussion and seminar)	3	1.5
Student – Teacher interaction	3	1.5
Total	**10**	**5**
Practical		
Attendance (Refer Table – XII)	2	
Based on Practical Records, Regular viva voce, etc.	3	
Total	**5**	

Table- II: Guidelines for the allotment of marks for attendance

Percentage of Attendance	Theory	Practical
95 – 100	4	2
90 – 94	3	1.5
85 – 89	2	1
80 – 84	1	0.5
Less than 80	0	0

10.2.1 Sessional Exams

Two Sessional exams shall be conducted for each theory / practical course as per the schedule fixed by the college(s). The scheme of question paper for theory and practical Sessional examinations is given below. The average marks of two Sessional exams shall be computed for internal assessment as per the requirements given in tables – X.

Sessional exam shall be conducted for 30 marks for theory and shall be computed for 15 marks. Similarly Sessional exam for practical shall be conducted for 40 marks and shall be computed for 10 marks.

Question paper pattern for theory Sessional examinations
For subjects having University examination

I. Multiple Choice Questions (MCQs) = 10 x 1 = 10

OR

Objective Type Questions (5 x 2) = 05 x 2 = 10

(Answer all the questions)

I. Long Answers (Answer 1 out of 2) = 1 x 10 = 10

II. Short Answers (Answer 2 out of 3) = 2 x 5 = 10

Total = 30 marks

For subjects having Non University Examination

I. Long Answers (Answer 1 out of 2) = 1 x 10 = 10

II. Short Answers (Answer 4 out of 6) = 4 x 5 = 20

Total = 30 marks

Question paper pattern for practical sessional examinations I. Synopsis	=	10
II. Experiments	=	25
III. Viva voce	=	05
Total	=	40 marks

11. Promotion and award of grades

A student shall be declared PASSand eligible for getting gradein a course of B.Pharm.program if he/she secures at least 50% marks in that particular course including internal assessment.For example, to be declared as PASS and to get grade, the student has to secure a minimum of 50 marks for the total of 100 including continuous mode of assessment and end semester theory examination and has to secure a minimum of 25 marks for the total 50 including internal assessment and end semester practical examination.

12. Carry forward of marks

In case a studentfails to secure the minimum 50% in any Theory or Practical course as specified in 12, then he/she shall reappear for the end semester examinationof that course. However, his/her marks of the Internal Assessmentshallbe carried overand he/she shall be entitled for grade obtained by him/her on passing.

13. Improvement of internal assessment

A student shall have the opportunity to improvehis/her performance only oncein the Sessional exam component of the internal assessment. The re-conduct of the Sessional exam shall be completed before the commencement of next end semester theory examinations.

14. Re-examination of end semester examinations

Reexamination ofend semester examinationshall be conducted as per the schedule given in table XIII. The exact dates of examinations shall be notified from time to time.

Table- III: Tentative schedule of end semester examinations

Semester	For Regular Candidates	For Failed Candidates
I, III, V and VII	November / December	May / June
II, IV, VI and VIII	May / June	November / December

Question paper pattern for end semester theory examinations

For 75 marks paper

I. Multiple Choice Questions(MCQs) = 20 x 1 = 20
OR
OR Objective
Type Questions (10 x 2) = 10 x 2
= 20
(Answer all the questions)

II. Long Answers (Answer 2 out of 3) = 2 x 10 = 20

III. Short Answers (Answer 7 out of 9) = 7 x 5 = 35

Total = 75 marks

For 50 marks paper

I. Long Answers (Answer 2 out of 3) = 2 x 10 = 20

II. Short Answers (Answer 6 out of 8) = 6 x 5 = 30

Total = 50 marks

For 35 marks paper

I. Long Answers (Answer 1 out of 2) = 1 x 10 =10

II. Short Answers (Answer 5 out of 7) = 5 x 5 = 25

Total = 35 marks

Question paper pattern for end semester practical examinations

I. Synopsis = 5

II. Experiments = 25

III. Viva voce = 5

Total = 35 marks

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15. Academic Progression:

No student shall be admitted to any examination unless he/she fulfills the norms given in 6. Academic progression rules are applicable as follows:

A student shall be eligible to carry forward all the courses of I, II and III semesters till the IV semester examinations. However, he/she shall not be eligible to attend the courses of V semester until all the courses of I and II semesters are successfully completed.

A student shall be eligible to carry forward all the courses of III, IV and V semesters till the VI semester examinations. However, he/she shall not be eligible to attend the courses of VII semester until all the courses of I, II, III and IV semesters are successfully completed.

A student shall be eligible to carry forward all the courses of V, VI and VII semesters till the VIII semester examinations. However, he/she shall not be eligible to get the course completion certificate until all the courses of I, II, III, IV, V and VI semesters are successfully completed.

A student shall be eligible to get his/her CGPA upon successful completion of the courses of I to VIII semesters within the stipulated time period as per the norms specified in 26.

A lateral entry student shall be eligible to carry forward all the courses of III, IV and V semesters till the VI semester examinations. However, he/she shall not be eligible to attend the courses of VII semester until all the courses of III and IV semesters are successfully completed.

A lateral entry student shall be eligible to carry forward all the courses of V, VI and VII semesters till the VIII semester examinations. However, he/she shall not be eligible to get the course completion certificate until all the courses of III, IV, V and VI semesters are successfully completed.

A lateral entry student shall be eligible to get his/her CGPA upon successful completion of the courses of III to VIII semesters within the stipulated time period as per the norms specified in 26.

Any student who hasgiven more than 4 chances for successful completion of I / III semester courses and more than 3 chances for successful completion of II / IV semester courses shall be permitted to attend V / VII semester classes ONLY during the subsequent academic year as the case may be. In simpler terms there shall NOT be any ODD BATCH for any semester.

Note: Grade ABshould be considered as failed and treated as one head for deciding academic progression. Such rules are also applicable for those students who fail to register for examination(s) of any course in any semester.

16. Grading of performances

16.1 Letter grades and grade points allocations:

Based on the performances, each student shall be awarded a final letter grade at the end of the semester for each course. The letter grades and their corresponding grade points are given in below Table.

Table – IV: Letter grades and grade points equivalent to Percentage of marks and performances

Percentage of Marks Obtained	Letter Grade	Grade Point	Performance
90.00 – 100	AA	10	Outstanding
80.00 – 89.99	AB	9	Excellent
70.00 – 79.99	BB	8	Good

60.00 – 69.99	BC	7	Fair
50.00 – 59.99	CC	6	Average
Less than 50	FF	0	Fail
Absent	Ab	0	Fail

A learner who remains absent for any end semester examination shall be assigned a letter grade of ABand a corresponding grade point of zero. He/she should reappear for the said evaluation/examination in due course.

17. The Semester grade point average (SGPA)

The performance of a student in a semester is indicated by a number called 'Semester Grade Point Average' (SGPA). The SGPA is the weighted average of the grade points obtainedin all the courses by the student during the semester. For example, if a student takes five courses(Theory/Practical) in a semester with credits C1, C2, C3, C4 and C5 and the student's grade pointsin these courses are G1, G2, G3, G4 and G5, respectively, and then students' SGPA is equal to:

$$\mathbf{SGPA} = \frac{C_1G_1 + C_2G_2 + C_3G_3 + C_4G_4 + C_5G_5}{C_1 + C_2 + C_3 + C_4 + C_5}$$

The SGPA is calculated to two decimal points.It should be noted that, the SGPA for any semester shall take into consideration the F and ABSgrade awarded in that semester. For example if a learner has a F or ABS grade in course 4, theSGPA shall then be computed as:

$$\mathbf{SGPA} = \frac{C_1G_1 + C_2G_2 + C_3G_3 + C_4 * \text{ZERO} + C_5G_5}{C_1 + C_2 + C_3 + C_4 + C_5}$$

18. Cumulative Grade Point Average (CGPA)

The CGPA is calculated with the SGPA of all the VIII semesters to two decimal points and is indicated in final grade report card/final transcript showing the grades of all VIII semesters and their courses. The CGPA shall reflect the failed status in case of F grade(s),till the course(s) is/are passed. When the course(s)is/are passed by obtaining a pass grade on subsequent examination(s) the CGPA shall only reflect the new grade and not the fail grades earned earlier. The CGPA is calculated as:

$$\mathbf{CGPA} = \frac{C_1S_1 + C_2S_2 + C_3S_3 + C_4S_4 + C_5S_5 + C_6S_6 + C_7S_7 + C_8S_8}{C_1 + C_2 + C_3 + C_4 + C_5 + C_6 + C_7 + C_8}$$

where $C_1, C_2, C_3,\ldots$ is the total number of credits for semester I,II,III,…. and $S_1, S_2, S_3,\ldots$is the SGPA of semester I,II,III,…. .

19. Declaration of class

The class shall be awarded on the basis of CGPA as follows:

First Class with Distinction	= CGPA of. 7.50 to 10.00
First Class	= CGPA of 6.00 to 7.49
Second Class	= CGPA of 5.00 to 5.99
Pass Class	< CGPA of 5.00

20. Project work

All the students shall undertake a project under the supervision of a teacher and submit a report. The area of the project shall directly relate any one of the elective subject opted by the student in semester VIII. The project shall be carried out in group not exceeding 5 in number. The project report shall be submitted in triplicate (typed & bound copy not less than 25 pages).

The internal and external examiner appointed by the University shall evaluate the project at the time of the Practical examinations of other semester(s). Students shall be evaluated in groups for four hours (i.e., about half an hour for a group of five students). The projects shall be evaluated as per the criteria given below.

Evaluation of Dissertation Book:

Objective(s) of the work done	15 Marks
Methodology adopted	20 Marks
Results and Discussions	20 Marks
Conclusions and Outcomes	20 Marks
Total	**75 Marks**

Evaluation of Presentation:

Presentation of work	25 Marks
Communication skills	20 Marks
Question and answer skills	30 Marks
Total	**75 Marks**

Explanation: The 75 marks assigned to the dissertation book shall be same for all the students in a group. However, the 75 marks assigned for presentation shall be awarded based on the performance of individual students in the given criteria.

21. Industrial training (Desirable)

Every candidate shall be required to work for at least 150 hours spread over four weeks in a Pharmaceutical Industry/Hospital. It includes Production unit, Quality Control department, Quality Assurance department, Analytical laboratory, Chemical manufacturing unit, Pharmaceutical R&D, Hospital (Clinical Pharmacy), Clinical Research Organization, Community Pharmacy, etc. After the Semester – VI and before the commencement of Semester – VII, and shall submit satisfactory report of such work and certificate duly signed by the authority of training organization to the head of the institute.

22. Practice School

In the VII semester, every candidate shall undergo practice school for a period of 150 hours evenly distributed throughout the semester. The student shall opt any one of the domains for practice school declared by the program committee from time to time.

At the end of the practice school, every student shall submit a printed report (in triplicate) on the practice school he/she attended (not more than 25 pages). Along with the exams of semester VII, the report submitted by the student, knowledge and skills acquired by the student through practice school shall be evaluated by the subject experts at college leveland grade point shall be awarded.

23. Award of Ranks

Ranks and Medals shall be awarded on the basis of final CGPA. However, candidates who fail in one or more courses during the B.Pharm program shall not be eligible for award of ranks.Moreover, the candidates should have completed the B. Pharm program in minimum prescribed number of years, (four years) for the award of Ranks.

24 Award of degree

Candidates who fulfill the requirements mentioned above shall be eligible for award of degree during the ensuing convocation.

25. Duration for completion of the program of study

The duration for the completion of the program shall be fixed as double the actual duration of the program and the students have to pass within the said period, otherwise they have to get fresh Registration.

26. Re-admission after break of study

Candidate who seeks re-admission to the program after break of study has to get the approval from the university by paying a condonation fee.

No condonation is allowed for the candidate who has more than 2 years of break up period and he/she has to rejoin the program by paying the required fees.

25. Extra Credit:

An extra credit is to be offered to a student for achievements in co-curricular and extra–curricular activities. This credit shall not be counted while considering the minimum credits for completing the program. The activities and appropriate weight (points) to be allocated to award an extra credit are broadly classified as per the tables below:

Extracurricular Activities (25 Marks maximum out of 100 marks) includes

Sr. No.	Name of the activity	Maximum Marks
1	Participation in any Sport	10
2	Participation in any Cultural event	10
3	Community and Extension activities	10
4	Any achievement includes award, prize or special recognition for event	05

Co-Curricular Activities (75 marks maximum out of 100 marks) include

Sr. No.	Name of the activity	Maximum Marks
1	Completion of certificate SWAYAM course (minimum 4 weeks)	20 (10 marks/course, given for maximum two course)
2	Participation in Conference/Seminar/training program/Teach fest etc. (without Paper) includes	10

	a. State/University/Zonal level: 03	
	b. National : 05	
	c. International conference : 07	
3	Presentation in Conference/Seminar/training program/Teach fest etc. includes	15
	a. Oral Presentation: 10	
	b. Poster Presentation: 05	
4	Publication in Peer-reviewed Journal (indexed in Web of Science and Scopus):	20
	a. First paper publication: 10	
	b. Second paper publication:10	
5	Participation in other technical events like model preparation, Rangoli, Painting, Pharma receipt, dare to sell etc.: (05)	05
6	Any achievement includes award, prize or special recognition for any technical events mentioned above: (05)	05

Activities are broadly classified into Extra-curricular and Co-curricular. All suggested activities are counted as total sum of 100 marks.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

BACHELOR OF PHARMACY PROGRAMME
SEMESTER-1
SCHEME OF TEACHING (for [A] group students)

Course Code	Name of the Course	No. of hours	Tutorial	Credit Points
BPH1001	Human Anatomy and Physiology-I (Theory)	3	1	4
BPH1002	Pharmaceutical Analysis-I (Theory)	3	1	4
BPH1003	Pharmaceutics-I (Theory)	3	1	4
BPH1004	Pharmaceutical Inorganic Chemistry (Theory)	3	1	4
BPH1005	Remedial Biology (Theory)*	2	-	2
HS1001	Communication skills (Theory)*	2	-	2
BPH1001P	Human Anatomy and Physiology-I (Practical)	4	-	2
BPH1002P	Pharmaceutical Analysis-I (Practical)	4	-	2
BPH1003P	Pharmaceutics-I (Practical)	4	-	2
BPH1004P	Pharmaceutical Inorganic Chemistry (Practical)	4	-	2
BPH1005P	Remedial Biology (Practical)*	2	-	1
HS1001P	Communication skills (Practical)*	2	-	1
HS 101B - HS106B	A Course From Liberal Arts **	2	-	2
Total		**38**	**4**	**32**

SCHEME OF EVALUATION

Course Code	Name of the Course	Marks Distribution			
		University (End Semester Exam)	Institute		Total
			Sessional Exams	Continuous Mode	
BPH1001	Human Anatomy and Physiology-I (Theory)	75	15	10	100
BPH1002	Pharmaceutical Analysis-I (Theory)	75	15	10	100
BPH1003	Pharmaceutics-I (Theory)	75	15	10	100
BPH1004	Pharmaceutical Inorganic Chemistry (Theory)	75	15	10	100
BPH1005	Remedial Biology (Theory)*	35	10	5	50
HS1001	Communication skills (Theory)*	35	10	5	50
BPH1001P	Human Anatomy and Physiology-I (Practical)	35	10	5	50
BPH1002P	Pharmaceutical Analysis-I (Practical)	35	10	5	50
BPH1003P	Pharmaceutics-I (Practical)	35	10	5	50
BPH1004P	Pharmaceutical Inorganic Chemistry (Practical)	35	10	5	50
BPH1005P	Remedial Biology (Practical)*	15	5	5	25
HS1001P	Communication skills (Practical)*	15	5	5	25
HS 101B - HS106B	A Course From Liberal Arts **	70	30		100

Total	610	240	850

SCHEME OF TEACHING (for [B] group students)

Course Code	Name of the Course	No. of hours	Tutorial	Credit Points
BPH1001	Human Anatomy and Physiology-I (Theory)	3	1	4
BPH1002	Pharmaceutical Analysis-I (Theory)	3	1	4
BPH1003	Pharmaceutics-I (Theory)	3	1	4
BPH1004	Pharmaceutical Inorganic Chemistry (Theory)	3	1	4
MTH1001	Remedial Mathematics (Theory)*	2	-	2
HS1001	Communication skills (Theory)*	2	-	2
BPH1001P	Human Anatomy and Physiology-I (Practical)	4	-	2
BPH1002P	Pharmaceutical Analysis-I (Practical)	4	-	2
BPH1003P	Pharmaceutics-I (Practical)	4	-	2
BPH1004P	Pharmaceutical Inorganic Chemistry (Practical)	4	-	2
HS1001P	Communication skills (Practical)*	2	-	1
HS 101B - HS106B	A Course From Liberal Arts **	2	-	2
Total		**36**	**4**	**31**

SCHEME OF EVALUATION

Course Code	Name of the Course	Marks Distribution			
		University (End Semester Exam)	Institute		Total
			Sessional Exams	Continuous Mode	
BPH1001	Human Anatomy and Physiology-I (Theory)	75	15	10	100
BPH1002	Pharmaceutical Analysis-I (Theory)	75	15	10	100
BPH1003	Pharmaceutics-I (Theory)	75	15	10	100
BPH1004	Pharmaceutical Inorganic Chemistry (Theory)	75	15	10	100
MTH1001	Remedial Mathematics (Theory)*	35	10	5	50
HS1001	Communication skills (Theory)*	35	10	5	50
BPH1001P	Human Anatomy and Physiology-I (Practical)	35	10	5	50
BPH1002P	Pharmaceutical Analysis-I (Practical)	35	10	5	50
BPH1003P	Pharmaceutics-I (Practical)	35	10	5	50
BPH1004P	Pharmaceutical Inorganic Chemistry (Practical)	35	10	5	50
HS1001P	Communication skills (Practical)*	15	5	5	25
HS 101B - HS106B	A Course From Liberal Arts **	70	30		100

Total	**595**	**230**	**825**

SCHEME OF TEACHING (for [AB] group students)
Semester-I

Course Code	Name of the Course	No. of hours	Tutorial	Credit Points
BPH1001	Human Anatomy and Physiology-I (Theory)	3	1	4
BPH1002	Pharmaceutical Analysis-I (Theory)	3	1	4
BPH1003	Pharmaceutics-I (Theory)	3	1	4
BPH1004	Pharmaceutical Inorganic Chemistry (Theory)	3	1	4
HS1001	Communication Skills (Theory)*	2	-	2
BPH1001P	Human Anatomy and Physiology-I (Practical)	4	-	2
BPH1002P	Pharmaceutical Analysis-I (Practical)	4	-	2
BPH1003P	Pharmaceutics-I (Practical)	4	-	2
BPH1004P	Pharmaceutical Inorganic Chemistry (Practical)	4	-	2
HS1001P	Communication Skills (Practical)*	2	-	1
HS 101B - HS106B	A Course From Liberal Arts **	2	-	2
Total		**34**	**4**	**29**

SCHEME OF EVALUATION

Course Code	Name of the Course	Marks Distribution			
		University (End Semester Exam)	Institute		Total
			Sessional Exams	Continuous Mode	
BPH1001	Human Anatomy and Physiology-I (Theory)	75	15	10	100
BPH1002	Pharmaceutical Analysis-I (Theory)	75	15	10	100
BPH1003	Pharmaceutics-I (Theory)	75	15	10	100
BPH1004	Pharmaceutical Inorganic Chemistry (Theory)	75	15	10	100
HS1001	Communication skills (Theory)*	35	10	5	50
BPH1001P	Human Anatomy and Physiology-I (Practical)	35	10	5	50
BPH1002P	Pharmaceutical Analysis-I (Practical)	35	10	5	50
BPH1003P	Pharmaceutics-I (Practical)	35	10	5	50
BPH1004P	Pharmaceutical Inorganic Chemistry (Practical)	35	10	5	50
HS1001P	Communication Skills (Practical)*	15	5	5	25
HS 101B - HS106B	A Course From Liberal Arts **	70	30		100
Total		**560**	**215**		**775**

*** Non University Examination (NUE)**

* Liberal arts elective courses:

Elective Course		Credits		Total hours	Examination scheme (Practical)		
Code	**Name**	**Theory**	**Practical**	**--**	**Internal**	**External**	**Total**
HS101	Liberal Arts - Painting	--	2	2	50	50	100
HS102	Liberal Arts - Photography	--	2	2	50	50	100
HS103	Liberal Arts - Sculpting	--	2	2	50	50	100
HS104	Liberal Arts - Pottery and Ceramic Art	--	2	2	50	50	100
HS105	Liberal Arts - Media and Graphic Design	--	2	2	50	50	100
HS106	Liberal Arts - Arts and Craft	--	2	2	50	50	100
Total		**--**	**2**	**2**	**50**	**50**	**100**
Total credits: 2, Total marks: 100							

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
BACHELOR OF PHARMACY PROGRAMME
SEMESTER-II

SCHEME OF TEACHING

Course Code	Name of the Course	No. of hours	Tutorial	Credit Points
BPH2001	Human Anatomy and physiology-II (Theory)	3	1	4
BPH2002	Pharmaceutical Organic Chemistry-I (Theory)	3	1	4
BPH2003	Pharmaceutical Engineering (Theory)	3	1	4
BPH2004	Computer Applications in Pharmacy (Theory)*	3	-	3
BPH2001P	Human Anatomy and Physiology-II (Practical)	4	-	2
BPH2002P	Pharmaceutical Organic Chemistry-I (Practical)	4	-	2
BPH2003P	Pharmaceutical Engineering (Practical)	4	-	2
BPH2004P	Computer Applications in Pharmacy (Practical)*	2	-	1
HS121.01 B	English Language and Literature	2	-	2
	Total	**28**	**3**	**24**

SCHEME OF EVALUATION

Course Code	Name of the Course	Marks Distribution			
		University (End Semester Exam)	Institute		Total
			Sessional Exams	Continuous Mode	
BPH2001	Human Anatomy and Physiology-II (Theory)	75	15	10	100
BPH2002	Pharmaceutical Organic Chemistry-I (Theory)	75	15	10	100
BPH2003	Pharmaceutical Engineering (Theory)	75	15	10	100
BPH2004	Computer Applications in Pharmacy (Theory)*	50	15	10	75
BPH2001P	Human Anatomy and Physiology-II (Practical)	35	10	05	50
BPH2002P	Pharmaceutical Organic Chemistry-I (Practical)	35	10	05	50
BPH2003P	Pharmaceutical Engineering (Practical)	35	10	05	50
BPH2004P	Computer Applications in Pharmacy (Practical)*	15	05	05	25
HS121.01 B	English Language and Literature	70	30		100
	Total	**465**	**185**		**650**

*** Non University Examination (NUE)**

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
BACHELOR OF PHARMACY PROGRAMME

SEMESTER-III

SCHEME OF TEACHING

Course Code	Name of the Course	No. of hours	Tutorial	Credit Points
BPH3001	Pharmaceutical Organic Chemistry-II (Theory)	3	1	4
BPH3002	Physical Pharmaceutics-I (Theory)	3	1	4
BPH3003	Pharmaceutical Microbiology (Theory)	3	1	4
BPH3004	Pathophysiology (Theory)	3	1	4
BPH3005	Environmental Sciences (Theory)*	3	-	3
BPH3001P	Pharmaceutical Organic Chemistry-II (Practical)	4	-	2
BPH3002P	Physical Pharmaceutics-I (Practical)	4	-	2
BPH3003P	Pharmaceutical Microbiology (Practical)	4	-	2
---	DHSS elective**	2	-	2
---	University elective***	2	-	2
	Total	**31**	**4**	**29**

SCHEME OF EVALUATION

Course Code	Name of the Course	Marks Distribution			
		University (End Semester Exam)	Institute		Total
			Sessional Exams	Continuous Mode	
BPH3001	Pharmaceutical Organic Chemistry-II (Theory)	75	15	10	100
BPH3002	Physical Pharmaceutics-I (Theory)	75	15	10	100
BPH3003	Pharmaceutical Microbiology (Theory)	75	15	10	100
BPH3004	Pathophysiology (Theory)	75	15	10	100
BPH3005	Environmental Sciences (Theory)*	50	15	10	75
BPH3001P	Pharmaceutical Organic Chemistry-II (Practical)	35	10	05	50
BPH3002P	Physical Pharmaceutics-I (Practical)	35	10	05	50
BPH3003P	Pharmaceutical Microbiology (Practical)	35	10	05	50
---	DHSS elective**	70	30		100
---	University elective***	70	30		100
	Total	**595**	**230**		**825**

*** Non University Examination (NUE)**

** DHSS elective courses:

Elective Course		Credits		Total hours	Examination scheme (Practical)		
Code	Name	Theory	Practical	--	Internal	External	Total
HS131 B	Philosophy	2	--	2	30	70	100
HS122 B	Values and Ethics	2	--	2	30	70	100
Total credits: 2, Total marks: 100							

*** University elective courses:

Elective Course		Credits		Total hours	Examination scheme (Theory/Practical)		
Code	Name	Theory	Practical	--	Internal	External	Total
CE281.01	Art Of Programming	2	-	2	30	70	100
CL281.01	Environmental Sustainability & Climate Change	2	-	2	30	70	100
EE283	Python for Electrical Engineers	2	-	2	30	70	100
IT281.01	ICT Resources & Multimedia	2	-	2	30	70	100
ME281.01	Engineering Drawing	2	-	2	30	70	100
PH233.01	Fundamentals Of Packaging	2	-	2	30	70	100
PD260.01	Basic Laboratory Techniques	2	-	2	30	70	100
NR251.01	First Aid & Life Support	2	-	2	30	70	100
PT191.01	Health Promotion & Fitness	2	-	2	30	70	100
CA224	Introduction To Web Designing	2	-	2	30	70	100
BM231	Banking & Insurance	2	-	2	30	70	100
EC281.01	Introduction To MATLAB Programming	2	-	2	30	70	100
PD261	Astro Physics, Space and Cosmos-1 (ASC-1)	2	-	2	30	70	100
Total		**2**	**-**	**2**	**30**	**70**	**100**
Total credits: 2, Total marks: 100							

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

BACHELOR OF PHARMACY PROGRAMME

SEMESTER-IV

SCHEME OF TEACHING

Course Code	Name of the Course	No. of hours	Tutorial	Credit Points
BPH4001	Pharmaceutical Organic Chemistry-III (Theory)	3	1	4
BPH4002	Biochemistry (Theory)	3	1	4
BPH4003	Physical Pharmaceutics-II (Theory)	3	1	4
BPH4004	Pharmacology-I (Theory)	3	1	4
BPH4005	Pharmacognosy-I (Theory)	3	1	4
BPH4002P	Biochemistry (Practical)	4	-	2
BPH4003P	Physical Pharmaceutics-II (Practical)	4	-	2
BPH4004P	Pharmacology-I (Practical)	4	-	2
BPH4005P	Pharmacognosy-I (Practical)	4	-	2
HS133 B	Creativity Problem solving and Innovation	2	-	2
---	University elective*	2	-	2
	Total	**35**	**5**	**32**

SCHEME OF EVALUATION

Course Code	Name of the Course	Marks Distribution			
		University (End Semester Exam)	Institute		Total
			Sessional Exams	Continuous Mode	
BPH4001	Pharmaceutical Organic Chemistry-III (Theory)	75	15	10	100
BPH4002	Biochemistry (Theory)	75	15	10	100
BPH4003	Physical Pharmaceutics-II (Theory)	75	15	10	100
BPH4004	Pharmacology-I (Theory)	75	15	10	100
BPH4005	Pharmacognosy-I (Theory)	75	15	10	100
BPH4002P	Biochemistry (Practical)	35	10	05	50
BPH4003P	Physical Pharmaceutics-II (Practical)	35	10	05	50
BPH4004P	Pharmacology-I (Practical)	35	10	05	50
BPH4005P	Pharmacognosy-I (Practical)	35	10	05	50
HS133 B	Creativity Problem solving and Innovation	70	30		100
---	University elective*	70	30		100
	Total	**655**	**245**		**900**

* University elective courses:

Elective Course		Credits		Total hours	Examination scheme (Theory/Practical)		
Code	**Name**	**Theory**	**Practical**	**--**	**Internal**	**External**	**Total**
EC282.01	Prototyping Electronics with Arduino	2	-	2	30	70	100
CE282.01	Web Designing	2	-	2	30	70	100
CL282.01	Basics Of Environmental Impact Assessment	2	-	2	30	70	100
EE286	Computer Programming for Electrical Engineering	2	-	2	30	70	100
IT282.01	Internet Technology & Web Design	2	-	2	30	70	100
ME282.01	Material Science	2	-	2	30	70	100
PH238.01	Cosmetics In Daily Life	2	-	2	30	70	100
NR261.01	Life Style Diseases & Management	2	-	2	30	70	100
PT192.01	Occupational Health & Ergonomics	2	-	2	30	70	100
CA225	Programming The Internet	2	-	2	30	70	100
BM241	Health Care Management	2	-	2	30	70	100
PD262	Astro Physics, Space and Cosmos-2 (ASC-2)	2	-	2	30	70	100
Total		**2**	**-**	**2**	**30**	**70**	**100**
Total credits: 2, Total marks: 100							

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

BACHELOR OF PHARMACY PROGRAMME

SEMESTER-5

SCHEME OF TEACHING

Course Code	Name of the Course	No. of hours	Tutorial	Credit Points
BPH5001	Medicinal Chemistry-I (Theory)	3	1	4
BPH5002	Industrial Pharmacy-I (Theory)	3	1	4
BPH5003	Pharmacology-II (Theory)	3	1	4
BPH5004	Pharmacognosy-II (Theory)	3	1	4
BPH5005	Pharmaceutical Jurisprudence (Theory)	3	1	4
BPH5001P	Medicinal Chemistry-I (Practical)	4		2
BPH5002P	Industrial Pharmacy-I (Practical)	4	-	2
BPH5003P	Pharmacology-II (Practical)	4	-	2
BPH5004P	Pharmacognosy-II (Practical)	4	-	2
HS124.01B	Professional Communication	2	-	2
Total		**33**	**5**	**30**

SCHEME OF EVALUATION

Course code	Name of the course	Marks Distribution			
		University (End Semester Exam)	Sessional Exams		Total
			Marks	Continuous Mode	
BPH5001	Medicinal Chemistry-I (Theory)	75	15	10	100
BPH5002	Industrial Pharmacy-I (Theory)	75	15	10	100
BPH5003	Pharmacology-II (Theory)	75	15	10	100
BPH5004	Pharmacognosy-II (Theory)	75	15	10	100
BPH5005	Pharmaceutical Jurisprudence (Theory)	75	15	10	100
BPH5001P	Medicinal Chemistry-I (Practical)	35	10	5	50
BPH5002P	Industrial Pharmacy-I (Practical)	35	10	5	50
BPH5003P	Pharmacology-II (Practical)	35	10	5	50
BPH5004P	Pharmacognosy-II (Practical)	35	10	5	50
HS124.01B	Professional Communication	70	30	-	100
Total		585	145	70	800

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

BACHELOR OF PHARMACY PROGRAMME

SEMESTER-6

SCHEME OF TEACHING

Course Code	Name of the Course	No. of hours	Tutorial	Credit Points
BPH6001	Medicinal Chemistry-II (Theory)	3	1	4
BPH6002	Pharmacology-III (Theory)	3	1	4
BPH6003	Herbal Drug Technology (Theory)	3	1	4
BPH6004	Industrial Pharmacy-II (Theory)	3	1	4
BPH6005	Pharmaceutical Biotechnology (Theory)	3	1	4
BPH6006	Instrumental Methods of Analysis (Theory)	3	1	4
BPH6002P	Pharmacology-III (Practical)	4	-	2
BPH6003P	Herbal Drug Technology (Practical)	4	-	2
BPH6006P	Instrumental Methods of Analysis (Practical)	4	-	2
HS 134B	Contributor Personality Development	2	-	2
Total		**32**	**6**	**32**

SCHEME OF EVALUATION

Course code	Name of the course	Marks Distribution			
		University (End Semester Exam)	Sessional Exams		Total
			Marks	Continuous Mode	
BPH6001	Medicinal Chemistry-II (Theory)	75	15	10	100
BPH6002	Pharmacology-III (Theory)	75	15	10	100
BPH6003	Herbal Drug Technology (Theory)	75	15	10	100
BPH6004	Industrial Pharmacy-II (Theory)	75	15	10	100
BPH6005	Pharmaceutical Biotechnology (Theory)	75	15	10	100
BPH6006	Instrumental Methods of Analysis (Theory)	75	15	10	100
BPH6002P	Pharmacology-III (Practical)	35	10	5	50
BPH6003P	Herbal Drug Technology (Practical)	35	10	5	50
BPH6006P	Instrumental Methods of Analysis (Practical)	35	10	5	50
HS 134B	Contributor Personality Development	70	30	-	100
Total		625	150	75	**850**

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

BACHELOR OF PHARMACY PROGRAMME

SEMESTER-7

SCHEME OF TEACHING

Course Code	Name of the Course	No. of hours	Tutorial	Credit Points
BPH7001	Medicinal Chemistry-III (Theory)	3	1	4
BPH7002	Quality Assurance (Theory)	3	1	4
BPH7003	Biopharmaceutics and Pharmacokinetics (Theory)	3	1	4
BPH7004	Pharmacy Practice (Theory)	3	1	4
BPH7005	Novel Drug Delivery System (Theory)	3	1	4
BPH7006	Practice School*	12	-	6
BPH7007	Entrepreneurship for Pharma professionals (Theory)*	1	-	1
BPH7001P	Medicinal Chemistry-III (Practical)	4	-	2
BPH7007P	Entrepreneurship for Pharma professionals (Project)	2	-	2
Total		**34**	**5**	**31**

* The subject experts at college level shall conduct examination.

SCHEME OF EVALUATION

Course code	Name of the course	Marks Distribution			
		University (End Semester Exam)	Sessional Exams		Total
			Marks	Continuous Mode	
BPH7001	Medicinal Chemistry-III (Theory)	75	15	10	100
BPH7002	Quality Assurance (Theory)	75	15	10	100
BPH7003	Biopharmaceutics and Pharmacokinetics (Theory)	75	15	10	100
BPH7004	Pharmacy Practice (Theory)	75	15	10	100
BPH7005	Novel Drug Delivery System (Theory)	75	15	10	100
BPH7006	Practice School*	125	-	25	150
BPH7007	Entrepreneurship for pharma professionals (Theory)*	-	50	-	50
BPH7001P	Medicinal Chemistry-III (Practical)	35	10	5	50
BPH7007P	Entrepreneurship for pharma professionals (Project)	80	20	-	100
Total		615	155	80	**850**

* The subject experts at college level shall conduct examination.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

BACHELOR OF PHARMACY PROGRAMME

SEMESTER-8

SCHEME OF TEACHING

Course Code	Name of the course	No. of hours	Tutorial	Credit Points
BPH8001	Biostatistics and Research Methodology (Theory)	3	1	4
BPH8002	Social and Preventive Pharmacy (Theory)	3	1	4
BPH8003	Pharma Marketing Management (Theory)	(Any two courses) 3+3=6	1+1=2	4+4=8
BPH8004	Pharmaceutical Regulatory Science (Theory)			
BPH8005	Pharmacovigilance (Theory)			
BPH8006	Quality Control and Standardization of Herbals (Theory)			
BPH8007	Computer Aided Drug Design (Theory)			
BPH8008	Cell and Molecular Biology (Theory)			
BPH8009	Cosmetic Science (Theory)			
BPH8010	Pharmacological Screening Methods (Theory)			
BPH8011	Advanced Instrumentation Techniques (Theory)			
BPH8012	Dietary Supplements and Nutraceuticals (Theory)			
BPH8013	Pharmaceutical Product Development (Theory)			
BPH8014	Project Work	12	-	6
	Total	**24**	**4**	**22**

SCHEME OF EVALUATION

<table>
<tr><th rowspan="3">Course Code</th><th rowspan="3">Name of the course</th><th colspan="4">Marks Distribution</th></tr>
<tr><th rowspan="2">University (End Semester Exam)</th><th colspan="2">Sessional Exams</th><th rowspan="2">Total</th></tr>
<tr><th>Marks</th><th>Continuous Mode</th></tr>
<tr><td>BPH8001</td><td>Biostatistics and Research Methodology (Theory)</td><td>75</td><td>10</td><td>15</td><td>100</td></tr>
<tr><td>BPH8002</td><td>Social and Preventive Pharmacy (Theory)</td><td>75</td><td>10</td><td>15</td><td>100</td></tr>
<tr><td>BPH8003</td><td>Pharma Marketing Management (Theory)</td><td rowspan="11">75 + 75
= 150</td><td rowspan="11">10 + 10
= 20</td><td rowspan="11">15 + 15
= 30</td><td rowspan="11">100 +100
=200</td></tr>
<tr><td>BPH8004</td><td>Pharmaceutical Regulatory Science (Theory)</td></tr>
<tr><td>BPH8005</td><td>Pharmacovigilance (Theory)</td></tr>
<tr><td>BPH8006</td><td>Quality Control and Standardization of Herbals (Theory)</td></tr>
<tr><td>BPH8007</td><td>Computer Aided Drug Design (Theory)</td></tr>
<tr><td>BPH8008</td><td>Cell and Molecular Biology (Theory)</td></tr>
<tr><td>BPH8009</td><td>Cosmetic Science (Theory)</td></tr>
<tr><td>BPH8010</td><td>Pharmacological Screening Methods (Theory)</td></tr>
<tr><td>BPH8011</td><td>Advanced Instrumentation Techniques (Theory)</td></tr>
<tr><td>BPH8012</td><td>Dietary Supplements and Nutraceuticals (Theory)</td></tr>
<tr><td>BPH8013</td><td>Pharmaceutical Product Development (Theory)</td></tr>
<tr><td>BPH8014</td><td>Project Work</td><td>150</td><td>-</td><td>-</td><td>150</td></tr>
<tr><td colspan="2">Total</td><td>450</td><td>40</td><td>60</td><td>550</td></tr>
</table>

Table-IX: Semester wise credits distribution

Semester	Credit Points
I	**29/31$/32#**
II	24
III	29
IV	32
V	30
VI	32
VII	31
VIII	22
Student can select a certificate course from SWAYAM, NPTEL, Etc. prescribe by Various Government Agencies	02*
Total credit points for the program	**229/231$/232#**

* The credit points assigned for a selection of certificate course/s from SWAYAM, NPTEL, Etc . prescribe by Various Government Agencies and submit a copy of course completion certificate, shall be given by the Principals of the colleges and the same shall be submitted to the University. The criteria to acquire this credit point shall be defined by the colleges from time to time.

$ Applicable ONLY for the students studied Physics/Chemistry/Botany/Zoology at HSC and appearing for Remedial Mathematics course.

Applicable ONLY for the students studied Mathematics/Physics/Chemistry at HSC and appearing for Remedial Biology course.

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme

SYLLABI

(Semester – 1)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH1001: Human Anatomy and Physiology-I (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to human body	10
	Cellular level of organization	
	Tissue level of organization	
2	Integumentary system	10
	Skeletal system	
	Joints	
3	Nervous system	10
4	Peripheral nervous system	8
5	Endocrine system	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Introduction to human body, Cellular level of organization, Tissue level of organization** Definition and scope of anatomy and physiology, levels of structural organization and body systems, basic life processes, homeostasis, basic anatomical terminology Structure and functions of cell, transport across cell membrane, cell division, cell junctions. General principles of cell communication, intracellular signalling pathway activation by extracellular signal molecule, Forms of intracellular signalling: a) Contact-dependent b) Paracrine c) Synaptic d) Endocrine Classification of tissues, structure, location and functions of epithelial, muscular and nervous and connective tissues	**10 Hours**	**22.22%**
2.	**Integumentary system, Skeletal system, Joints** Structure and functions of skin Divisions of skeletal system, types of bone, salient features and functions of bones of axial and appendicular skeletal system Organization of skeletal muscle, physiology of muscle contraction, neuromuscular junction Structural and functional classification, types of joints movements and its articulation	**10 Hours**	**22.22%**

3.	**Nervous system** Organization of nervous system, neuron, neuroglia, classification and properties of nerve fibre, electrophysiology, action potential, nerve impulse, receptors, synapse, neurotransmitters. Central nervous system: Meninges, ventricles of brain and cerebrospinal fluid. Structure and functions of brain (cerebrum, brain stem, cerebellum), spinal cord (gross structure, functions of afferent and efferent nerve tracts, reflex activity)	**10 Hours**	**22.22%**
4.	**Peripheral nervous system** Classification of peripheral nervous system: Structure and functions of sympathetic and parasympathetic nervous system. Origin and functions of spinal and cranial nerves.	**8 Hours**	**17.77%**
5.	**Endocrine system** Classification of hormones, mechanism of hormone action, structure and functions of pituitary gland, thyroid gland, parathyroid gland, adrenal gland, pancreas, pineal gland, thymus and their disorders.	**7 Hours**	**15.55%**

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe gross morphology, structure and functions of various organs of the human body.
CO2	write various homeostatic mechanisms and their imbalances.
CO3	identify, draw and differentiate various tissues and organs of different systems of human body.
CO4	summarize coordinated working pattern of different organs of each system

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	3	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH1001P: Human Anatomy and Physiology-I (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the Practical
1	Study of compound microscope.
2	Microscopic study of epithelial and connective tissue
3	Microscopic study of muscular and nervous tissue
4	Identification of axial bones
5	Identification of appendicular bones
6	To study the integumentary and special senses using
7	To study the nervous system using specimen, models,
8	To study the endocrine system using specimen, models,
9	To demonstrate the general neurological examination
10	To demonstrate the function of olfactory nerve
11	To examine the different types of taste.
12	To demonstrate the visual acuity
13	To demonstrate the reflex activity
14	Recording of body temperature
15	To demonstrate positive and negative feedback

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	distinguish various tissues by observing morphology and structure through microscopy
CO2	identify various organs of the different body systems and describe their functions
CO3	identify various bones
CO4	record pulse rate and blood pressure
CO5	analyse the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	3	3	-	-	-	-	-	-	-
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

- **Textbook:**
    1. Essentials of Medical Physiology by K. Sembulingam and P. Sembulingam. Jaypee brother's medical publishers, New Delhi.
    2. Anatomy and Physiology in Health and Illness by Kathleen J.W. Wilson, Churchill, Livingstone, New York
    3. Physiological basis of Medical Practice-Best and Tailor. Williams & Wilkins Co,Riverview, MI USA
    4. Text book of Medical Physiology- Arthur C,Guyton andJohn.E. Hall. Miamisburg, OH, U.S.A
    5. Principles of Anatomy and Physiology by Tortora Grabowski. Palmetto, GA, U.S.A.
    6. Textbook of Human Histology by Inderbir Singh, Jaypee brother‘s medical publishers, New Delhi.
    7. Textbook of Practical Physiology by C.L. Ghai, Jaypee brother‘s medical publishers, New Delhi.
    8. Practical workbook of Human Physiology by K. Srinageswari and Rajeev Sharma, Jaypee brother‘s medical publishers, New Delhi.
- **Reference book:**
    1. Physiological basis of Medical Practice-Best and Tailor. Williams & Wilkins Co, Riverview, MI USA
    2. Textbook of Medical Physiology- Arthur C, Guyton and John. E. Hall. Miamisburg, OH, U.S.A.
    3. Human Physiology (vol 1 and 2) by Dr. C.C. Chatterrje,Academic Publishers Kolkata
- **Web Materials:**
    1. http:// www.visiblebody.com/
    2. http://www.getbodysmart.com/
    3. http:// www.innerbody.com
    4. http:// libguides.middlesex.mass.edu/
    5. http://wps.aw.com/bc_marieb_ehap_8/
    6. http://scioly.org/wiki/index.php/Anatomy_and_Physiology

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH1002: Pharmaceutical Analysis (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Pharmaceutical analysis	10
	Errors	
2	Acid base titration	10
	Non aqueous titration	
3	Precipitation titrations	10
	Complexometric titration	
	Gravimetry	
4	Redox titrations	8
5	Electrochemical methods of analysis	7
	Conductometry	
	Potentiometry	
	Polarography	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Pharmaceutical analysis, Errors**	**10 Hours**	**22.22%**
	Definition and scope Different techniques of analysis Methods of expressing concentration Primary and secondary standards. Preparation and standardization of various molar and normal solutions-Oxalic acid, sodium hydroxide, hydrochloric acid, sodium thiosulphate, sulphuric acid, potassium permanganate and ceric ammonium sulphate. Sources of errors, types of errors, methods of minimizing errors, accuracy, precision and significant figures.		
2.	**Acid base titration, Non-aqueous titration**	**10 Hours**	**22.22%**
	Theories of acid base indicators, classification of acid base titrations and theory involved in titrations of strong, weak, and very weak acids and bases, neutralization curves. Solvents, acidimetry and alkalimetry titration and estimation of Sodium benzoate and Ephedrine HCl.		

3.	**Precipitation titrations, Complexometric titration, Gravimetry** Mohr‘s method, Volhard‘s, Modified. Classification, metal ion indicators, masking and demasking reagents, estimation of Magnesium sulphate, and calcium gluconate. Principle and steps involved in gravimetric analysis. Purity of the precipitate: co-precipitation and post precipitation, Estimation of barium sulphate.	**10 Hours**	**22.22%**
4.	**Redox titrations** (a) Concepts of oxidation and reduction (b) Types of redox titrations (Principles and applications) Cerimetry, Iodimetry, Iodometry, Bromatometry, Dichrometry, Titration with potassium iodate.	**8 Hours**	**17.77%**
5.	**Electrochemical methods of analysis, Conductometry, Potentiometry, Polarography** Introduction, Conductivity cell, Conductometric titrations, applications. Electrochemical cell, construction and working of reference (Standard hydrogen, silver chloride electrode and calomel electrode) and indicator electrodes (metal electrodes and glass electrode), methods to determine end point of potentiometric titration and applications. Principle, Ilkovic equation, construction and working of dropping mercury electrode and rotating platinum electrode, applications	**7 Hours**	**15.55%**

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe principle and applications of various types of volumetric and electrochemical methods of pharmaceutical analysis.
CO2	describe principle and applications of gravimetric analysis.
CO3	perform calculations necessary to prepare desired concentration of standard, test solutions and calculation related to titration curve for various types of volumetric methods of pharmaceutical analysis.
CO4	describe various errors to be involved in pharmaceutical analysis, their sources and proposing mitigation strategies for analytical errors.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH1002P: Pharmaceutical Analysis (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the Unit
1	Preparation and standardization of
	(a) Sodium hydroxide
	(b) Sulphuric acid
	(c) Sodium thiosulfate
	(d) Potassium permanganate
	(e) Ceric ammonium sulphate
2	Assay of the following compounds along with
	(a) Ammonium chloride by acid base titration
	(b) Ferrous sulphate by Cerimetry
	(c) Copper sulphate by Iodometry
	(d) Calcium gluconate by complexometry
	(e) Hydrogen peroxide by Permanganometry
	(f) Sodium benzoate by non-aqueous titration
	(g) Sodium Chloride by precipitation titration
3	Determination of Normality by electro-analytical methods
	(a) Conductometric titration of strong acid against strong base
	(b) Conductometric titration of strong acid and weak acid against strong base
	(c) Potentiometric titration of strong acid against strong base

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	prepare and standardise various titration solutions.
CO2	perform various experimental task for volumetric and electrochemical titration.
CO3	handle and operate various laboratory instruments for electrochemical analysis.
CO4	describe principle and various terminologies related to volumetric and electrochemical analysis.
CO5	analyse the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	-	3	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	3
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

- ❖ **Textbook/ Reference book**

1. A.H. Beckett & J.B. Stenlake's, Practical Pharmaceutical Chemistry Vol I & II, Stahlone
2. Press of University of London
3. A.I. Vogel, Textbook of Quantitative Inorganic analysis
4. P. Gundu Rao, Inorganic Pharmaceutical Chemistry
5. Bentley and Driver's Textbook of Pharmaceutical Chemistry
6. John H. Kennedy, Analytical chemistry principles
7. Indian Pharmacopoeia.

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH1003: Pharmaceutics-I (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Historical background and development of profession of pharmacy	10
	Dosage forms	
	Prescription	
	Posology	
2	Pharmaceutical calculations	10
	Powders	
	Liquid dosage forms	
3	Monophasic liquids	8
	Biphasic liquids	
	Suspensions	
	Emulsions	
4	Suppositories	8
	Pharmaceutical incompatibilities	
5	Semisolid dosage forms	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Historical background and development of profession of pharmacy, Dosage forms, Prescription, Posology**	**10 Hours**	**22.22%**
	History of profession of Pharmacy in India in relation to pharmacy education, industry and organization, Pharmacy as a career, Pharmacopoeias: Introduction to IP, BP, USP and Extra Pharmacopoeia. Introduction to dosage forms, classification, and definitions Definition, Parts of prescription, handling of Prescription and Errors in prescription. Definition, Factors affecting posology. Pediatric dose calculations based on age, body weight and body surface area		
2.	**Pharmaceutical calculations, Powders, Liquid dosage forms.**	**10 Hours**	**22.22%**
	Weights and measures – Imperial & Metric system, Calculations involving percentage solutions, alligation, proof spirit and isotonic solutions based on freezing point and molecular weight.		

	Definition, classification, advantages and disadvantages, Simple & compound powders – official preparations, dusting powders, effervescent, efflorescent, and hygroscopic powders, eutectic mixtures. Geometric dilutions. Advantages and disadvantages of liquid dosage forms. Excipients used in formulation of liquid dosage forms. Solubility enhancement techniques		
3.	**Monophasic liquids, Biphasic liquids, suspensions, Emulsions**	**8 Hours**	**17.77%**
	Definitions and preparations of Gargles, Mouthwashes, Throat Paint, Eardrops, Nasal drops, Enemas, Syrups, Elixirs, Liniments and Lotions. Definition, advantages and disadvantages, classifications, Preparation of suspensions; Flocculated and Deflocculated suspension & stability problems and methods to overcome. Definition, classification, emulsifying agent, test for the identification of type of Emulsion, Methods of preparation & stability problems and methods to overcome.		
4.	**Suppositories, Pharmaceutical incompatibilities**	**8 Hours**	**17.77%**
	Definition, types, advantages and disadvantages, types of bases, methods of preparations. Displacement value & its calculations, evaluation of suppositories. Definition, classification, physical, chemical, and therapeutic incompatibilities with examples.		
5.	**Semisolid dosage forms**	**7 Hours**	**15.55%**
	Definitions, classification, mechanisms, and factors influencing dermal penetration of drugs. Preparation of ointments, pastes, creams and gels. Excipients used in semi solid dosage forms. Evaluation of semi solid dosages forms.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	summarize the history of profession of pharmacy.
CO2	describe the basics of different dosage forms, pharmaceutical incompatibilities, and compute pharmaceutical calculations.
CO3	interpret and analyse the prescription.
CO4	describe methodology for preparation of various conventional dosage forms and rationalize them.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	3	3	-	3	-	-
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	3	-	-	3	-	3
CO4	3	-	-	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH1003P: Pharmaceutics-I (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Syrups
	Syrup IP
	Paracetamol paediatric syrup
2	Elixirs
	Piperazine citrate elixir
	Paracetamol paediatric elixir
3	Linctus
	(a) Simple Linctus BPC
4	Solutions
	(a) Strong solution of ammonium acetate
	(b) Cresol with soap solution
5	Suspensions
	(a) Calamine lotion
	(b) Magnesium Hydroxide mixture
	(c) Emulsions
	(d) Turpentine Liniment
	(e) Liquid paraffin emulsion
6	Powders and Granules
	(a) ORS powder (WHO)
	(b) Effervescent granules
	(c) Dusting powder
7	Suppositories
	(a) Glycero gelatin suppository
	(b) Soap glycerin suppository
8	Semisolids
	(a) Sulphur ointment
	(b) Non staining iodine ointment with methyl salicylate
	(c) Bentonite gel

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	interpret the formula and prepare solid dosage forms.
CO2	interpret the formula and prepare semi solid dosage forms.
CO3	interpret the formula and prepare liquid dosage forms.
CO4	analyse the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. H.C. Ansel et al., Pharmaceutical Dosage Form and Drug Delivery System, Lippincott Williams and Walkins, New Delhi.
2. Carter S.J., Cooper and Gunn‘s-Dispensing for Pharmaceutical Students, CBS publishers, New Delhi.
3. M.E. Aulton, Pharmaceutics, The Science& Dosage Form Design, Churchill Livingstone, Edinburgh.
4. Indian pharmacopoeia.
5. British pharmacopoeia.
6. Lachmann. Theory and Practice of Industrial Pharmacy,Lea& Febiger Publisher, TheUniversity of Michigan.
7. Alfonso R. Gennaro Remington. The Science and Practice of Pharmacy, Lippincott Williams, New Delhi.
8. Carter S.J., Cooper and Gunn‘s. Tutorial Pharmacy, CBS Publications, New Delhi.
9. E.A. Rawlins, Bentley‘s Text Book of Pharmaceutics, English Language Book Society, Elsevier Health Sciences, USA.
10. Isaac Ghebre Sellassie: Pharmaceutical Pelletization Technology, Marcel Dekker, INC, New York.
11. Dilip M. Parikh: Handbook of Pharmaceutical Granulation Technology, Marcel Dekker, INC, New York.
12. Francoise Nieloud and Gilberte Marti-Mestres: Pharmaceutical Emulsions and Suspensions, Marcel Dekker, INC, New York.

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH1004: Pharmaceutical Inorganic Chemistry (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Impurities in pharmaceutical substances	10
	General methods of preparation	
	Prescription	
	Posology	
2	Acids, Bases and Buffers	10
	Major extra and intracellular electrolytes	
	Dental products	
3	Gastrointestinal agents	10
	Acidifiers	
	Antacid	
	Cathartics	
	Antimicrobials	
4	Miscellaneous compounds Expectorants	8
	Emetics	
	Haematinics	
	Poison and Antidote	
	Astringents	
5	Radiopharmaceuticals	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Impurities in pharmaceutical substances, General Method of preparations.**	10 Hours	22.22%
	History of Pharmacopoeia, Sources and types of impurities, principle involved in the limit test for Chloride, Sulphate, Iron, Arsenic, Lead and Heavy metals, modified limit test for Chloride and Sulphate. Assay for the compounds superscripted with asterisk (*), properties and medicinal uses of inorganic compounds belonging to the following classes		
2.	**Acids, Bases and Buffers, Major extra and intracellular electrolytes, Dental products**	10 Hours	22.22%
	Acids, Bases and Buffers: Buffer equations and buffer capacity in general, buffers in pharmaceutical systems, preparation, stability, buffered isotonic		

	solutions, measurements of tonicity, calculations, and methods of adjusting isotonicity. **Major extra and intracellular electrolytes**: Functions of major physiological ions, Electrolytes used in the replacement therapy: Sodium chloride*, Potassium chloride, Calcium gluconate* and Oral Rehydration Salt (ORS), Physiological acid base balance. **Dental products** : Dentifrices, role of fluoride in the treatment of dental caries, Desensitizing agents, Calcium carbonate, Sodium fluoride, and Zinc eugenol cement.		
3.	**Gastrointestinal agents, Acidifiers, Antacid, Cathartics, Antimicrobials.**	**10 Hours**	**22.22%**
	Acidifiers: Ammonium chloride* and Dil. HCl **Antacid:**Ideal properties of antacids, combinations of antacids,Sodium Bicarbonate*, Aluminum hydroxide gel, Magnesium hydroxide mixture **Cathartics:** Magnesium sulphate, Sodium orthophosphate, Kaolin and Bentonite **Antimicrobials**: Mechanism, classification, Potassium permanganate, Boric acid, Hydrogen peroxide*, Chlorinated lime*, Iodine and its preparations		
4.	**Miscellaneous compounds Expectorants, Emetics, Haematinics, Poison and Antidote, Astringents**	**8 Hours**	**17.77%**
	Expectorants: Potassium iodide, mmonium chloride*. **Emetics**: Copper sulphate*, Sodium potassium tartarate **Haematinics:** Ferrous sulphate*, Ferrous gluconate **Poison and Antidote:** Sodium hiosulphate*, Activated charcoal, Sodium nitrite333 **Astringents**: Zinc Sulphate, Potash Alum		
5.	**Radiopharmaceuticals**	**7 Hours**	**15.55%**
	Radio activity, Measurement of radioactivity, Properties of α, β, γ radiations, Half-life, radio isotopes and study of radio isotopes - Sodium iodide I^{131}, Storage conditions, precautions & pharmaceutical application of radioactive substances.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	summarize the history and basics of pharmaceutical inorganic chemistry.
CO2	classify various sources of contamination in pharmaceuticals
CO3	describe limit test and its significance
CO4	interpret monograph of selected inorganic pharmaceutical compounds
CO5	describe basics of radio pharmaceuticals and their therapeutic as well as diagnostic applications.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	3
CO5	3	-	-	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH1004P: Pharmaceutical Inorganic Chemistry (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Limit tests for following ions
	(a) Limit test for Chlorides and Sulphates
	(b) Modified limit test for Chlorides and Sulphates
	(c) Limit test for Iron
	(d) Limit test for Heavy metals
	(e) Limit test for Lead
	(f) Limit test for Arsenic
2	Identification test
	(a) Magnesium
	(b) Ferrous hydroxide
	(c) Sodium sulphate
	(d) Calcium bicarbonate
	(e) Copper gluconate
	(f) Sulphate
3	Test for purity
	(a) Swelling power of Bentonite
	(b) Neutralizing capacity of aluminium hydroxide gel
	(c) Determination of potassium iodate and iodine
4	Preparation of inorganic pharmaceuticals
	(a) Boric acid
	(b) Potash alum
	(c) Ferrous
	(d) Sulphate

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	perform limit test as per the methods given in IP.
CO2	identify given inorganic compounds through chemical tests.
CO3	perform quantitative analysis of selected inorganic compounds.
CO4	prepare inorganic pharmaceuticals following pharmacopeial procedures.
CO5	analyse the problem, communicate suggested solution, and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	3	-	-	-	-	-	-	-	-
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. A.H. Beckett & J.B. Stenlake's, Practical Pharmaceutical Chemistry Vol I & II, Stahlone Press of University of London, 4 th edition.
2. A.I. Vogel, Textbook of Quantitative Inorganic analysis
3. P. Gundu Rao, Inorganic Pharmaceutical Chemistry, 3 rd Edition
4. M.L Schroff, Inorganic Pharmaceutical Chemistry
5. Bentley and Driver's Textbook of Pharmaceutical Chemistry
6. Anand & Chatwal, Inorganic Pharmaceutical Chemistry

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH1005: Remedial Biology (Theory)

Total hours: 30 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Living world	7
	Morphology of Flowering plants	
2	Body fluids and circulation	7
	Digestion and Absorption	
	Breathing and respiration	
3	Excretory products and their elimination	7
	Neural control and coordination	
	Chemical coordination and regulation	
	Human reproduction	
4	Plants and mineral nutrition	5
	Photosynthesis	
5	Plant respiration	4
	Plant growth and development	
	Cell - The unit of life	
	Tissues	

A. Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Living world, Morphology of Flowering plants.**	**7 Hours**	**23.33%**
	Definition and characters of living organisms, Diversity in the living world, Binomial nomenclature, Five kingdoms of life and basis of classification. Salient features of Monera, Potista, Fungi, Animalia and Plantae, Virus. Morphology of different parts of flowering plants – Root, stem, inflorescence, flower, leaf, fruit, seed, General Anatomy of Root, stem, leaf of monocotyledons & Dicotylidones.		
2.	**Body fluids and circulation, Digestion and Absorption, Breathing and respiration.**	**7 Hours**	**23.33%**
	Composition of blood, blood groups, coagulation of blood, Composition and functions of lymp, Human circulatory system, Structure of human heart and blood vessels, Cardiac cycle, cardiac output and ECG. Human alimentary canal and digestive glands, Role of digestive enzymes, Digestion, absorption and assimilation of digested food.		

	Human respiratory system, Mechanism of breathing and its regulation, Exchange of gases, transport of gases and regulation of respiration, Respiratory volumes.		
3.	**Excretory products and their elimination, Neural control and coordination, Chemical coordination and regulation, Human reproduction**	**7 Hours**	**23.33%**
	Modes of excretion, Human excretory system- structure and function, Urine formation, Rennin angiotensin system. Definition and classification of nervous system, Structure of a neuron, Generation and conduction of nerve impulse, Structure of brain and spinal cord, Functions of cerebrum, cerebellum, hypothalamus and medulla oblongata. Endocrine glands and their secretions, Functions of hormones secreted by endocrine glands. Parts of female reproductive system, Parts of male reproductive system, Spermatogenesis and Oogenesis, Menstrual cycle.		
4.	**Plants and mineral nutrition, Photosynthesis**	**5 Hours**	**16.66%**
	Essential mineral, macro and micronutrients Nitrogen metabolism, Nitrogen cycle, biological nitrogen fixation. Autotrophic nutrition, photosynthesis, Photosynthetic pigments, Factors affecting photosynthesis.		
5.	**Plant respiration, Plant growth and development, Cell - The unit of life, Tissues**	**4 Hours**	**13.33%**
	Respiration, glycolysis, fermentation (anaerobic). Phases and rate of plant growth, Condition of growth, Introduction to plant growth regulators. Structure and functions of cell and cell organelles. Cell division. Definition, types of tissues, location and functions.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	identify diversity of living organism and its characeteristic
CO2	understand various systems of human body.
CO3	define essential requirement of plants nutrition, plant respiration, plant growth and development
CO4	identify types of cells and tissues

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH1005P: Remedial Biology (Practical)

Total hours: 30 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Introduction to experiments in biology
	(a) Morphology of Flowering plants
	(b) Study of Microscope
	(c) Section cutting techniques
	(d) Mounting and staining
	(e) Permanent slide preparation
2	Study of cell and its inclusions
3	Study of Stem, Root, Leaf and its modifications
4	Detailed study of frog by using computer models
5	Microscopic study and identification of tissues
6	Identification of bones
7	Determination of blood group
8	Determination of blood pressure
9	Determination of tidal volume

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	differentiate various cells and tissues through histological examination.
CO2	identify various types of cells inclusion.
CO3	assess blood pressure, blood group and tidal volume.
CO4	identify and recognize different bones of human body.
CO5	analyse the problem, communicate suggested solution, and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

- ❖ **Reference book:**

1. Practical human anatomy and physiology. by S.R.Kale and R.R.Kale.

2. A Manual of pharmaceutical biology practical by S.B.Gokhale, C.K.Kokate and S.P.Shriwastava.

3. Biology practical manual according to National core curriculum. Biology forum of Karnataka. Prof .M.J.H.Shafi

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
HS1001: Communication Skills (Theory)

Total hours: 30 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Communication Skills	7
	Barriers to communication	
	Perspectives in Communication	
2	Elements of Communication	7
	Communication Styles	
3	Basic Listening Skills	7
	Effective Written Communication	
	Writing Effectively	
4	Interview Skills	5
5	Group discussion	4

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Communication Skills, Barriers to communication, Perspectives in Communication.**	**7 Hours**	**23.33%**
	Introduction, Definition, The Importance of Communication, The Communication Process – Source, Message, Encoding, Channel, Decoding, Receiver, Feedback, Context. Physiological Barriers, Physical Barriers, Cultural Barriers, Language Barriers, Gender Barriers, Interpersonal Barriers, Psychological Barriers, Emotional barriers. Introduction, Visual Perception, Language, Other factors affecting our perspective - Past Experiences, Prejudices, Feelings, Environment.		
2.	**Elements of Communication, Communication Styles:**	**7 Hours**	**23.33%**
	Introduction, Face to Face Communication - Tone of Voice, Body Language (Non-verbal communication), Verbal Communication, Physical Communication. Introduction, The Communication Styles Matrix with example for each - Direct Communication Style, Spirited Communication Style, Systematic Communication Style, Considerate Communication Style.		
3.	**Basic Listening Skills, Writing Effectively, Effective Written Communication**	**7 Hours**	**23.33%**

	Introduction, Self-Awareness, Active Listening, becoming an Active Listener, Listening in Difficult Situations. Introduction, When and When Not to Use Written Communication - Complexity of the Topic, Amount of Discussion Required, Shades of Meaning, Formal Communication. Subject Lines, Put the Main Point First, Know Your Audience, Organization of the Message.		
4.	**Interview Skills, Giving Presentations**	**5 Hours**	**16.66%**
	Purpose of an interview, Do's and Dont's of an interview. Dealing with Fears, Planning your Presentation, Structuring Your Presentation, Delivering Your Presentation, Techniques of Delivery.		
5.	**Group Discussion**	**4 Hours**	**13.33%**
	Introduction, Communication skills in group discussion, Do's and Don'ts of group discussion.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	Understand the behavioural needs for a Pharmacist to function effectively in the areas of pharmaceutical operation
CO2	Identify Leadership qualities and essential skills.
CO3	Summarize Verbal and Non-Verbal communication effectively.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	3	-	-	3
CO2	-	-	-	-	3	-	-	3	-	-	-
CO3	-	-	-	-	-	-	-	3	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
HS1001P: Communication Skills (Practical)

Total hours: 30 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Basic communication covering the following topics
	(a) Meeting People
	(b) Asking Questions
	(c) Making Friends
	(d) What did you do?
	(e) Do‘s and Dont‘s
2	Pronunciations covering the following topics
	(a) Pronunciation (Consonant Sounds)
	(b) Pronunciation (Vowel Sounds)
3	Advanced Learning
	(a) Listening Comprehension / Direct and Indirect Speech
	(b) Figures of Speech
	(c) Effective Communication Writing Skills
	(d) Effective Writing
	(e) Interview Handling Skills
	(f) E-Mail etiquette
	(g) Presentation Skills

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	Identify Basic communication skill.
CO2	Differentiate Consonant and Vowel Sounds.
CO3	Learn Writing Skills, Interview Handling Skills and Presentation Skills.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	-	-	-	-	-	-	3	-	-	3
CO2	-	-	-	-	-	-	-	3	-	-	-
CO3	-	-	-	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Basic communication skills for Technology, Andreja. J. Ruther Ford, 2 nd Edition, Pearson Education, 2011
2. Communication skills, Sanjay Kumar, Pushpalata, 1 st Edition, Oxford Press, 2011
3. Organizational Behaviour, Stephen .P. Robbins, 1 st Edition, Pearson, 2013
4. Brilliant- Communication skills, Gill Hasson, 1 st Edition, Pearson Life, 2011
5. The Ace of Soft Skills: Attitude, Communication and Etiquette for success, Gopala
6. Swamy Ramesh, 5 thEdition, Pearson, 2013 7. Developing your influencing skills, Deborah Dalley, Lois Burton, Margaret, Green hall, 1st Edition Universe of Learning LTD, 2010
7. Communication skills for professionals, Konar nira, 2 ndEdition, New arrivals
8. PHI, 2011 9. Personality development and soft skills, Barun K Mitra, 1 st Edition, Oxford Press, 10. 2011
9. Soft skill for everyone, Butter Field, 1st Edition, Cengage Learning india pvt.ltd, 2011
10. Soft skills and professional communication, Francis Peters SJ, 1 st Edition, Mc Graw 12. Hill Education, 2011
11. Effective communication, John Adair, 4 th Edition, Pan Mac Millan,2009
12. Bringing out the best in people, Aubrey Daniels, 2 nd Edition, Mc Graw Hill, 1999

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
MTH1001: Remedial Mathematics (Theory)

Total hours: 30 Hr

Outline of the course:

No.	Unit	Minimum number of hours
1	Sets, Relations and Functions	6
2	Trigonometry	8
3	Limit, Continuity and Differentiation	8
4	Integration	8
Total		**30**

Detailed Syllabus:

No.	Unit details	Contact hours	Approx. Weightage %
1	**Sets, Relations and Functions**	6	20
	Sets, Number systems (Real and Complex numbers) Cartesian Product of sets, Relation Functions and their types Types of function: One-One, Many-One, into, onto Classification of function : Algebraic , Transcendental-exponential and logarithmic functions Visualization of graphs of standard functions Application of logarithms in pharmaceutical computations		
2	**Trigonometry**	8	26.66
	Measurement of angle and trigonometric functions Trigonometric -ratios, addition, subtraction and transformation formulae, Trigonometric -ratios of multiple, submultiple, Allied and certain angles.		
3	**Limit, Continuity and Differentiation**	8	26.66
	Limits of functions and evaluation Continuity of functions Differentiation Derivatives of Standard Functions and evaluation Derivative as a Rate of Change Visualization of graphs of Continuous and Differentiable functions		

4	**Integration** Indefinite Integrals (Primitives / Anti derivatives) Primitives of Standard Functions Methods of Integration Definite Integral Integration as Area under the curve	8	26.66

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	understand basic concepts of functions of single variable and characteristics (types) of function through plots. Solution of equations
CO2	understand the algebra of matrices, basic concept of Statistics, computing descriptive statistics.
CO3	understand the concept of Integration and differentiation for future need

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	-	-	-	-	-	-	-	-	-	3
CO2	-	-	-	-	-	-	-	-	-	-	3
CO3	-	-	-	-	-	-	-	-	-	-	3

Recommended study materials:

1. Kreyszig, Erwin. Advanced engineering mathematics. John Wiley & Sons, 2010.
2. Stewart, James. "Calculus: Early Transcendentals, 6E." Belmont, CA: Thompson Brooks/Cole (2006).
3. Wylie, C. R., and L. C. Barrett. "Advanced Engineering Mathematics." McGraw-Hill,1982
4. Greenberg, Michael D. Advanced engineering mathematics. Prentice-Hall, 1988.
5. Thomas, G. B., and R. L. Finney. "Calculus with Analytic Geometry (9th Edition), 1996.", Addison Wesley Publishing.
6. Stewart, James, Lothar Redlin, and Saleem Watson. Algebra and trigonometry. Nelson Education, 2015.

A COURSE FROM LIBERAL ARTS (HS101B -HS106B)

Contact Hours: 30

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS101-HS106	A Course from Liberal Arts	02	02	--	--	50	50	100

II. Course Objectives

To help learners to

- recognize the nature of aesthetic values and explore elements of arts and aesthetics with reference to personal, cultural and civic sphere
- connect art and aesthetics with Science and Technology to understand and extend research and innovation for a society

III. Courses

Students may select **any one course** from the ones given below. However, depending upon the group size and availability of resource person(s), student's selection may be finalized by Faculty of Management Studies.

Sr. No.	Course Code	Course Title(s)	Credits
1.	HS101	Painting	02
2.	HS102	Photography	
3.	HS103	Sculpting	
4.	HS104	Pottery and Ceramic Art	
5.	HS105	Media and Graphic Design	
6.	HS106	Art and Craft	

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It may also run in the workshop mode.

V. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 50 marks for internal evaluation and 50 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	10	10
2	Performance/ Activities	-	10	10
3	Project	-	25	25
4	Attendance	-	05	05
			Total	**50**

External Evaluation

University Practical examination will be for 50 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	50	50
			Total	**50**

VI. Learning Outcomes

At the end of the course, students will have developed the ability to enjoy, interact with and perform arts and aesthetics; and will have developed the ability and creativity to transfer sense of design and innovation in science and technology.

PAINTING (HS101 B)

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS101 B	Painting	02	02	--	--	50	50	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**An Introduction toPainting** • *An Introduction to Painting* • *Principles of Composition* • *Medium and Techniques of Painting* • *History of Painting: Folk Indian Painting / Western Painting* • *2D and 3D Painting*	2
2	**Drawing from Nature and Object** • *Objects of Drawing: Nature and Manmade / Artificial Objects* • *Drawing Still / Live Objects* • *Drawing from Memory* • *Drawing from Life*	4
3	**Colour Design and Colour Value** • ***Color Theory****:* *Color wheel (primary/secondary, complementary), transparency/opacity, hue, value (intensity, brightness), chroma (saturation, purity) & temperature (warm/cold)* • ***Color Contrast & Attributes****:* *Interaction, harmony, psychology/mood, culture & expression* • ***Media Characteristics & Surfaces****:* *Acrylic, oil, paper, wood & canvas (primed/unprimed)*	6
4	**Composition and Perspective** • ***Composition:*** *Space, movement, balance, asymmetry, rhythm, shapes, proportion & lighting* • *Perspective: An approximate reproduction* • ***Types of Perspectives:*** *Linear Perspective, One-point Perspective, Two-point Perspective, Three-point Perspective, Four-Point Perspective*	6

5	**Figure Drawing and Proportion** • *Proportions of the Human Body* • *Three views– Anterior (front), Lateral (side) and Posterior (back)* • *Fundamental Proportion – The Big Three*	4
6	**Sketching** • *Sketching and Freehand* • *Sketching Techniques* • *Sketch and Drawing Medium*	4
7	**Contemporary Issues in Painting** • *Contemporary Indian Art* • *Pioneers of Contemporary Indian Art* • *Contemporary Issues in Painting*	4
	Total Hours	**30**

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops(each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 50 marks for internal evaluation and 50 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	10	10
2	Performance/ Activities	-	10	10
3	Project	-	25	25
4	Attendance	-	05	05
			Total	**50**

External Evaluation

University Practical examination will be for 50 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	50	50
			Total	**50**

V. Major Learning Outcomes

At the end of the course, a student will

1. have cultivated a sense of creativity.
2. beappreciative of art history, art criticism and aesthetics.
3. be able to recognize the elements of arts in painting.
4. have better cognizance and association of meaning of colors, shapes, and composition.

5. be able to acknowledge the principles of painting as in design and colors, concept, medium and formats.
6. have instantaneous painting experience about designing, lights, shades and colors and such other important aspects.

PHOTOGRAPHY (HS102 B)

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS102 B	Photography	02	02	--	--	50	50	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**An Introduction to Photography** • *Art, Design and Visualization* • *Basics of Photography and Various Types of Photography* • *Basics of Post Production* • *A Brief History of Photography:* *Early Experiments and Later Developments*	**03**
2	**Camera and Operating System** • *Role of Camera in the Photography* • *Types of Camera* *Pin-hole, box, folding, large and medium format cameras, single lens reflex (SLR) and twin lens reflex (TLR), miniature, subminiature and instant camera* • *Principal Parts of Photographic Camera* *Lens, Aperture, Shutters, various types and their functions, focal plane shutter and in-between the lens shutter, shutter synchronization, self-timer* • *Types of Lenses* *Single (meniscus), achromatic, symmetrical and unsymmetrical lenses, telephoto, zoom, macro, supplementary and fish-eye lenses* • *Different Models of Camera, their Features and Operating Systems* • *Camera and Size of the Image, Speed and Power of Lens*	**05**
3	**Light and Shade** • *Reflection and refraction of light* • *Dispersion of light through a glass prism, lenses* • *Colour Filters:* *Different kinds, Red, yellow, green, neutral density, half filters, filter factor, colour correction filter* • *Photographic Light Sources:* *Natural source, the Sun, nature and intensity of the sunlight at different times of the day, different weather conditions* • *Artificial light sources:*	**10**

	Nature, intensity of different types of light sources used in photography namely; (i) Photo flood lamp, (ii) Spot light, (iii) Halogen lamp, Barn doors and snoot, lighting stands • *Flash unit: Bulb flash and Electronic flash, main components, electronic flash units, studio flash, slave unit, multiple flash, computer flash, x-contact, exposure table*	
4	**Composition** • *Different kinds of image formations* • *Principal focus and focal length of the lens* • *Depth of field, angle of view and perspective* • *Perspective and composition* • *Rules of composition*	**09**
5	**Contemporary Issues in Photography** • *Present Day Photography* • *Contemporary Photographers and their Contributions* • *Major Issues in Contemporary Photography*	**03**
	Total Hours	**30**

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 50 marks for internal evaluation and 50 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	10	10
2	Performance/ Activities	-	10	10
3	Project	-	25	25
4	Attendance	-	05	05
			Total	**50**

External Evaluation

University Practical examination will be for 50 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	50	50
			Total	**50**

V. Learning Outcomes

At the end of the course, students will be able to

- Understand, appreciate and demonstrate innovative approach, beauty and acute acumen in the area of photography
- Develop photography skills and become familiar with the functions and importance of the visual elements of nature and artificial objects
- Become independent thinkers who will contribute inventively and critically to culture through the making of art photography
- Have thorough understanding and acute sense of light and shade, composition, and presentation of a piece of an art
- Experiment and Represent the cultivated sense and skills in Photography to the mass
- Prepare an impressive portfolio encompassing holistic approach to art and other the areas of study

SCULPTING (HS103 B)

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS103 B	Sculpting	02	02	--	--	50	50	100

II. Course Objectives

- To promote creativity and aesthetic sense pertaining to Sculpting by introducing them to the history of sculpting, its basic concepts and contemporary techniques and issues
- To help the students understand and develop the skill of sketching and drawing from life, natural and manmade objects and structures using various means like pencil, pen, ink, crayon, chalk, colour etc.
- To help them understand methods and materials of sculpture i.e. clay, plaster, cement, wood, stone, bronze, enlarging and reducing devises, welding torch etc.
- To help the students develop the sense of structure, and understand how forms achieve their structural unity through adherence to principles of physical nature of material being observed and studied (e.g. Plants, insects, minerals etc)
- To introduce the basic visual elements of 2-D and 3-D designs with emphasis on fundamentals of two and three-dimensional designs
- To acquaint the learners with various perspectives to draw and mould a sculpture
- To make the learners understand the colour theory and its practical usages
- To provide the students a sound background of the traditional and representational form in sculpture and enable them to develop their own vision

III. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**An Introduction to Sculpture** • *What Sculpture is and how it is different from other Arts* • *Basic elements, techniques, and history of sculpture* • *Form, Space and Basic Shapes in Sculpture* • *Casting materials, moldmaking, basic wood cutting and shaping, metal work, Welding, assembling, adhesives, plaster, mixing*	**06**

2	**Fundamental Principles of Sculpture** • *Basic Principles of Aesthetics in Sculpture* • *Visual Principles – balance, sequence, weight, and structural dynamics in sculpture* • *Structural Principles and communicative possibilities of sculpture*	**10**
3	**Process of Modeling** • *Additive and reductive processes* • *Major Techniques of Sculpture: Modeling, Carving, Pointing* • *Materials used in Modeling* • *Clay Modeling and Carving*	**11**
4	**Contemporary Issues in Sculpture** • *Sculpture and Present Day Context* • *Contemporary Sculptors and their Contribution* • *Major Contemporary Issues in Sculpture*	**03**
	Total Hours	**30**

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

V. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 50 marks for internal evaluation and 50 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	10	10
2	Performance/ Activities	-	10	10
3	Project	-	25	25
4	Attendance	-	05	05
			Total	**50**

External Evaluation

University Practical examination will be for 50 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	50	50
			Total	**50**

VI. Learning Outcomes

Upon successful completion of this course, students will be able to:

a. Incorporate basic principles of aesthetics into sculpture
b. Understand and apply basic concepts, styles and latest techniques of sculpting
c. Explore traditional and experimental materials and design for sculpture
d. Maintain a sketchbook of ideas and drawings to work out art project and to document coursework and discussions
e. Understand the latest jargons, and develop collaborative skills to exhilarate the speed of accomplishing the piece of art
f. Make their portfolio rich by accomplishing projects given during course
g. Become familiar with varied key sculptural techniques and formal ideas through hands-on workshops and experimentation with a variety of materials and three-dimensional assignments
h. Present their work with greater impact and confidence for future prospects
i. Get benefitted into other subjects of their study by developing broader and all-inclusive approach to learning

POTTERY AND CERAMIC ART (HS104 B)

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS104 B	Pottery and Ceramic Art	02	02	--	--	50	50	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**An Introduction to Clay** • *Elements and Materials of Ceramic Art and Shape* • *Prehistoric Cultures* • *Basics of Pinching, Slabbing and Coiling* • *Potter's wheel, centring the clay, forming bowls and cylinders* • *Trimming and burnishing on the wheel* • *Sketching the Pottery Models*	**5**
2	**Types and Techniquesof Making Pottery** • *Types of Pottery:* *Porcelain Pottery, Earthenware Pottery, Stoneware Pottery* • *Techniques of Pottery:* *Hand-Built Pottery: Pinch, Coil, Slab* *Wheel-Thrown Pottery*	**7**
3	**Methods of Making Pottery** • *Coil Method* • *Pinch Method* • *Slab Method*	**8**
4	**Decorating the Clays** • *Different Methods of Decoration* • *Textures in Pottery* • *Colours, Painting, Carving, Glazing etc. in Pottery*	**7**
5	**Contemporary Issues in Pottery and Ceramic Art** • *Present Day Pottery and Ceramic Art* • *Place of Pottery and Ceramic Art in Contemporary Art Society* • *Major Practitioners of Contemporary Pottery and Ceramic Art and their Contributions*	**3**
	Total Hours	**30**

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 50 marks for internal evaluation and 50 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	10	10
2	Performance/ Activities	-	10	10
3	Project	-	25	25
4	Attendance	-	05	05
			Total	**50**

External Evaluation

University Practical examination will be for 50 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	50	50
			Total	**50**

V. Learning Outcomes

At the end of the course, the learners will

- Have basic understanding of clay and glaze composition and formulation with emphasis on hand-built ceramic forms.
- Have explored a variety of hand-building methods including extended pinch, slab built and extruded forms
- have learned firing and glazing methods for stoneware clay
- have learned how finishing and decorating contribute or detract from the intention as an artist
- Finally, a student will also have developed a sense of appreciation regarding how a unified, coherent form that is finely crafted is beautiful in its own right

MEDIA AND GRAPHIC DESIGN (HS105 B)

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS105 B	Media and Graphic Design	02	02	--	--	50	50	100

II. Course Objectives

- To encourage/ foster creativity among the students
- To introduce students to the fundamentals of graphic designs
- To cultivate / spawn awareness among students about the significance of art and designs, art criticism and aesthetics
- To help the students understand the meanings of concept, designs, shapes, colors, print and medium
- To give the students first-hand experience of working on Graphic Software
- To develop in students an understanding of major issues, techniques and aspects of designs and print

III. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**An Introduction to Media and Graphic Design** • *Creating Art, Art in Context and Art as Inquiry* • *History of Graphic Design* • *Constructional, Representational, and Simplification Drawing*	03
2	**Layout and Design** • *Layout, Design and Aesthetics* • *Elements of Design* • *Principles of Design:* *Harmony, Balance, Rhythm, Perspective, Emphasis, Orientation, Repetition and Proportion* • *Impact/function of Design* • *Indigenous design practices* • *Role of design in the changing social scenario*	07
3	**Form and Space** • *Types of Forms: Man-made, Nature* • *Types of Space: Negative and Positive* • *Composition of Form and Space to create Layout* • *Exploring Creativity*	06
4	**Computer Graphics** • *An Introduction to Graphic Software* • *Flash, Coreldraw, Illustrator and Photoshop* • *Pre-press Process*	04

5	**Fonts** • *Construction of Type* • *Anatomy of Type* • *Visual Language* • *Creating Logo and Symbol*	**04**
6	**Basic Print Media** • *An Introduction to Press and its Development Phases* • *Types of Press* • *Types of Printing Technologies* • *Post-press Processes*	**03**
7	**Contemporary Issues in Graphic Design** • *Present Day Graphic Designs* • *Contemporary Designers and their Contribution* • *Major Contemporary Issues in Graphic Design*	**03**
	Total Hours	**30**

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

V. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 50 marks for internal evaluation and 50 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	10	10
2	Performance/ Activities	-	10	10
3	Project	-	25	25
4	Attendance	-	05	05
			Total	**50**

External Evaluation

University Practical examination will be for 50 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	50	50
			Total	**50**

VI. Learning Outcomes

At the end of the course, a student will

- have cultivated a sense of creativity.
- be appreciative of art and designs, art criticism and aesthetics.
- be able to recognize the elements of arts in graphic design.

- have better cognizance and association with the meaning of designs, shapes, colors, print and medium.
- be able to design graphics using computer softwares like Photoshop, CorelDraw, and Illustrator.

ART AND CRAFT (HS106 B)

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS106 B	Art and Craft	02	02	--	--	50	50	100

II. Course Objectives

- To encourage/ foster creativity among the students
- To enable the students to work through the process of bringing an idea from conception to realization
- To enable students to create artifacts that are visually expressive
- To cultivate / spawn awareness among students about the significance of art, craft and aesthetics
- To develop students' graphic skills which may work towards the realization of ideas in the creation of 2D and 3D
- To provide opportunities to students to be conversant with the use of a variety of materials, media, tools and equipments for Art and Craft

III. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**An Introduction to Art and Craft** • *Basic Design and Forms* • *Space and Geometry* • *Elements of Nature and Object*	**03**
2	**Paper Cutting** • *Study of Designs* • *Context of Space and Form* • *Types of Textures and Papers* • *Principles of Paper Cutting*	**05**
3	**Design from Nature** • *Nature as a Source of Design* • *Principles of Designing Natural Object* • *Decorative Forms* • *Cutting, Collaging, Embossing and Itching*	**08**
4	**Card Board Modelling** • *Principles of Form and Space* • *Dimensions of Space and Shape* • *Process of Modeling and Decoration*	**06**
5	**Print Making** • *An Introduction to Print Making* • *A Brief History of Print Making* • *Types of Print Making* • *Processing of Print Making* • *Sketching and Drawing*	**05**

6	**Contemporary Issues in Art and Craft** • *Present-day Art and Craft* • *Using the Waste for making the Best* • *Contemporary Issues in Art and Craft*	**03**
	Total Hours	**30**

IV. Instruction Method and Pedagogy

Teaching will be practical - based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

V. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 50 marks for internal evaluation and 50 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	10	10
2	Performance/ Activities	-	10	10
3	Project	-	25	25
4	Attendance	-	05	05
			Total	**50**

External Evaluation

University Practical examination will be for 50 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	50	50
			Total	**50**

VI. Learning Outcomes

At the end of the course, students will be

- aware about the significance of art, craft and aesthetics.
- able to create artifacts that are visually expressive.
- able to lead the ideas from conceptualization with reference to the 2D and 3D model making.
- conversant with the use of a variety of materials, media, tools and equipments for creative Art and Craft.

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme

SYLLABI

(Semester – 2)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH2001: Human Anatomy and Physiology-II (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Body fluids and blood	10
	Lymphatic system	
2	Cardiovascular system	10
3	Digestive system	6
	Respiratory system	
4	Respiratory system	10
	Urinary system	
5	Reproductive system	9
	Introduction to genetics	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Body fluids and blood, Lymphatic system.**	**10 Hours**	**22.22%**
	Body fluids, composition, and functions of blood, hemopoeisis, formation of hemoglobin, anemia, mechanisms of coagulation, blood grouping, Rh factors, transfusion, its significance and disorders of blood, Reticulo endothelial system. Lymphatic organs and tissues, lymphatic vessels, lymph circulation and functions of lymphatic system.		
2.	**Cardiovascular system**	**10 Hours**	**22.22%**
	Heart – anatomy of heart, blood circulation, blood vessels, structure and functions of artery, vein and capillaries, elements of conduction system of heart and heartbeat, its regulation by autonomic nervous system, cardiac output, cardiac cycle. Regulation of blood pressure, pulse, electrocardiogram, and disorders of heart.		
3.	**Digestive system, Respiratory system**	**6 Hours**	**13.33%**
	Anatomy of GI Tract with special reference to anatomy and functions of stomach, (Acid production in the stomach, regulation of acid production through parasympathetic nervous system, pepsin role in protein digestion) small intestine and large intestine, anatomy and functions of salivary glands, pancreas and liver, movements of GIT, digestion and absorption of nutrients and disorders of GIT.		

	Anatomy of respiratory system with special reference to anatomy of lungs, mechanism of respiration, regulation of respiration.		
4.	**Respiratory system, Urinary system**	**10 Hours**	**22.22%**
	Lung Volumes and capacities transport of respiratory gases, artificial respiration, and resuscitation methods. Anatomy of urinary tract with special reference to anatomy of kidney and nephrons, functions of kidney and urinary tract, physiology of urine formation, micturition reflex and role of kidneys in acid base balance, role of RAS in kidney and disorders of kidney		
5.	**Reproductive system, Introduction to genetics**	**9 Hours**	**20%**
	Anatomy of male and female reproductive system, Functions of male and female reproductive system, sex hormones, physiology of menstruation, fertilization, spermatogenesis, oogenesis, pregnancy, and parturition. Chromosomes, genes and DNA, protein synthesis, genetic pattern of inheritance.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe gross morphology, structure, and functions of various organs of the human body.
CO2	identify, draw and differentiate various tissues and organs of different systems of human body.
CO3	describe structure and role of genetic material in protein synthesis as well as its function in inheritance.
CO4	summarize energetics in cell.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme
BPH2001P: HUMAN ANATOMY AND PHYSIOLOGY (Practical)

Total hours: 60 Hours

Outline of the course:

Sr. No.	Title of the unit
1.	Introduction to hemocytometry.
2.	Enumeration of white blood cell (WBC) count
3.	Enumeration of total red blood corpuscles (RBC) count
4.	Determination of bleeding time
5.	Determination of clotting time
6.	Estimation of hemoglobin content
7.	Determination of blood group.
8.	Determination of erythrocyte sedimentation rate (ESR).
9.	Determination of heart rate and pulse rate.
10.	Recording of blood pressure.
11.	Determination of tidal volume and vital capacity.
12.	Study of digestive, respiratory, cardiovascular systems, urinary and reproductive systems with the help of models, charts and specimens.
13.	Recording of basal mass index
14.	Study of family planning devices and pregnancy diagnosis test.
15.	Demonstration of total blood count by cell analyser
16.	Permanent slides of vital organs and gonads.

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	summarize functional characteristics of various systems
CO2	describe the fundamental physiological mechanism involves in demonstrated practical
CO3	interlinking various systems in terms of Feedback mechanisms and perform various tests related to blood cells counts which relates with the diagnosis of various disease conditions
CO4	identify and describe functionality of various devices for family planning and pregnancy diagnostic tests
C05	analyze the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	3	-	-
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Essentials of Medical Physiology by K. Sembulingam and P. Sembulingam. Jaypee brother‘s medical publishers, New Delhi.
2. Anatomy and Physiology in Health and Illness by Kathleen J.W. Wilson, Churchill Livingstone, New York
3. Physiological basis of Medical Practice-Best and Tailor. Williams & Wilkins Co,Riverview,MI USA
4. Text book of Medical Physiology- Arthur C,Guyton andJohn.E. Hall. Miamisburg, OH, U.S.A.
5. Principles of Anatomy and Physiology by Tortora Grabowski. Palmetto, GA, U.S.A.
6. Textbook of Human Histology by Inderbir Singh, Jaypee brother‘s medical publishers, New Delhi.
7. Textbook of Practical Physiology by C.L. Ghai, Jaypee brother‘s medical publishers, New Delhi.
8. Practical workbook of Human Physiology by K. Srinageswari and Rajeev Sharma, jaypee brother‘s medical publishers, New Delhi.

❖ **Reference book:**

1. Physiological basis of Medical Practice-Best and Tailor. Williams & Wilkins Co, Riverview, MI USA
2. Textbook of Medical Physiology- Arthur C, Guyton and John. E. Hall. Miamisburg, OH, U.S.A.
3. Human Physiology (vol 1 and 2) by Dr. C.C. Chatterrje, Academic Publishers Kolkata

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH2002: Pharmaceutical Organic Chemistry-I (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Classification, nomenclature, and isomerism	7
2	Alkanes, Alkenes and Conjugated dienes	10
3	Alkyl halides	10
	Alcohols	
4	Carbonyl compounds (Aldehydes and ketones)	10
5	Carboxylic acids	8
	Aliphatic amines	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Classification, nomenclature, and isomerism**	**7 Hours**	**15.55%**
	Classification of Organic Compounds Common and IUPAC systems of nomenclature of organic compounds (up to 10 Carbons open chain and carbocyclic compounds) Structural isomerism in organic compounds		
2.	**Alkanes*, Alkenes* and Conjugated dienes***	**10 Hours**	**22.22%**
	SP3 hybridization in alkanes, Halogenation of alkanes, uses of paraffins. Stabilities of alkenes, SP 2 hybridization in alkenes E1 and E2 reactions – kinetics, order of reactivity of alkyl halides, rearrangement of carbocations, Saytzeffs orientation and evidence. E1 verses E2 reactions, Factors affecting E1 and E2 reactions. Ozonolysis, electrophilic addition reactions of alkenes, Markownikoff‘s orientation, free radical addition reactions of alkenes, Anti Markownikoff‘s orientation. Stability of conjugated dienes, Diel-Alder, electrophilic addition, free radical addition reactions of conjugated dienes, allylic rearrangement		
3.	**Alkyl halides*, Alcohols***	**10 Hours**	**22.22%**
	SN1 and SN2 reactions - kinetics, order of reactivity of alkyl halides, stereochemistry, and rearrangement of carbocations. SN1 versus SN2 reactions, Factors affecting SN1 and SN2 reactions Structure and uses of ethylchloride, Chloroform, trichloroethylene, tetrachloroethylene, dichloromethane, tetrachloromethane and iodoform. Qualitative tests, Structure and uses of Ethyl alcohol, chlorobutanol, Cetosteryl alcohol, Benzyl alcohol, Glycerol, Propylene glycol		

4.	**Carbonyl compounds* (Aldehydes and ketones)**	**10 Hours**	**17.77%**
	Nucleophilic addition, Electromeric effect, aldol condensation, Crossed Aldol condensation, Cannizzaro reaction, Crossed Cannizzaro reaction, Benzoin condensation, Perkin condensation, qualitative tests, Structure and uses of Formaldehyde, Paraldehyde, Acetone, Chloral hydrate, Hexamine, Benzaldehyde, Vanilin, Cinnamaldehyde.		
5.	**Carboxylic acids*,** Aliphatic amines*	**8 Hours**	**17.77%**
	Acidity of carboxylic acids, effect of substituents on acidity, inductive effect and qualitative tests for carboxylic acids, amide and ester Structure and Uses of Acetic acid, Lactic acid, Tartaric acid, Citric acid, Succinic acid. Oxalic acid, Salicylic acid, Benzoic acid, Benzyl benzoate, Dimethyl phthalate, Methyl salicylate and Acetyl salicylic acid. Basicity, effect of substituent on Basicity. Qualitative test, Structure and uses of Ethanolamine, Ethylenediamine, Amphetamine.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	assign nomenclature to structure as per IUPAC system.
CO2	identify the type of isomerism of the organic compound.
CO3	write the reaction with their reactivity, stability, and orientation.
CO4	enumerate the preparations, reactions and uses of important organic compounds.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH2002P: PHARMACEUTICAL ORGANIC CHEMISTRY -I (Practical)

Total hours: 60 Hours

Outline of the course:

Sr. No.	Title of the unit
1.	Systematic qualitative analysis of unknown organic compounds like
	Preliminary test: Color, odour, aliphatic/aromatic compounds, saturation and unsaturation, etc.
	Detection of elements like Nitrogen, Sulphur and Halogen by Lassaigne‘s test
	Solubility test
	Functional group test like Phenols, Amides/ Urea, Carbohydrates, Amines, Carboxylic acids, Aldehydes and Ketones, Alcohols, Esters, Aromatic and Halogenated Hydrocarbons, Nitro compounds and Anilides.
	Melting point/Boiling point of organic compounds
	Identification of the unknown compound from the literature using melting point/ boiling point.
	Preparation of the derivatives and confirmation of the unknown compound by melting point/ boiling point.
	Minimum 5 unknown organic compounds to be analyzed systematically.
2.	Preparation of suitable solid derivatives from organic compounds
3.	Construction of molecular models

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	identify the nature of organic compounds
CO2	prepare various derivatives of organic compounds
CO3	interpret the stereo models of organic compounds
C04	analyze the problem, communicate suggested solution and interprete the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	3	3	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	-	-	3	3	-	-	-	-	-	-	3
CO4	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Organic Chemistry by Morrison and Boyd
2. Organic Chemistry by I.L. Finar , Volume-I
3. Textbook of Organic Chemistry by B.S. Bahl & Arun Bahl.
4. Organic Chemistry by P.L.Soni
5. Practical Organic Chemistry by Mann and Saunders.
6. Vogel‘s text book of Practical Organic Chemistry
7. Advanced Practical organic chemistry by N.K.Vishnoi.
8. Introduction to Organic Laboratory techniques by Pavia, Lampman and Kriz.
9. Reaction and reaction mechanism by Ahluwaliah/Chatwal.

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH2003: Pharmaceutical Engineering (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Flow of fluids	10
	Size Reduction	
	Size Separation	
	Mixing	
2	Crystallization	10
	Evaporation	
	Heat Transfer	
3	Drying	10
	Distillation	
4	Filtration	8
	Centrifugation	
5	Plant location, industrial hazards, and plant safety	7
	Materials of pharmaceutical plant construction, Corrosion, and its prevention	
	Material handling systems	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Flow of fluids, Size Reduction, Size Separation, Mixing**	**10 Hours**	**22.22%**
	Types of manometers, Reynolds number and its significance, Bernoulli 's theorem and its applications, Energy losses, Orifice meter, Venturimeter, Pitot tube and Rotometer. Objectives, Mechanisms & Laws governing size reduction, factors affecting size reduction, principles, construction, working, uses, merits and demerits of Hammer mill, ball mill, fluid energy mill, Edge runner mill & end runner mill. Objectives, applications & mechanism of size separation, official standards of powders, sieves, size separation Principles, construction, working, uses, merits and demerits of Sieve shaker, cyclone separator, Air separator, Bag filter & elutriation tank. Objectives, applications & factors affecting mixing, Difference between solid and liquid mixing, mechanism of solid mixing, liquids mixing and semisolids mixing. Principles, Construction, Working, uses, Merits and Demerits of		

	Double cone blender, twin shell blender, ribbon blender, Sigma blade mixer, planetary mixers, Propellers, Turbines, Paddles & Silverson Emulsifier		
2.	**Crystallization, Evaporation, Heat Transfer**	**10 Hours**	**22.22%**
	Objectives, applications, & theory of crystallization. Solubility curves, principles, construction, working, uses, merits and demerits of Agitated batch crystallizer, Swenson Walker Crystallizer, Krystal crystallizer, Vacuum crystallizer. Caking of crystals, factors affecting caking & prevention of caking. Objectives, application, and factors influencing evaporation, differences between evaporation and other heat process. principles, construction, working, uses, merits and demerits of Steam jacketed kettle, horizontal tube evaporator, climbing film evaporator, forced circulation evaporator, multiple effect evaporator& Economy of multiple effect evaporator. Objectives, applications & Heat transfer mechanisms. Fourier 's law, Heat transfer by conduction, convection & radiation. Heat interchangers & heat exchangers		
3.	**Drying, Distillation**	**10 Hours**	**22.22%**
	Objectives, applications & mechanism of drying process, measurements& applications of Equilibrium Moisture content, rate of drying curve. principles, construction, working, uses, merits and demerits of Tray dryer, drum dryer spray dryer, fluidized bed dryer, vacuum dryer, freeze dryer. Objectives, applications & types of distillation. principles, construction, working, uses, merits and demerits of (lab scale and industrial scale) Simple distillation, preparation of purified water and water for injection BP by distillation, flash distillation, fractional distillation, distillation under reduced pressure, steam distillation & molecular distillation		
4.	**Filtration, Centrifugation**	**8 Hours**	**17.77%**
	Objectives, applications, Theories & Factors influencing filtration, filter aids, filter medias. Principle, Construction, Working, Uses, Merits and demerits of plate & frame filter, filter leaf, rotary drum filter, Meta filter & Cartridge filter, membrane filters and Seidtz filter. Objectives, principle & applications of Centrifugation, principles, construction, working, uses, merits and demerits of Perforated basket centrifuge, non-perforated basket centrifuge, semi continuous centrifuge & super centrifuge		
5.	**Plant location, industrial hazards and plant safety, Materials of pharmaceutical plant construction, Corrosion and its prevention, Material handling systems**	**7 Hours**	**15.55%**
	Plant Layout, utilities and services, Mechanical hazards, Chemical hazards, Fire hazards, explosive hazards, and their safety.		

	Factors affecting during materials selected for pharmaceutical plant construction, Theories of corrosion, types of corrosion and their prevention. Ferrous and nonferrous metals, inorganic, and organic nonmetals. Objectives & applications of Material handling systems, different types of conveyors such as belt, screw, and pneumatic conveyors.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	summarize various unit operations used in pharmaceutical industries with applications.
CO2	describe and suggest material handling techniques.
CO3	suggest and justify appropriate equipment of the unit operations including their principle, construction, working and specific applications.
CO4	describe preventive methods used for environmental pollution and corrosion control in pharmaceutical industries.
CO5	draw and comprehend significance of pharmaceutical plant lay out design.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	3	-
CO5	3	-	-	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme
BPH2003P: PHARMACEUTICAL ENGINEERING (Practical)

Total hours: 60 Hours

Outline of the course:

Sr. No.	Title of the unit
1.	Determination of radiation constant of brass, iron, unpainted and painted glass.
2.	Steam distillation – To calculate the efficiency of steam distillation.
3.	To determine the overall heat transfer coefficient by heat exchanger.
4.	Construction of drying curves (for calcium carbonate and starch).
5.	Determination of moisture content and loss on drying.
6.	Determination of humidity of air – i) From wet and dry bulb temperatures – use of Dew point method.
7.	Description of Construction working and application of Pharmaceutical Machinery such as rotary tablet machine, fluidized bed coater, fluid energy mill, de humidifier
8.	Size analysis by sieving – To evaluate size distribution of tablet granulations – Construction of various size frequency curves including arithmetic and logarithmic probability plots.
9.	Size reduction: To verify the laws of size reduction using ball mill and determining Kicks, Rittinger ‘s, Bond‘s coefficients, power requirement and critical speed of Ball Mill.
10.	Demonstration of colloid mill, planetary mixer, fluidized bed dryer, freeze dryer and such other major equipment.
11.	Factors affecting Rate of Filtration and Evaporation (Surface area, Concentration and Thickness/ viscosity
12.	To study the effect of time on the Rate of Crystallization.
13.	To calculate the uniformity Index for given sample by using Double Cone Blender.

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	perform the expriments involving unit operations like filtration, distillation, evaporation, drying, mixing, crystallization.
CO2	identify the equipment and carry out the experiments related to size reduction and size separation
CO3	describe the basic concepts of Heat transfer and HVAC (Humidity Ventilation and Air Conditioning).
CO4	analyze the problem, communicate suggested solution and interprete the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	3	-	-	-	-	-	-	-
CO2	3	-	3	3	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

- **Textbook:**
  1. Introduction to chemical engineering – Walter L Badger & Julius Banchero, Latest edition.
  2. Solid phase extraction, Principles, techniques and applications by Nigel J.K. Simpson- Latest edition.
  3. Unit operation of chemical engineering – Mcabe Smith, Latest edition.
  4. Pharmaceutical engineering principles and practices – C.V.S Subrahmanyam et al., Latest edition.
  5. Remington practice of pharmacy- Martin, Latest edition.
  6. Theory and practice of industrial pharmacy by Lachmann., Latest edition.
  7. Physical pharmaceutics- C.V.S Subrahmanyam et al., Latest edition.
  8. Cooper and Gunn‘s Tutorial pharmacy, S.J. Carter, Latest edition

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH2004: Computer Applications in Pharmacy (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Number system Concept of Information Systems and Software	9
2	Web technologies	9
3	Application of computers in Pharmacy	9
4	Bioinformatics	9
5	Chromatographic dada analysis (CDS), Laboratory Information management System (LIMS) and Text Information Management System (TIMS)	9

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Number system, Concept of Information Systems and Software** Binary number system, Decimal number system, Octal number system, Hexadecimal number systems, conversion decimal to binary, binary to decimal, octal to binary etc, binary addition, binary subtraction – One‘s complement, Two‘s complement method, binary multiplication, binary division. Information gathering, requirement and feasibility analysis, data flow diagrams, process specifications, input/output design, process life cycle, planning and managing the project	**9 Hours**	**20%**
2.	**Web technologies** :Introduction to HTML, XML,CSS and Programming languages, introduction to web servers and Server Products Introduction to databases, MYSQL, MS ACCESS, Pharmacy Drug database	**9 Hours**	**20%**
3.	**Application of computers in Pharmacy** Drug information storage and retrieval, Pharmacokinetics, Mathematical model in Drug design, Hospital and Clinical Pharmacy, Electronic Prescribing	**9 Hours**	**20%**

	and discharge (EP) systems, barcode medicine identification and automated dispensing of drugs, mobile technology and adherence monitoring Diagnostic System, Lab-diagnostic System, Patient Monitoring System, Pharma Information System		
4.	**Bioinformatics**	**9 Hours**	**20%**
	Introduction, Objective of Bioinformatics, Bioinformatics Databases, Concept of Bioinformatics, Impact of Bioinformatics in Vaccine Discovery		
5.	**Chromatographic dada analysis (CDS), Laboratory Information management System (LIMS) and Text Information Management System(TIMS)**	**9 Hours**	**20%**

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	apply knowledge of computer in pharmacy.
CO2	classify and explain various types of databases.
CO3	enlist and describe various applications of databases in pharmacy.
CO4	write advantages and disadvantages of computers in pharmacy management system.
CO5	explain various software's of computers used in pharmacy

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	-	3	3	-	-	-	-	-	-	3
CO2	-	-	-	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
CO5	3	-	-	3	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH2004P: COMPUTER APPLICATIONS IN PHARMACY (Practical)

Total hours: 30 Hours

Outline of the course:

Sr. No.	Title of the unit
1.	Design a questionnaire using a word processing package to gather information about a particular disease.
2.	Create a HTML web page to show personal information.
3.	Retrieve the information of a drug and its adverse effects using online tools
4.	Creating mailing labels Using Label Wizard, generating label in MS WORD
5.	Create a database in MS Access to store the patient information with the required fields Using access
6.	Design a form in MS Access to view, add, delete and modify the patient record in the database
7.	Generating report and printing the report from patient database
8.	Creating invoice table using – MS Access
9.	Drug information storage and retrieval using MS Access
10.	Creating and working with queries in MS Access
11.	Exporting Tables, Queries, Forms and Reports to web pages
12.	Exporting Tables, Queries, Forms and Reports to XML pages

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	generate database using varoius tools
CO2	program to retrive data from database
CO3	apply MS of acesse for storage and retriving drug information
CO4	export contents to web and xml pages
CO5	analyze the problem, communicate suggested solution and interprete the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	-	-	3	-	-	-	-	-	-	3
CO2	-	-	-	3	-	-	-	-	-	-	-
CO3	3	-	-	3	-	-	-	-	-	-	-
CO4	-	-	-	3	-	-	-	-	-	-	-
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Computer Application in Pharmacy – William E.Fassett –Lea and Febiger, 600 South Washington Square, USA, (215) 922-1330.
2. Computer Application in Pharmaceutical Research and Development –Sean Ekins –Wiley-Interscience, A John Willey and Sons, INC., Publication, USA
3. Bioinformatics (Concept, Skills and Applications) – S.C.Rastogi-CBS Publishers and Distributors, 4596/1- A, 11 Darya Gani, New Delhi – 110 002(INDIA)
4. Microsoft office Access - 2003, Application Development Using VBA, SQL Server, DAP and Infopath – Cary N.Prague – Wiley Dreamtech India (P) Ltd., 4435/7, Ansari Road, Daryagani, New Delhi - 110002

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
HS121.01 B: ENGLISH LANGUAGE AND LITERATURE

Total hours: 30 Hr

Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**Communicative English** • *Introduction to Communicative Grammar and Usage* • *Parts of Speech* • *Tenses and Moods* • *Reading Literature for English Language*	**08**
2	**Functional English** • *Introduction to Functional English* • *English for Personal and Social Use* • *English for Career and Professional Use*	**08**
3	**Literature Text and Appreciation** • *Introduction to Literature and Appreciation* • *Appreciation of Prose or Fiction* • *Appreciation of Poetry*	**08**
4	**Language, Literature and Contemporary Issues** • *Language, Culture and Society* • *Literature and contemporary issues*	**06**
	Total	**30**

Course Outcome (COs):

After completion of the course, the student would:

CO1	have developed the ability to communicate effectively.
CO2	be able to communicate message accurately, handle intercultural situation that require thoughtful communication, to use appropriate words and tones and so on.
CO3	be able to understand and demonstrate communicative and functional use of English language.
CO4	be able to appreciate literature and understand socio-cultural context.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	-	-	-	-	-	-	3	-	-	-
CO2	-	-	-	-	2	-	-	3	-	-	-
CO3	-	-	-	-	-	-	-	3	-	-	-
CO4	-	-	-	-	-	-	-	3	-	-	2

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

I. **Reference Books**

- Hurd Stella(2005), *Success with Languages* , Routledge
- John Eastwood (2002) *Oxford Practice Grammar,* Oxford
- LoiuseMullany& Peter Stockwell (2010),*Introduction to English Language*, Routledge
- Peter Brooker, Raman Saledan& Peter Widowson (2005), *Reader's Guide to Contemporary literary theory*, Pearson

Additional Reading

- http://www.ocr.org.uk/Images/72885-level-2-functional-skills-english-underpinning-skills-support-material-for-learners.pd

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme

SYLLABI

(Semester – 3)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH3001: PHARMACEUTICAL ORGANIC CHEMISTRY–II (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	**Benzene and its derivatives**	10
2	**Phenols & Aromatic Amines**	10
3	**Fats and Oils**	10
4	**Polynuclear hydrocarbons**	8
5	**Cyclo alkanes**	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Benzene and its derivatives**	**10 Hours**	**22.22%**
	Analytical, synthetic and other evidences in the derivation of structure of benzene, Orbital picture, resonance in benzene, aromatic characters, Huckel's rule Reactions of benzene - nitration, sulphonation, halogenation-reactivity, Friedelcrafts alkylation- reactivity, limitations, Friedelcrafts acylation. Substituents, effect of substituents on reactivity and orientation of mono substituted benzene compounds towards electrophilic substitution reaction Structure and uses of DDT, Saccharin, BHC and Chloramine		
2.	**Phenols & Aromatic Amines**	**10 Hours**	**22.22%**
	Phenols* - Acidity of phenols, effect of substituents on acidity, qualitative tests, Structure and uses of phenol, cresols, resorcinol, naphthols **Aromatic Amines* -** Basicity of amines, effect of substituents on basicity, and synthetic uses of aryl diazonium salts		
3.	**Fats and Oils**	**10 Hours**	**22.22%**
	Fatty acids – reactions Hdrolysis, Hydrogenation, Saponification and Rancidity of oils, Drying oils. Analytical constants – Acid value, Saponification value, Ester value, Iodine value, Acetyl value, Reichert Meissl (RM) value – significance and principle involved in their determination fluid.		
4.	**Polynuclear hydrocarbons**	**8 Hours**	**17.77%**
	Synthesis, reactions Structure and medicinal uses of Naphthalene, Phenanthrene, Anthracene, Diphenylmethane, Triphenylmethane and their derivatives		
	Cyclo alkanes		

5.	Stabilities – Baeyer‘s strain theory, limitation of Baeyer‘s strain theory, Coulson and Moffitt‘s modification, Sachse Mohr‘s theory (Theory of strainless rings), reactions of cyclopropane and cyclobutane only	**7 Hours**	**15.55%**

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	identify aromatic, anti-aromatic and non-aromatic compounds.
CO2	write the reaction with their reactivity, stability and orientation
CO3	enumerate the preperations, reactions and uses of important organic compounds
CO4	describe various methods for evaluation of fats and oils with significance.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	-

BPH3001P: PHARMACEUTICAL ORGANIC CHEMISTRY-II (Practical)

Total hours: 60 Hours

Outline of the course:

Sr. No.	Title of the unit
1.	**Experiments involving laboratory techniques**
2.	**Determination of following oil values (including standardization of reagents)**
3.	**Preparation of compounds**
	Benzanilide/Phenyl benzoate/Acetanilide from Aniline/ Phenol /Aniline by acylation reaction.
	2,4,6-Tribromo aniline/Para bromo acetanilide from Aniline/Acetanilide by halogenation (Bromination) reaction.
	5-Nitro salicylic acid/Meta di nitro benzene from Salicylic acid / Nitro benzene by nitration reaction.
	Benzoic acid from Benzyl chloride by oxidation reaction, Benzoic acid/ Salicylic acid from alkyl benzoate/ alkyl salicylate by hydrolysis reaction.
	1-Phenyl azo-2-napthol from Aniline by diazotization and coupling reactions.
	Benzil from Benzoin by oxidation reaction. Dibenzal acetone from Benzaldehyde by Claison Schmidt reaction Cinnammic acid from Benzaldehyde by Perkin reaction *P*-Iodo benzoic acid from *P*-amino benzoic acid.

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	apply various physicochemical techiques for synthesis of organic compound
CO2	evaluate fixed oils using compandial methods
CO3	perform synthesis of various intermediates.
CO4	analyze the problem, communicate suggested solution and interprete the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	3	3	-	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Organic Chemistry by Morrison and Boyd
2. Organic Chemistry by I.L. Finar , Volume-I
3. Textbook of Organic Chemistry by B.S. Bahl & Arun Bahl.
4. Organic Chemistry by P.L.Soni
5. Practical Organic Chemistry by Mann and Saunders.
6. Vogel‘s text book of Practical Organic Chemistry
7. Advanced Practical organic chemistry by N.K.Vishnoi.
8. Introduction to Organic Laboratory techniques by Pavia, Lampman and Kriz

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH3002: PHYSICAL PHARMACEUTICS-I (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	**Solubility of drugs**	10
2	**States of Matter and properties of matter and Physicochemical properties of drug molecules**	10
3	**Micromeritics**	10
4	**Complexation and protein binding**	8
5	**pH, buffers and Isotonic solutions**	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Solubility of drugs** Solubility expressions, mechanisms of solute solvent interactions, ideal solubility parameters, solvation & association, quantitative approach to the factors influencing solubility of drugs, Dissolution & drug release, diffusion principles in biological systems. Solubility of gas in liquids, solubility of liquids in liquids, (Binary solutions, ideal solutions) Raoult's law, real solutions, azeotropic mixtures, fractional distillation. Partially miscible liquids, Critical solution temperature and applications. Distribution law, its limitations and applications	**10 Hours**	**22.22%**
2.	**States of Matter and properties of matter and Physicochemical properties of drug molecules** State of matter, changes in the state of matter, latent heats, vapour pressure, sublimation critical point, eutectic mixtures, gases, aerosols – inhalers, relative humidity, liquid complexes, liquid crystals, glassy states, solid-crystalline, amorphous & polymorphism. Refractive index, optical rotation, dielectric constant, dipole moment, dissociation constant, determinations and applications	**10 Hours**	**22.22%**
3.	**Micromeretics** Particle size and distribution, average particle size, number and weight distribution, particle number, methods for determining particle size by (different methods), counting and separation method, particle shape, specific surface, methods for determining surface area, permeability, adsorption, derived properties of powders, porosity, packing arrangement, densities, bulkiness & flow properties.	**10 Hours**	**22.22%**
4.	**Complexation and protein binding**	**8 Hours**	**17.77%**

	Introduction, Classification of Complexation, Applications, methods of analysis, protein binding, Complexation and drug action, crystalline structures of complexes and thermodynamic treatment of stability constants.		
5.	**pH, buffers and Isotonic solutions**	**7 Hours**	**15.55%**
	Sorensen's pH scale, pH determination (electrometric and calorimetric), applications of buffers, buffer equation, buffer capacity, buffers in pharmaceutical and biological systems, buffered isotonic solutions.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe the fundamentals, significance and applications of solubility, dissolution and diffusion of drug.
CO2	summarize changes in states of matter and effects on relevant physicochemical properties of drug molecules with significance.
CO3	describe, determine the fundamental and derived properties of solid states.
CO4	state the types, applications and methods of preparation of various types of complexes.
CO5	describe the fundaments of buffer solutions and its applications in pharmaceuticals.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	-
C05	3	-	3	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH3002P: PHYSICAL PHARMACEUTICS-I (Practical)

Total hours: 60 Hours

Outline of the course:

Sr. No.	Title of the unit
1.	Determination the solubility of drug at room temperature
2.	Determination of pKa value by Half Neutralization/ Henderson Hassel Balch equation.
3.	Determination of Partition co- efficient of benzoic acid in benzene and water
4.	Determination of Partition co- efficient of Iodine in CCl4 and water.
5.	Determination of % composition of NaCl in a solution using phenol-water system by CST method
6.	Determination of particle size, particle size distribution using sieving method
7.	Determination of particle size, particle size distribution using Microscopic method
8.	Determination of bulk density, true density and porosity
9.	Determine the angle of repose and influence of lubricant on angle of repose
10.	Determination of stability constant and donor acceptor ratio of PABA-Caffeine complex by solubility method
11.	Determination of stability constant and donor acceptor ratio of Cupric-Glycine complex by pH titration method

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	determine the particle size distribution (sieving and microscopic method) and interpretation of obtained results.
CO2	determine solubility, critical solution temperature and partition co-efficient of drug through various approaches.
CO3	determine pKa and stability constant of given sample through mathematical and graphical method.
CO4	determine various derived properties of powder/ granules and significances of results obtained.
CO5	analyze the problem, communicate suggested solution and interprete the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	3	-	-	-	-	-	-	-	3
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Physical pharmacy by Alfred Martin
2. Experimental pharmaceutics by Eugene, Parott.
3. Tutorial pharmacy by Cooper and Gunn.
4. Stocklosam J. Pharmaceutical calculations, Lea &Febiger, Philadelphia.
5. Liberman H.A, Lachman C., Pharmaceutical Dosage forms, Tablets, Volume-1 to 3, MarcelDekkar Inc.
6. Liberman H.A, Lachman C, Pharmaceutical dosage forms. Disperse systems, volume 1, 2, 3. Marcel Dekkar Inc.
7. Physical pharmaceutics by Ramasamy C and ManavalanR.
8. Laboratory manual of physical pharmaceutics, C.V.S. Subramanyam, J. Thimma settee

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH3003: PHARMACEUTICAL MICROBIOLOGY (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Microbiology	10
2	Techniques in Microbiology	10
3	Sterilization techniques	10
4	Aseptic techniques	8
5	Pharmaceutical Microbiology	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Introduction to Microbiology**	10 Hours	22.22%
	Introduction, history of microbiology, its branches, scope and its importance. Introduction to Prokaryotes and Eukaryotes Study of ultra-structure and morphological classification of bacteria, nutritional requirements, raw materials used for culture media and physical parameters for growth, growth curve, isolation and preservation methods for pure cultures, cultivation of anaerobes, quantitative measurement of bacterial growth (total & viable count). Study of different types of phase contrast microscopy, dark field microscopy and electron microscopy		
2.	**Techniques in Microbiology**	10 Hours	22.22%
	Identification of bacteria using staining techniques (simple, Gram‘s &Acid fast staining) and biochemical tests (IMViC). Study of principle, procedure, merits, demerits and applications of Physical, chemical and mechanical method of sterilization. Evaluation of the efficiency of sterilization methods Equipment employed in large scale sterilization. Sterility indicators.		
3.	**Sterilization techniques**	10 Hours	22.22%
	Study of morphology, classification, reproduction/replication and cultivation of Fungi and Virus. Classification and mode of action of disinfectants Factors influencing disinfection, antiseptics and their evaluation. For bacteriostatic and bactericidal actions Evaluation of bactericidal & Bacteriostatic. Sterility testing of products (solids, liquids, ophthalmic and other sterile products) according to IP, BP and USP.		
4.	**Aseptic techniques**	8 Hours	17.77%
	Designing of aseptic area, laminar flow equipment; study of different sources of contamination in an aseptic area and methods of prevention, clean area classification. Principles and methods of different microbiological assay. Methods for standardization of antibiotics, vitamins and amino acids.		

	Assessment of a new antibiotic and testing of antimicrobial activity of a new substance. General aspects-environmental cleanliness.		
5.	**Pharmaceutical Microbiology**	**7 Hours**	**15.55%**
	Types of spoilage, factors affecting the microbial spoilage of pharmaceutical products, sources and types of microbial contaminants, assessment of microbial contamination and spoilage. Preservation of pharmaceutical products using antimicrobial agents, evaluation of microbial stability of formulations. Growth of animal cells in culture, general procedure for cell culture, Primary, established and transformed cell cultures. Application of cell cultures in pharmaceutical industry and research.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	compare and classify various kinds of microorganism and describe the methods for identification, cultivation and preservation of microorganism.
CO2	summarize the process of microbiological assay and sterility testing with rationale.
CO3	illustrate methods for standardization of biological products, role of disinfectants and their evaluation process.
CO4	draw the design of aseptic area and instruments .
CO5	describe methods to control contamination in aseptic area .

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	3	-	-	-	-	-	-	-
C05	3	-	3	-	-	-	-	-	-	-	3

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme
BPH3003P: PHARMACEUTICAL MICROBIOLOGY (Practical)

Total hours: 60 Hours

Outline of the course:

Sr. No.	Title of the unit
1.	Introduction and study of different equipment and processing, e.g., B.O.D. incubator, laminar flow, aseptic hood, autoclave, hot air sterilizer, deep freezer, refrigerator, microscopes used in experimental microbiology.
2.	Sterilization of glassware, preparation and sterilization of media.
3.	Sub culturing of bacteria and fungus. Nutrient stabs and slants preparations.
4.	Staining methods- Simple, Grams staining and acid-fast staining (Demonstration with practical).
5.	Isolation of pure culture of micro-organisms by multiple streak plate technique and other techniques
6.	Microbiological assay of antibiotics by cup plate method and other methods
7.	Motility determination by Hanging drop method.
8.	Sterility testing of pharmaceuticals
9.	Bacteriological analysis of water
10.	Biochemical test (IMViC reactions)
11.	Revision Practical Class

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	sterilize consumables to be used in microbial experiments
CO2	culture and sub-culture bacteria
CO3	isolate and differentiate microorganisms.
CO4	perform sterility testing and microbial assay of pharmaceutical products
CO5	describe applications of equipment used in microbiology laboratory.
CO6	analyze the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	3
CO5	3	-	-	-	-	-	-	-	-	-	-
CO6	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. W.B. Hugo and A.D. Russel: Pharmaceutical Microbiology, Blackwell Scientific publications, Oxford London.
2. Prescott and Dunn., Industrial Microbiology, 4th edition, CBS Publishers & Distributors, Delhi.
3. Pelczar, Chan Kreig, Microbiology, Tata McGraw Hill edn.
4. Malcolm Harris, Balliere Tindall and Cox: Pharmaceutical Microbiology.
5. Rose: Industrial Microbiology.
6. Probisher, Hinsdill et al: Fundamentals of Microbiology, 9th ed. Japan
7. Cooper and Gunn‘s: Tutorial Pharmacy, CBS Publisher and Distribution.
8. Peppler: Microbial Technology.
9. I.P., B.P., U.S.P.- latest editions.
10. Ananthnarayan: Textbook of Microbiology, Orient-Longman, Chennai
11. Edward: Fundamentals of Microbiology.
12. N.K.Jain: Pharmaceutical Microbiology, Vallabh Prakashan, Delhi
13. Bergeys manual of systematic bacteriology, Williams and Wilkins- A Waverly company

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH3004: PATHOPHYSIOLOGY (THEORY)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Basic principles of Cell injury and Adaptation and Basic mechanism involved in the process of inflammation and repair	10
2	Cardiovascular System, Respiratory system and Renal system	10
3	Haematological Diseases, Endocrine Disease, Nervous System, Gastrointestinal system	10
4	Disease of bones and joints, Principles of cancer, Diseases of bones and joints, Principles of Cancer	8
5	Infectious diseases and Sexually transmitted diseases	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Basic principles of Cell injury and Adaptation and Basic mechanism involved in the process of inflammation and repair**	**10 Hours**	**22.22%**
	Introduction, definitions, Homeostasis, Components and Types of Feedback systems, Causes of cellular injury**,** Pathogenesis (Cell membrane damage, Mitochondrial damage, Ribosome damage, Nuclear damage)**,** Morphology of cell injury – Adaptive changes (Atrophy, Hypertrophy, hyperplasia, Metaplasia, Dysplasia)**,** Cell swelling, Intra cellular accumulation, Calcification, Enzyme leakage and Cell Death Acidosis &Alkalosis, Electrolyte imbalance. Introduction, Clinical signs of inflammation, Different types of Inflammation, Mechanism of Inflammation – Alteration in vascular permeability and blood flow, migration of WBC‘s, Mediators of inflammation, Basic principles of wound healing in the skin, Pathophysiology of Atherosclerosis		
2.	**Cardiovascular System, Respiratory system and Renal system**	**10 Hours**	**22.22%**
	Hypertension, congestive heart failure, ischemic heart disease (angina, myocardial infarction, atherosclerosis and arteriosclerosis) Asthma, Chronic obstructive airways diseases Acute and chronic renal failure		
3.	**Haematological Diseases, Endocrine Disease, Nervous System, Gastrointestinal system**	**10 Hours**	**22.22%**

	Iron deficiency, megaloblastic anemia (Vit B12 and folic acid), sickle cell anemia, thalasemia, hereditary acquired anemia, hemophilia. Diabetes, thyroid diseases, disorders of sex hormones Epilepsy, Parkinson's disease, stroke, psychiatric disorders: depression, schizophrenia and Alzheimer's disease. Peptic Ulcer, Inflammatory bowel diseases, jaundice, hepatitis (A, B, C, D, E, F) alcoholic liver disease.		
4.	**Disease of bones and joints, Principles of cancer, Diseases of bones and joints, Principles of Cancer**	**8 Hours**	**17.77%**
	Rheumatoid arthritis, osteoporosis, and gout classification, etiology and pathogenesis of cancer Rheumatoid Arthritis, Osteoporosis, Gout Classification, etiology and pathogenesis of Cancer		
5.	**Infectious diseases and Sexually transmitted diseases**	**7 Hours**	**15.55%**
	Meningitis, Typhoid, Leprosy, Tuberculosis Urinary tract infections AIDS, Syphilis, Gonorrhea		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe risk factors, etiology and pathogenesis of the selected disease conditions
CO2	summarize signs and symptoms of selected disease conditions.
CO3	describe and differentiate the complications of selected diseases conditions.
CO4	narrate role of cell injury and inflammation in pathological conditions.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Vinay Kumar, Abul K. Abas, Jon C. Aster; Robbins &Cotran Pathologic Basis of Disease; South Asia edition; India; Elsevier; 2014.
2. Harsh Mohan; Text book of Pathology; 6th edition; India; Jaypee Publications; 2010.
3. Laurence B, Bruce C, Bjorn K.; Goodman Gilman's The Pharmacological Basis of
4. Therapeutics; 12th edition; New York; McGraw-Hill; 2011.

5. Best, Charles Herbert 1899-1978; Taylor, Norman Burke 1885-1972; West, John B (John Burnard); Best and Taylor‘s Physiological basis of medical practice; 12th ed; united states;
6. William and Wilkins, Baltimore;1991 [1990 printing].
7. Nicki R. Colledge, Brian R. Walker, Stuart H. Ralston;Davidson‘s Principles and Practice of Medicine; 21st edition; London; ELBS/Churchill Livingstone; 2010.
8. Guyton A, John. E Hall; Textbook of Medical Physiology; 12th edition; WB Saunders Company; 2010.
9. Joseph DiPiro, Robert L. Talbert, Gary Yee, Barbara Wells, L. Michael Posey;
10. Pharmacotherapy: A Pathophysiological Approach; 9th edition; London; McGraw-Hill Medical; 2014.
11. V. Kumar, R. S. Cotran and S. L. Robbins; Basic Pathology; 6th edition; Philadelphia; WB Saunders Company; 1997.
12. Roger Walker, Clive Edwards; Clinical Pharmacy and Therapeutics; 3rd edition; London; Churchill Livingstone publication; 2003.

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH3005: ENVIRONMENTAL SCIENCES (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	The Multidisciplinary nature of environmental studies.	15
2	Concept of an ecosystem.	15
3	Environmental Pollution.	15

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**The Multidisciplinary nature of environmental studies**	**15 Hours**	**33.33%**
	Natural Resources Renewable and non-renewable resources: Natural resources and associated problems Forest resources; b) Water resources; c) Mineral resources; d) Food resources; e) Energy resources; f) Land resources: Role of an individual in conservation of natural resources		
2.	**Concept of an ecosystem.**	**15 Hours**	**33.33%**
	Structure and function of an ecosystem. Introduction, types, characteristic features, structure and function of the ecosystems: Forest ecosystem; Grassland ecosystem; Desert ecosystem; Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)		
3.	**Environmental Pollution.**	**15 Hours**	**33.33%**
	Air pollution; Water pollution; Soil pollution		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	summarize basic knowledge about the environment and its allied problems.
CO2	narrate various Environmental Pollution like Air pollution; Water pollution and Soil pollution.
CO3	describe basic concept, structure and function of an ecosystem.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	-	-	-	-	-	-	-	-	3	-
CO2	-	-	-	-	-	-	-	-	-	3	-
CO3	-	-	-	-	-	-	-	-	-	3	-

Recommended Study Material:

❖ **Textbook:**

1. Y.K. Sing, Environmental Science, New Age International Pvt, Publishers, Bangalore
2. Agarwal, K.C. 2001 Environmental Biology, Nidi Publ. Ltd. Bikaner.
3. Bharucha Erach, The Biodiversity of India, Mapin Pu blishing Pvt. Ltd., Ahmedabad – 380 013, India,
4. Brunner R.C., 1989, Hazardous Waste Incineration, McGraw Hill Inc. 480p
5. Clark R.S., Marine Pollution, Clanderson Press Oxford
6. Cunningham, W.P. Cooper, T.H. Gorhani, E & Hepworth, M.T. 2001, Environmental Encyclopedia, Jaico Publ. House, Mumbai, 1196p
7. De A.K., Environmental Chemistry, Wiley Eastern Ltd. Down of Earth, Centre for Science and Environment

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
HS122B: VALUES AND ETHICS

Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	**Introduction to Values and Ethics** • *Need, Relevance and Significance of Values and Ethics : General* • *Concept and Meaning of Values and Ethics*	06
2	**Elements and Principles of Values** • *Universal & Personal Values* • *Social, Civic & Democratic Values* • *Adaptation Models & Methods of Values*	08
3	**Applied Ethics** • *Universal Code of Ethics* • *Professional Ethics* • *Organizational Ethics* • *Ethical Leadership* • *Domain Specific Ethics*	08
4	**Value, Ethics & Global Issues** • *Cross-Cultural Issues* • *Role of Ethics & Values in Sustainability* • *Case Studies*	08
	Total	30

Course Outcome (COs):

After completion of the course, the student would:

CO1	Understand the concepts and mechanics of values and ethics.
CO2	Understand the significance of value and ethical inputs in and get motivated to apply them in their life and profession.
CO3	Understand the significance of value and ethical inputs in and get motivated to apply them in social, global and civic issues.
CO4	Develop their responsibility towards society.
CO5	Comprehend their own core values and adhere to those values at their workplace.
CO6	Practice Ethical Leadership.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	-	-	-	-	-	-	-	-	-	2
CO2	-	2	-	-	-	-	-	3	-	-	-
CO3	-	3	-	-	-	3	-	-	2	2	-
CO4	-	-	-	-	3	-	-	3	2	-	-
CO5	-	-	-	-	-	-	3	2	-	-	-
CO6	-	-	-	-	3	-	-	2	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Reference Books / Reading

- Human Values and Ethics in Workplace, United Nations Settlement Program, 2006. (http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
- Ethics for Everyone, Arthur Dorbin, 2009. (http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf) .
- Values and Ethics for 21st Century, BBVA. (https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf) www.ethics.org

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme

SYLLABI

(Semester – 4)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH4001: PHARMACEUTICAL ORGANIC CHEMISTRY-III (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Stereo isomerism	10
2	Geometrical isomerism	10
3	Heterocyclic compounds	10
4	Synthesis, reactions and medicinal uses of following compounds/derivatives	8
5	Reactions of synthetic importance	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Stereo isomerism**	**10 Hours**	**22.22%**
	Optical isomerism – Optical activity, enantiomerism, diastereoisomerism, meso compounds Elements of symmetry, chiral and achiral molecules DL system of nomenclature of optical isomers, sequence rules, RS system of nomenclature of optical isomers Reactions of chiral molecules Racemic modification and resolution of racemic mixture. Asymmetric synthesis: partial and absolute		
2.	**Geometrical isomerism**	**10 Hours**	**22.22%**
	Nomenclature of geometrical isomers (Cis Trans, EZ, Syn Anti systems) Methods of determination of configuration of geometrical isomers. Conformational isomerism in Ethane, n-Butane and Cyclohexane. Stereo isomerism in biphenyl compounds (Atropisomerism) and conditions for optical activity. Stereospecific and stereo selective reactions		
3.	**Heterocyclic compounds**	**10 Hours**	**22.22%**
	Nomenclature and classification Synthesis, reactions and medicinal uses of following compounds/derivatives Pyrrole, Furan, and Thiophene - Relative aromaticity, reactivity and Basicity of pyrrole		
4.	**Synthesis, reactions and medicinal uses of following compounds/derivatives**	**8 Hours**	**17.77%**
	Pyrazole, Imidazole, Oxazole and Thiazole. Pyridine, Quinoline, Isoquinoline, Acridine and Indole. Basicity of pyridine Synthesis and medicinal uses of Pyrimidine, Purine, azepines and their derivatives		
5.	**Reactions of synthetic importance**	**7 Hours**	**15.55%**
	Metal hydride reduction ($NaBH_4$ and $LiAlH_4$), Clemmensen reduction, Birch reduction, Wolff Kishner reduction. Oppenauer-oxidation and Dakin reaction.		

	Beckmanns rearrangement and Schmidt rearrangement. Claisen-Schmidt condensation		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	depict methods of preparation & reaction of organic compounds
CO2	summarize the stereo chemical aspects of organic compounds and stereo chemical reaction
CO3	suggest various reaction of synthetic importance.
CO4	write the nomenclature, properties and medicinal use of heterocyclic compounds

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

- **Textbook:**

1. Organic chemistry by I.L. Finar, Volume-I & II.
2. A textbook of organic chemistry – Arun Bahl, B.S. Bahl.
3. Heterocyclic Chemistry by Raj K. Bansal

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH4002: BIOCHEMISTRY (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Carbohydrate metabolism and Biological oxidation	10
2	Lipid metabolism and Amino acid metabolism	10
3	Nucleic acid metabolism and genetic information transfer	10
4	Biomolecules and Bioenergetics	8
5	Enzymes	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Carbohydrate metabolism and Biological oxidation**	**10 Hours**	**22.22%**
	1. Carbohydrate metabolism • Glycolysis – Pathway, energetics and significance Citric acid cycle-Pathway, energetics and significance • HMP shunt and its significance; Glucose-6-Phosphate dehydrogenase (G6PD) deficiency • Glycogen metabolism Pathways and glycogen storage diseases (GSD) Gluconeogenesis- Pathway and its significance • Hormonal regulation of blood glucose level and Diabetes mellitus **2. Biological oxidation** • Electron transport chain (ETC) and its mechanism. • Oxidative phosphorylation & its mechanism and substrate level phosphorylation • Inhibitors ETC and oxidative phosphorylation/Uncouplers		
2.	**Lipid metabolism and Amino acid metabolism**	**10 Hours**	**22.22%**
	1. Lipid metabolism • β-Oxidation of saturated fatty acid (Palmitic acid) • Formation and utilization of ketone bodies; ketoacidosis De novo synthesis of fatty acids (Palmitic acid) • Biological significance of cholesterol and conversion of cholesterol into bile acids, steroid hormone and vitamin D • Disorders of lipid metabolism: Hypercholesterolemia, atherosclerosis, fatty liver and obesity **2. Amino acid metabolism**		

	• General reactions of amino acid metabolism: Transamination, deamination & decarboxylation, urea cycle and its disorders • Catabolism of phenylalanine and tyrosine and their metabolic disorders (Phenyketonuria, Albinism, alkeptonuria, tyrosinemia) • Synthesis and significance of biological substances; 5-HT, melatonin, dopamine, noradrenaline, adrenaline Catabolism of heme; hyperbilirubinemia and jaundice		
3.	**Nucleic acid metabolism and genetic information transfer**	**10 Hours**	**22.22%**
	• Biosynthesis of purine and pyrimidine nucleotides • Catabolism of purine nucleotides and Hyperuricemia and Gout Disease Organization of mammalian genome • Structure of DNA and RNA and their functions • DNA replication (semi conservative model) • Transcription or RNA synthesis • Genetic code, Translation or Protein synthesis and inhibitors		
4.	**Biomolecules and Bioenergetics**	**8 Hours**	**17.77%**
	1. Biomolecules • Introduction, classification, chemical nature and biological role of carbohydrate, lipids, nucleic acids, amino acids and proteins. **2. Bioenergetics** • Concept of free energy, endergonic and exergonic reaction, Relationship between free energy, enthalpy and entropy; Redox potential. Energy rich compounds; classification; biological significances of ATP and cyclic AMP		
5.	**Enzymes**	**7 Hours**	**15.55%**
	• Introduction, properties, nomenclature and IUB classification of enzymes Enzyme kinetics (Michaelis plot, Line Weaver Burke plot) • Enzyme inhibitors with examples • Regulation of enzymes: enzyme induction and repression, allosteric enzymes regulation • Therapeutic and diagnostic applications of enzymes and isoenzymes Coenzymes –Structure and biochemical functions		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe biochemical aspects of cell metabolism, importance of enzyme and enzymatic reactions.
CO2	summarize metabolic pathway of important biomolecules.
CO3	summarize role of DNA and RNA in protein synthesis.
CO4	describe enzymatic reaction and its application in drug metabolism.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH4002P: Biochemistry (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Qualitative analysis of carbohydrates (Glucose, Fructose, Lactose, Maltose, Sucrose and starch)
2	Identification tests for Proteins (albumin and Casein)
3	Quantitative analysis of reducing sugars (DNSA method) and Proteins (Biuret method)
4	Qualitative analysis of urine for abnormal constituents
5	Determination of blood creatinine
6	Determination of blood sugar
7	Determination of serum total cholesterol
8	Preparation of buffer solution and measurement of pH
9	Study of enzymatic hydrolysis of starch
10	Determination of Salivary amylase activity
11	Study the effect of Temperature on Salivary amylase activity.
12	Study the effect of substrate concentration on salivary amylase activity.

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	identify and discriminate the given sample of carbohydrate and protein
CO2	determine carbohydrates, creatinine, proteins and cholesterols in given samples
CO3	study the effect of external variables on enzyme activity
CO4	prepare buffer solution and measure pH
CO5	analyze the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	3	-	-	-	-	-	-	-	3
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Principles of Biochemistry by Lehninger.
2. Harper‘s Biochemistry by Robert K. Murry, Daryl K. Granner and Victor W. Rodwell.
3. Biochemistry by Stryer.
4. Biochemistry by D. Satyanarayan and U.Chakrapani
5. Textbook of Biochemistry by Rama Rao.
6. Textbook of Biochemistry by Deb.
7. Outlines of Biochemistry by Conn and Stumpf
8. Practical Biochemistry by R.C. Gupta and S. Bhargavan.
9. Introduction of Practical Biochemistry by David T. Plummer. (3rd Edition)
10. Practical Biochemistry for Medical students by Rajagopal and Ramakrishna.
11. Practical Biochemistry by Harold Varley.

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme
BPH4003: PHYSICAL PHARMACEUTICS-II (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Drug stability	10
2	Rheology and Deformation of solids	10
3	Coarse dispersion	10
4	Surface and interfacial phenomenon	8
5	Colloidal dispersions	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Drug stability** Reaction kinetics: zero, pseudo-zero, first & second order, units of basic rate constants, determination of reaction order. Physical and chemical factors influencing the chemical degradation of pharmaceutical product: temperature, solvent, ionic strength, dielectric constant, specific & general acid base catalysis, Simple numerical problems. Stabilization of medicinal agents against common reactions like hydrolysis & oxidation. Accelerated stability testing in expiration dating of pharmaceutical dosage forms. Photolytic degradation and its prevention.	**10 Hours**	**22.22%**
2.	**Rheology and Deformation of solids** Newtonian systems, law of flow, kinematic viscosity, effect of temperature, non-Newtonian systems, pseudoplastic, dilatants, plastic, thixotropy, thixotropy in formulation, determination of viscosity, capillary, falling Sphere, rotational viscometers Plastic and elastic deformation, Heckel equation, Stress, Strain, Elastic Modulus	**10 Hours**	**22.22%**
3.	**Coarse dispersion** Suspension, interfacial properties of suspended particles, settling in suspensions, formulation of suspensions. Emulsions and theories of emulsification, microemulsion and multiple emulsions; Physical stability of emulsions, preservation of emulsions, rheological properties of emulsions, phase equilibria and emulsion formulation	**10 Hours**	**22.22%**
	Surface and interfacial phenomenon		

4.	Liquid interface, surface & interfacial tensions, surface free energy, measurement of surface & interfacial tensions, spreading coefficient, adsorption at liquid interfaces, surface active agents, HLB Scale, solubilisation, detergency, adsorption at solid interface	**8 Hours**	**17.77%**
5.	**Colloidal dispersions**	**7 Hours**	**15.55%**
	Classification of dispersed systems & their general characteristics, size & shapes of colloidal particles, classification of colloids & comparative account of their general properties. Optical, kinetic & electrical properties. Effect of electrolytes, coacervation, peptization & protective action.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	describe reaction rate kinetic orders, factors influencing to accelerated stability studies.
CO2	explain Newtonian, non-Newtonian systems, viscosity determination, viscometers and thixotropy.
CO3	describe properties, stability, rheological consideration, formulation of suspension and emulsion.
CO4	narrate surface tension, interfacial tension, spreading coefficient, HLB, adsorption at solid-solid and solid- liquid interface

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH4003P: Physical Pharmaceutics-I (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Determination of surface tension of given liquids by drop count and drop weight method
2	Determination of HLB number of a surfactant by saponification
3	Determination of Freundlich and Langmuir constants using activated
4	Determination of critical micellar concentration of surfactants
5	Determination of viscosity of liquid using Ostwald‘s viscometer
6	Determination sedimentation volume with effect of different
7	Determination sedimentation volume with effect of different concentration of single suspending agents
8	Determination of viscosity of semisolid by using Brookfield
9	Determination of reaction rate constant first order.
10	Determination of reaction rate constant second order
11	Accelerated stability studies

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	determine viscosity and surface tension of given liquids at room temperature.
CO2	determine spreading co-efficient and critical micelle concentration of given sample of liquids.
CO3	analyse physical stability parameters of suspension and emulsion.
CO4	compute reaction rate constant and order of reaction through numerical and graphical method.
CO5	analyse the problem, communicate suggested solution, and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	3	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	3

CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

- **Textbook:**

1. Physical Pharmacy by Alfred Martin, Sixth edition
2. Experimental pharmaceutics by Eugene, Parott.
3. Tutorial pharmacy by Cooper and Gunn.
4. Stocklosam J. Pharmaceutical calculations, Lea & Febiger, Philadelphia.
5. Liberman H.A, Lachman C., Pharmaceutical Dosage forms, Tablets, Volume-1 to 3,
    a. Marcel Dekkar Inc.
6. Liberman H.A, Lachman C, Pharmaceutical dosage forms. Disperse systems, volume
    a. 1, 2, 3. Marcel Dekkar Inc.
7. Physical Pharmaceutics by Ramasamy C, and Manavalan R

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH4004: PHARMACOLOGY-I (THEORY)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	General Pharmacology	10
2	General Pharmacology	10
3	Pharmacology of peripheral nervous system	10
4	Pharmacology of central nervous system	8
5	Pharmacology of central nervous system	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**General Pharmacology** Introduction to Pharmacology- Definition, historical landmarks and scope of pharmacology, nature and source of drugs, essential drugs concept and routes of drug administration, Agonists, antagonists (competitive and non-competitive), spare receptors, addiction, tolerance, dependence, tachyphylaxis, idiosyncrasy, allergy. Pharmacokinetics- Membrane transport, absorption, distribution, metabolism and excretion of drugs. Enzyme induction, enzyme inhibition, kinetics of elimination	**10 Hours**	**22.22%**
2.	**General Pharmacology** Pharmacodynamics- Principles and mechanisms of drug action. Receptor theories and classification of receptors, regulation of receptors. drug receptors interactions signal transduction mechanisms, G-protein–coupled receptors, ion channel receptor, transmembrane enzyme linked receptors, transmembrane JAK-STAT binding receptor and receptors that regulate transcription factors, dose response relationship, therapeutic index, combined effects of drugs and factors modifying drug action. Adverse drug reactions. Drug interactions (pharmacokinetic and pharmacodynamic) Drug discovery and clinical evaluation of new drugs -Drug discovery phase, preclinical evaluation phase, clinical trial phase, phases of clinical trials and pharmacovigilance.	**10 Hours**	**22.22%**
3.	**Pharmacology of peripheral nervous system** Organization and function of ANS. Neurohumoral transmission,co-transmission and classification of neurotransmitters.	**10 Hours**	**22.22%**

	Parasympathomimetics, Parasympatholytics, Sympathomimetics, sympatholytics. Neuromuscular blocking agents and skeletal muscle relaxants (peripheral). Local anesthetic agents. Drugs used in myasthenia gravis and glaucoma		
4.	**Pharmacology of central nervous system** Neurohumoral transmission in the CNS, special emphasis on importance of various neurotransmitters like with GABA, Glutamate, Glycine, serotonin, dopamine. General anesthetics and pre-anesthetics Sedatives, hypnotics and centrally acting muscle relaxants. Anti-epileptics Alcohols and disulfiram	**8 Hours**	**17.77%**
5.	**Pharmacology of central nervous system** Psychopharmacological agents: Antipsychotics, antidepressants, anti-anxiety agents, anti-manics and hallucinogens. Drugs used in Parkinsons disease and Alzheimer‘s disease. CNS stimulants and nootropics Opioid analgesics and antagonists Drug addiction, drug abuse, tolerance and dependence	**7 Hours**	**15.55%**

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	summarize general concept of pharmacology with various pharmacokinetics parameters.
CO2	compare pharmacological actions of different categories of drugs at macromolecular levels.
CO3	describe pharmacology of central nervous system agents
CO4	describe pharmacology of peripheral nervous system agents

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	3	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH4004P Pharmacology-I (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Introduction to experimental pharmacology.
2	Commonly used instruments in experimental pharmacology.
3	Study of common laboratory animals
4	Maintenance of laboratory animals as per CPCSEA guidelines
5	Common laboratory techniques. Blood withdrawal, serum and plasma separation, anesthetics and euthanasia used for animal studies
6	Study of different routes of drugs administration in mice/rats
7	Study of effect of hepatic microsomal enzyme inducers on the phenobarbitone sleeping time in mice
8	Effect of drugs on ciliary motility of frog oesophagus
9	Effect of drugs on rabbit eye
10	Effects of skeletal muscle relaxants using rota-rod apparatus
11	Effect of drugs on locomotor activity using actophotometer
12	Anticonvulsant effect of drugs by MES and PTZ method
13	Study of stereotype and anti-catatonic activity of drugs on rats/mice
14	Study of anxiolytic activity of drugs using rats/mice
15	Study of local anesthetics by different methods

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	comprehend the functional roles of CPCSEA in animal experiments.
CO2	perform basic laboratory techniques for administration of drugs and withdrawal of biological fluid samples.
CO3	evaluate effect of drugs through appropriate assembly and/or procedure on muscle relaxation, locomotor activity, catatonic activity, sleep (anxiety), eye
CO4	quantify the agonism of drug by performing *in vitro* experiments.
CO5	analyze the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	3	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Rang H. P., Dale M. M., Ritter J. M., Flower R. J., Rang and Dale‘s
2. Pharmacology,Churchil Livingstone Elsevier
3. Katzung B. G., Masters S. B., Trevor A. J., Basic and clinical pharmacology, Tata Mc Graw-Hill
4. Goodman and Gilman‘s, The Pharmacological Basis of Therapeutics
5. Marry Anne K. K., Lloyd Yee Y., Brian K. A., Robbin L.C., Joseph G. B., Wayne A. K., Bradley R.W., Applied Therapeutics, The Clinical use of Drugs, The Point Lippincott Williams & Wilkins
6. Mycek M.J, Gelnet S.B and Perper M.M. Lippincott‘s Illustrated Reviews-Pharmacology
7. K. D.Tripathi. Essentials of Medical Pharmacology, JAYPEE Brothers Medical Publishers (P) Ltd, New Delhi.
8. Sharma H. L., Sharma K. K., Principles of Pharmacology, Paras medical publisher
9. Modern Pharmacology with clinical Applications, by Charles R.Craig& Robert,
10. Ghosh MN. Fundamentals of Experimental Pharmacology. Hilton & Company, Kolkata.
11. Kulkarni SK. Handbook of experimental pharmacology. Vallabh Prakashan,

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH4005: PHARMACOGNOSY-I (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Pharmacognosy, Classification of drugs, Quality control of Drugs of Natural Origin.	10
2	Cultivation, Collection, Processing and storage of drugs of natural origin and Conservation of medicinal plants	10
3	Plant tissue culture	7
4	Pharmacognosy in various systems of medicine and Introduction to secondary metabolites	10
5	Study of biological source, chemical nature and uses of drugs of natural origin containing drugs	8

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Introduction to Pharmacognosy, Classification of drugs, Quality control of Drugs of Natural Origin.**	**10 Hours**	**22.22%**
	Definition, history, scope and development of Pharmacognosy 1. Sources of Drugs – Plants, Animals, Marine & Tissue culture 2. Organized drugs, unorganized drugs (dried latex, dried juices, dried extracts, gums and mucilages, oleoresins and oleo- gum -resins). Alphabetical, morphological, taxonomical, chemical, pharmacological, chemo and sero taxonomical classification of drugs Adulteration of drugs of natural origin. Evaluation by organoleptic, microscopic, physical, chemical and biological methods and properties. Adulteration of drugs of natural origin. Evaluation by organoleptic, microscopic, physical, chemical and biological methods and properties.		
2.	**Cultivation, Collection, Processing and storage of drugs of natural origin and Conservation of medicinal plants**	**10 Hours**	**22.22%**
	Cultivation and Collection of drugs of natural origin. Factors influencing cultivation of medicinal plants. Plant hormones and their applications. Polyploidy, mutation and hybridization with reference to medicinal plants		
3.	**Plant tissue culture**	**7 Hours**	**15.55%**
	Historical development of plant tissue culture, types of cultures, Nutritional requirements, growth and their maintenance. Applications of plant tissue culture in pharmacognosy Edible vaccines		
4.	**Pharmacognosy in various systems of medicine and Introduction to secondary metabolites**	**10 Hours**	**22.22%**

	Role of Pharmacognosy in allopathy and traditional systems of medicine namely, Ayurveda, Unani, Siddha, Homeopathy and Chinese systems of medicine. Definition, classification, properties and test for identification of Alkaloids, Glycosides, Flavonoids, Tannins, Volatile oil and Resins		
5.	**Study of biological source, chemical nature and uses of drugs of natural origin containing drugs**	**8 Hours**	**17.77%**
	Plant Products: Fibers - Cotton, Jute, Hemp Hallucinogens, Teratogens, Natural allergens Primary metabolites: General introduction, detailed study with respect to chemistry, sources, preparation, evaluation, preservation, storage, therapeutic used and commercial utility as Pharmaceutical Aids and/or Medicines for the following Primary metabolites: Carbohydrates: Acacia, Agar, Tragacanth, Honey Proteins and Enzymes: Gelatin, casein, proteolytic enzymes (Papain, bromelain, serratiopeptidase, urokinase, streptokinase, pepsin). Lipids (Waxes, fats, fixed oils): Castor oil, Chaulmoogra oil, Wool Fat, Bees Wax Marine Drugs: Novel medicinal agents from marine sources		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	summarize the history, scope and development of Pharmacognosy
CO2	describe factors affecting cultivation, collection, processing and storage of drugs
CO3	apply knowledge of applications of plant tissue culture in pharmacognosy
CO4	classify different types of phytoconstituents and its importance
CO5	summarize and justify role of Pharmacognosy in allopathy as well as traditional systems of medicine
CO6	analyze the experimental data and interpret the presence of adulteration in crude drugs

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	3	-
CO4	3	-	-	-	-	-	-	-	-	-	-
CO5	3	-	-	-	-	-	-	-	-	-	3
C06	3	-	3	-	-	-	-	-	-	-	3

Bachelor of Pharmacy Programme
BPH4005P: Pharmacognosy-I (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Analysis of crude drugs by chemical tests: (i) Tragaccanth (ii) Acacia (iii)Agar (iv) Gelatin (v) starch (vi) Honey (vii) Castor oil
2	Determination of stomatal number and index
3	Determination of vein islet number, vein islet termination and
4	Determination of size of starch grains, calcium oxalate crystals by eye piece micrometer
5	Determination of Fiber length and width
6	Determination of number of starch grains by Lycopodium spore method
7	Determination of Ash value
8	Determination of Extractive values of crude drugs
9	Determination of moisture content of crude drugs
10	Determination of swelling index and foaming

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	identify the presence of phytoconsituents in crude drugs by chemical tests.
CO2	perform quantitative microscopic determination.
CO3	perform proximate analysis of herbal drugs.
CO4	analyze the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	3	3	-	-	-	-	-	-	-
CO3	3	-	3	3	-	-	-	-	-	-	-
CO4	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

- **Textbook:**

1. W.C.Evans, Trease and Evans Pharmacognosy, 16th edition, W.B. Sounders & Co., London, 2009.
2. Tyler, V.E., Brady, L.R. and Robbers, J.E., Pharmacognosy, 9th Edn., Lea and Febiger, Philadelphia, 1988.
3. Textbook of Pharmacognosy by T.E. Wallis
4. Mohammad Ali. Pharmacognosy and Phytochemistry, CBS Publishers & Distribution, New Delhi.
5. Textbook of Pharmacognosy by C.K. Kokate, Purohit, Gokhlae (2007), 37th Edition, Nirali Prakashan, New Delhi.
6. Herbal drug industry by R.D. Choudhary (1996), Ist Edn, Eastern Publisher, New Delhi.
7. Essentials of Pharmacognosy, Dr.SH.Ansari, IInd edition, Birla publications, New Delhi, 2007
8. Practical Pharmacognosy: C.K. Kokate, Purohit, Gokhlae
9. Anatomy of Crude Drugs by M.A. Iyengar

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
HS133 B: Creativity, Problem Solving and Innovation

Total hours: 30 Hr

Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	**Introduction to Creativity, Problem Solving and Innovation** • *Definitions of Creativity and Innovation* • *Need for Problem Solving and Innovation* • *Scope of Creativity in various Domains* • *Types and Styles of Thinking* • Strategies to develop Creativity, Problem Solving and Innovation skills	**06**
2	**Questioning, Learning and Visualization** • *Strategy and Methods of Questioning* • *Asking the Right Questions* • *Strategy of Learning and its Importance* • *Sources and Methods of Learning* • *Purpose and Value of Creativity Education in real life* • *Visualization strategies - Making thoughts Visible* • *Mind Mapping and Visualizing Thinking*	**06**
3	**Creative Thinking and Problem Solving** • Creative Thinking and its need • Strategy of Thinking Fluency • Generating all Possibilities • SCAMPER Technique • Divergent Vs Convergent Thinking • Lateral Vs Vertical Thinking • Fusion of Ideas for Problem Solving • Applying strategies for Problem Solving	**06**
4	**Logic, Language and Reasoning** • *Basic Concepts of Logic* • *Statement Vs Sentence* • *Premises Vs Conclusion* • *Concept of an Argument* • *Functions of Language: Informative, Expressive and Directive* • *Inductive Vs Deductive Reasoning* • *Critical Thinking & Creativity* • *Moral Reasoning*	**06**
5	**Contemporary Issues and Practices in Creativity and Problem Solving** • *Cognitive Research Trust Thinking for Creatively Solving Problems* • *Case Study on Contemporary Issues and Practices in Creativity and Problem Solving*	**06**
	Total	**30**

Course Outcome (COs):
After completion of the course, the student would;

CO1	Demonstrate creativity in their day to day activities and academic output
CO2	Solve personal, social and professional problems with a positive and an objective mindset
CO3	Think creatively and work towards problem solving in a strategic way
CO4	Initiate new and innovative practices in their chosen field of profession

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	3	-	-	-	-	-	3	-	-	-
CO2	-	-	2	-	-	-	-	3	2	2	-
CO3	-	-	2	-	3	-	-	3	-	-	-
CO4	-	-	-	-	-	-	-	3	-	-	2

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Books / Reference Books / Reading

Text Books

I. R Keith Sawyer, Zig Zag, The Surprising Path to Greater Creativity, Jossy-Bass Publication 2013

II. Michael Michalko, Crackling Creativity, The Secrets of Creative Genus, Ten Speed Press 2001

Reference Books

1. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
2. Edward De Beno, De Beno’s Thinking Course, Revised Edition, Pearson Publication 1994
3. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
4. Tony Buzan, How to Mind Map, Thorsons Publication 2002
5. Scott Berkum, The Myths of Innovation, Expended and revised edition, Berkun Publication 2010
6. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
7. Ira Flatow, The all Laughed, Harper Publication 1992
8. Paul Sloane, Des MacHale & M.A. DiSpezio, The Ultimate Lateral & Critical Thinking

Puzzle book, Sterling Publication 2002

Additional Readings

1. Keith Sawer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
2. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
3. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
4. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
5. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
6. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
7. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
8. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994
9. Soctt G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
10. Donald J Treffinger, scott G Isaksen, K Brain stead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
11. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
12. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
13. Howard Gardner, Creating minds, Basic Books Publication 1993
14. Mihaly Csikzentmihalyi, Creativity–Flow and Psychology of Discovery and Invention, Harper Publication 1996
15. Martin Gerdner, W. H., Ahal Insight, Freeman Publication 1978
16. Paul Sloane, Test Your Lateral Thinking IQ, Sterling Publication1994
17. Paul Sloane & Des Machale Intriguing, Lateral Thinking Puzzles, Sterling Publication 1996

Articles / Videos / Other Suggested Materials

- Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme

SYLLABI

(Semester – 5)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH5001: Medicinal Chemistry-I (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Medicinal Chemistry	10
2	Drugs acting on Autonomic Nervous System	10
3	Cholinergic neurotransmitters	10
4	Drugs acting on Central Nervous System	8
5	Drugs acting on Central Nervous System	7

Detailed Syllabus:

Course Content: **Study of the development of the following classes of drugs, Classification, mechanism of action, uses of drugs mentioned in the course, Structure activity relationship of selective class of drugs as specified in the course and synthesis of drugs superscripted (*)**

Sr. No.	UNIT	Hours	Weight age (%)
1.	**Introduction to Medicinal Chemistry, History and development of medicinal chemistry, Physicochemical properties in relation to biological action, Drug metabolism**	**10 Hours**	**22.22%**
	• **Introduction to Medicinal Chemistry** • **History and development of medicinal chemistry** • **Physicochemical properties in relation to biological action** Ionization, Solubility, Partition Coefficient, Hydrogen bonding, Protein binding, Chelation, Bioisosterism, Optical and Geometrical isomerism. • **Drug metabolism** Drug metabolism principles- Phase I and Phase II. Factors affecting drug metabolism including stereo chemical aspects.		
2.	**Drugs acting on Autonomic Nervous System:**	**10 Hours**	**22.22%**
	Drugs acting on Autonomic Nervous System **Adrenergic Neurotransmitters:** Biosynthesis and catabolism of catecholamine. Adrenergic receptors (Alpha & Beta) and their distribution. • **Sympathomimetic agents: SAR of Sympathomimetic agents** **Direct acting:** Nor-epinephrine, Epinephrine, Phenylephrine*, Dopamine, Methyldopa, Clonidine, Dobutamine, Isoproterenol,		

	Terbutaline, Salbutamol*, Bitolterol, Naphazoline, Oxymetazoline and Xylometazoline. • **Indirect acting agents:** Hydroxyamphetamine, Pseudoephedrine, Propylhexedrine. • **Agents with mixed mechanism:** Ephedrine, Metaraminol. • **Adrenergic Antagonists:** • **Alpha adrenergic blockers:** Tolazoline*, Phentolamine, Phenoxybenzamine, Prazosin, Dihydroergotamine, Methysergide. • **Beta adrenergic blockers:** SAR of beta blockers, Propranolol*,Metibranolol, Atenolol, Betazolol, Bisoprolol, Esmolol, Metoprolol, Labetolol, Carvedilol.		
3.	**Cholinergic neurotransmitters, Parasympathomimetic agents:** **Cholinergic neurotransmitters:** Biosynthesis and catabolism of acetylcholine. Cholinergic receptors (Muscarinic & Nicotinic) and their distribution. • **Parasympathomimetic agents: SAR of Parasympathomimetic agents** • **Direct acting agents:** Acetylcholine, Carbachol*, Bethanechol, Methacholine, Pilocarpine. • **Indirect acting/ Cholinesterase inhibitors (Reversible & Irreversible):** Physostigmine, Neostigmine*, Pyridostigmine, Edrophonium chloride, Tacrine hydrochloride, Ambenonium chloride, Isofluorphate, Echothiophate iodide, Parathione, Malathion. • **Cholinesterase reactivator:** Pralidoxime chloride. • **Cholinergic Blocking agents: SAR of cholinolytic agents** • **Solanaceous alkaloids and analogues:** Atropine sulphate, Hyoscyamine sulphate, Scopolamine hydrobromide, Homatropine hydrobromide, Ipratropium bromide*. • **Synthetic cholinergic blocking agents:** Tropicamide, Cyclopentolate hydrochloride, Clidinium bromide, Dicyclomine hydrochloride*, Glycopyrrolate, Methantheline bromide, Propantheline bromide, Benztropine mesylate, Orphenadrine citrate, Biperidine hydrochloride, Procyclidine hydrochloride*, Tridihexethyl chloride, Isopropamide iodide, Ethopropazine hydrochloride.	**10 Hours**	**22.22%**
4.	**Drugs acting on Central Nervous System** **A. Sedatives and Hypnotics:** **Benzodiazepines:** SAR of Benzodiazepines, Chlordiazepoxide, Diazepam*, Oxazepam, Chlorazepate, Lorazepam, Alprazolam, Zolpidem **Barbiturtes:** SAR of barbiturates, Barbital*, Phenobarbital, Mephobarbital, Amobarbital, Butabarbital, Pentobarbital, Secobarbital **Miscelleneous:** Amides & imides: Glutethmide.	**8 Hours**	**17.77%**

	Alcohol & their carbamate derivatives: Meprobomate, Ethchlorvynol. Aldehyde & their derivatives: Triclofos sodium, Paraldehyde.		
	B. Antipsychotics **Phenothiazeines:** SAR of Phenothiazeines - Promazine hydrochloride, Chlorpromazine hydrochloride*, Triflupromazine, Thioridazine hydrochloride Piperacetazine hydrochloride, Prochlorperazine maleate, Trifluoperazine hydrochloride. **Ring Analogues of Phenothiazeines:** Chlorprothixene, Thiothixene, Loxapine succinate, Clozapine. **Fluro buterophenones:** Haloperidol, Droperidol, Risperidone. **Beta amino ketones:** Molindone hydrochloride. **Benzamides:** Sulpieride. **C. Anticonvulsants:** SAR of Anticonvulsants**,** mechanism of anticonvulsant action **Barbiturates**: Phenobarbitone, Methabarbital. **Hydantoins:** Phenytoin*, Mephenytoin, Ethotoin **Oxazolidine diones:** Trimethadione, Paramethadione **Succinimides:** Phensuximide, Methsuximide, Ethosuximide* **Urea and monoacylureas**: Phenacemide, Carbamazepine* **Benzodiazepines**: Clonazepam **D. Miscellaneous:** Primidone, Valproic acid , Gabapentin, Felbamate		
5.	**Drugs acting on Central Nervous System** **General anesthetics:** **Inhalation anesthetics:** Halothane*, Methoxyflurane, Enflurane, Sevoflurane, Isoflurane, Desflurane. **Ultra short acting barbitutrates:** Methohexital sodium*, Thiamylal sodium, Thiopental sodium. **Dissociative anesthetics:** Ketamine hydrochloride.* **Narcotic and non-narcotic analgesics** **Morphine and related drugs:** SAR of Morphine analogues, Morphine sulphate, Codeine, Meperidine hydrochloride, Anilerdine hydrochloride, Diphenoxylate hydrochloride, Loperamide hydrochloride, Fentanyl citrate*, Methadone hydrochloride*, Propoxyphene hydrochloride, Pentazocine, Levorphanol tartarate. **Narcotic antagonists:** Nalorphine hydrochloride, Levallorphan tartarate, Naloxone hydrochloride. **Anti-inflammatory agents:** Sodium salicylate, Aspirin, Mefenamic acid*, Meclofenamate, Indomethacin, Sulindac, Tolmetin, Zomepriac, Diclofenac, Ketorolac, Ibuprofen*, Naproxen, Piroxicam, Phenacetin, Acetaminophen, Antipyrine, Phenylbutazone.	**7 Hours**	**15.55%**

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	corelate physicochemical properties of drug in relation to biological activity.
CO2	describe the chemistry and structure activity relationship of drugs with respect to their pharmacological activity.
CO3	illustrate the drug metabolic pathways and its impact on adverse effect and therapeutic effect of drugs.
CO4	suggest, categorise and outline the chemical synthetic pathways of selected drugs.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	3	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH5001P: Medicinal Chemistry-I (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the practical
1	**Preparation of drugs and intermediates**
	1,3-pyrazole
	1,3-oxazole
	Benzimidazole
	Benztriazole
	2,3- diphenyl quinoxaline
	Benzocaine
	Phenytoin
	Phenothiazine
	Barbiturate
2	**Assay of drugs**
	Chlorpromazine
	Phenobarbitone
	Atropine
	Ibuprofen
	Aspirin
	Furosemide
3	**Determination of Partition coefficient for drugs**

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	perform synthesis of drugs and its intermediates.
CO2	perform Assay of drugs and intermediates using appropriate analytical methods
CO3	determine Partition coefficient of medicinal compounds
CO4	apply software tools to draw structures, reactions and mechanism
CO5	analyze the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	3	3	-	-	-	-	3	-	-	-
CO2	3	3	3	3	-	-	-	3	-	-	-
CO3	3	3	3	3	-	-	-	3	-	-	-
CO4	3	3	3	-	-	-	-	3	-	-	-
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Wilson and Griswold‘s Organic medicinal and Pharmaceutical Chemistry.
2. Foye‘s Principles of Medicinal Chemistry.
3. Burger‘s Medicinal Chemistry, Vol. I to IV.
4. Introduction to principles of drug design- Smith and Williams.
5. Remington‘s Pharmaceutical Sciences.
6. Martindale‘s extra pharmacopoeia.
7. Graham L. Petrick‘s An introduction to Medicinal Chemistry
8. D. Sriram and P.Yogeshwari‘s Medicinal Chemistry
9. Camile G. Wermuth‘s The Practice of Medicinal Chemistry
10. Kadam, Mahadik and Bothra‘s Principle of Medicinal Chemistry (Vol I and II)

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH5002: Industrial Pharmacy-I (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Preformulation Studies	7
	Physical properties	
	Chemical Properties	
2	Tablets	10
	Liquid orals	
3	Capsules	8
	Pellets	
4	Parenteral Products	10
	Ophthalmic Preparations	
5	Cosmetics	10
	Pharmaceutical Aerosols	
	Packaging Materials Science	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Preformulation Studies, Physical properties, Chemical Properties**	**7 Hours**	**15.55%**
	Introduction to preformulation, goals and objectives, study of physicochemical characteristics of drug substances. Physical form (crystal & amorphous), particle size, shape, flow properties, solubility profile (pKa, pH, partition coefficient), polymorphism Hydrolysis, oxidation, reduction, racemisation, polymerization BCS classification of drugs Application of preformulation considerations in the development of solid, liquid oral and parenteral dosage forms and its impact on stability of dosage forms.		
2.	**Tablets, Liquid orals**	**10 Hours**	**22.22%**
	Introduction, ideal characteristics of tablets, classification of tablets. Excipients, Formulation of tablets, granulation methods, compression and processing problems. Equipment and tablet tooling.		

	Tablet coating: Types of coating, coating materials, formulation of coating composition, methods of coating, equipment employed and defects in coating. Quality control tests: In process and finished product tests Formulation and manufacturing consideration of solutions, suspensions and emulsions; Filling and packaging; evaluation of liquid orals official in pharmacopoeia		
3.	**Capsules, Pellets**	**8 Hours**	**17.77%**
	Hard gelatin capsules: Introduction, Extraction of gelatin and production of hard gelatin capsule shells. Size of capsules, Filling, finishing and special techniques of formulation of hard gelatin capsules. In process and final product quality control tests for capsules. Soft gelatin capsules: Nature of shell and capsule content, size of capsules, importance of base adsorption and minimum/gram factors, production, in process and final product quality control tests. Packing, storage and stability testing of soft gelatin capsules Introduction, formulation requirements, Palletization process, equipment for manufacture of pellets		
4.	**Parenteral Products, Ophthalmic Preparations**	**10 Hours**	**22.22%**
	Definition, types, advantages and limitations. Preformulation factors and essential requirements, vehicles, additives, importance of isotonicity Production procedure, production facilities and controls. Formulation of injections, sterile powders, emulsions, suspensions, large volume parenterals and lyophilized products, Sterilization. Containers and closures selection, filling and sealing of ampoules, vials and infusion fluids. Quality control tests. Introduction, formulation considerations; formulation of eye drops, eye ointments and eye lotions; methods of preparation; labeling, containers; evaluation of ophthalmic preparations		
5.	**Cosmetics, Pharmaceutical Aerosols, Packaging Materials Science**	**10 Hours**	**22.22%**
	Formulation and preparation of the following cosmetic preparations: lipsticks, shampoos, cold cream and vanishing cream, toothpastes, hair dyes and sunscreens. Definition, propellants, containers, valves, types of aerosol systems; formulation and manufacture of aerosols; Evaluation of aerosols; Quality control and stability studies. Materials used for packaging of pharmaceutical products, factors influencing choice of containers, legal and official requirements for containers, stability aspects of packaging materials, quality control tests.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	appraise physicochemical characteristics of drug substances and suggest the application
CO2	describe the various pharmaceutical dosage forms and their manufacturing techniques
CO3	rationalise development of various dosage forms
CO4	suggest methods for assessing quality attributes of pharmaceutical dosage forms
CO5	describe and rationalize need base selection of packaging materials.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	-
CO5	3	-	3	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH 5002P: Industrial Pharmacy-I (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Preformulation study for prepared granules
2	Preparation and evaluation of Paracetamol tablets
3	Preparation and evaluation of Aspirin tablets
4	Coating of tablets
5	DRC of acetylcholine using frog rectus abdominis muscle.
6	Preparation of Calcium Gluconate injection
7	Preparation of Ascorbic Acid injection
8	Preparation of Paracetamol Syrup
9	Preparation of Eye drops
10	Preparation of Pellets by extrusion spheronization technique
11	Preparation of Creams (cold / vanishing cream)
12	Evaluation of Glass containers

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	carry out preformulation studies to improve the performance of drug and development of dosage form.
CO2	formulate and sterile, solid, liquid and semisolid dosage forms and evaluate them for their quality using available resources in given timeline.
CO3	operate instrumentation /equipment used for preparation, evaluation and coating of tablets/granules as per standards
CO4	conduct the quality control test of the marketed tablets and capsules as per procedure and specification in I.P.
CO5	evaluate the packaging material as per pharmacopoeial method and ensure its compatibility and safety.
CO6	analyze the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	3	3	-	-	-	-	-	-	-	3
CO2	3	3	3	3	-	-	-	-	-	-	3
CO3	3	-	-	3	-	-	-	-	-	-	3
CO4	3	3	3	3	-	-	-	-	3	-	-
CO5	3	3	3	-	-	-	-	-	3	-	-
C06	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Pharmaceutical dosage forms - Tablets, volume 1 -3 by H.A. Liberman, Leon Lachman &J.B.Schwartz
2. Pharmaceutical dosage form - Parenteral medication vol- 1&2 by Liberman & Lachman
3. Pharmaceutical dosage form disperse system VOL-1 by Liberman & Lachman
4. Modern Pharmaceutics by Gilbert S. Banker & C.T. Rhodes, 3rd Edition
5. Remington: The Science and Practice of Pharmacy, 20th edition Pharmaceutical Science (RPS)
6. Theory and Practice of Industrial Pharmacy by Liberman & Lachman
7. Pharmaceutics- The science of dosage form design by M.E.Aulton, Churchill livingstone, Latest edition
8. Introduction to Pharmaceutical Dosage Forms by H. C.Ansel, Lea &Febiger, Philadelphia, 5thedition, 2005
9. Drug stability - Principles and practice by Cartensen & C.J. Rhodes, 3rd Edition, Marcel Dekker Series, Vol 107.
10. Pharmaceutical Packaging Technology edited by D.A. Dean, E. R. Evans, Taylor and Francis, New York
11. Indian Pharmacopoeia, published by Indian Pharmacopoeial commission, Ghaziabad.
12. United State Pharmacopoeia, Indian edition, United State Pharmacopoeial convention INC.
13. British Pharmacopoeia, British Pharmacopoeia commission office, U.K.
14. Handbook of Preformulations, S. K. Niazi, Informa Healthcare, New York.
15. Pharmaceutical Preformulations & Formulation, edited by Marks Gibson, Interpharm/CRC, Boca Raton, Florida, USA.

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH5003: Pharmacology-II (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Pharmacology of drugs acting on cardiovascular system	07
2	Pharmacology of drugs acting on cardiovascular system Pharmacology of drugs acting on urinary system	10
3	Autocoids and related drugs	08
4	Pharmacology of drugs acting on endocrine system	10
5	Pharmacology of drugs acting on endocrine system Bioassay	10

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weight age (%)
1.	**Pharmacology of drugs acting on cardiovascular system**	**7 Hours**	**15.55%**
	a. Introduction to hemodynamic and electrophysiology of heart. b. Drugs used in congestive heart failure c. Anti-hypertensive drugs. d. Anti-anginal drugs. e. Anti-arrhythmic drugs. f. Anti-hyperlipidemic drugs.		
2.	**Pharmacology of drugs acting on cardio vascular system, Pharmacology of drugs acting on urinary system**	**10 Hours**	**22.22%**
	a. Drug used in the therapy of shock. b. Hematinics, coagulants and anticoagulants. c. Fibrinolytics and anti-platelet drugs d. Plasma volume expanders a. Diuretics b. Anti-diuretics		
3.	**Autocoids and related drugs**	**8 Hours**	**17.77%**
	a. Introduction to autacoids and classification b. Histamine, 5-HT and their antagonists. c. Prostaglandins, Thromboxanes and Leukotrienes. d. Angiotensin, Bradykinin and Substance P. e. Non-steroidal anti-inflammatory agents f. Anti-gout drugs g. Antirheumatic drugs		
	Pharmacology of drugs acting on endocrine system		

4.	a. Basic concepts in endocrine pharmacology. b. Anterior Pituitary hormones- analogues and their inhibitors. c. Thyroid hormones- analogues and their inhibitors. d. Hormones regulating plasma calcium level- Parathormone, Calcitonin and Vitamin-D. d. Insulin, Oral Hypoglycemic agents and glucagon. e. ACTH and corticosteroids.	**10 Hours**	**22.22%**
5.	**Pharmacology of drugs acting on endocrine system, Bioassay** a. Androgens and Anabolic steroids. b. Estrogens, progesterone and oral contraceptives. c. Drugs acting on the uterus. a. Principles and applications of bioassay. b. Types of bioassay c. Bioassay of insulin, oxytocin, vasopressin, ACTH, d-tubocurarine	**10 Hours**	**22.22%**

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	classify and provide specific examples for drugs affecting the Cardiovascular system, urinary system, endocrine system and autacoids
CO2	describe and compare mechanism of action, adverse effects and contraindications for drugs affecting on cardiovascular system, urinary system, endocrine system and autacoids
CO3	illustrate clinical conditions with suggestion and justify the appropriate use of drugs in the particular disease conditions
CO4	summarize and suggest the assay of drugs *in-vitro* and *in-vivo.*

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH5003P: Pharmacology-II (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Introduction to in-vitro pharmacology and physiological salt solutions.
2	Effect of drugs on isolated frog heart.
3	Effect of drugs on blood pressure and heart rate of dog.
4	Study of diuretic activity of drugs using rats/mice
5	DRC of acetylcholine using frog rectus abdominis muscle.
6	Effect of physostigmine and atropine on DRC of acetylcholine using frog rectus
7	Bioassay of histamine using guinea pig ileum by matching method
8	Bioassay of oxytocin using rat uterine horn by interpolation method
9	Bioassay of serotonin using rat fundus strip by three point bioassay
10	Bioassay of acetylcholine using rat ileum/colon by four point bioassay
11	Determination of PA2 value of prazosin using rat anococcygeus muscle (by Schilds plot method)
12	Determination of PD2 value using guinea pig ileum
13	Anti-inflammatory activity of drugs using carrageenan induced paw-edema model
14	Analgesic activity of drug using central and peripheral methods

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	perform isolated tissue based experiments
CO2	perform various receptor activity using isolated tissue preparation
CO3	demonstrate screening model based experiments
CO4	quantify drugs using various *in-vivo* and *in-vitro* experiments
CO5	analyze the problem, communicate suggested solution and interprete the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	3	-	-	-	-	-	-	-	-	3
CO2	3	3	-	-	-	-	-	-	-	-	3
CO3	3	3	-	-	-	-	-	-	-	-	3
CO4	3	3	-	-	-	-	-	-	-	-	3
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Rang H. P., Dale M. M., Ritter J. M., Flower R. J., Rang and Dale‘s Pharmacology, Churchil Livingstone Elsevier
2. Katzung B. G., Masters S. B., Trevor A. J., Basic and clinical pharmacology, Tata Mc Graw-Hill.
3. Goodman and Gilman‘s, The Pharmacological Basis of Therapeutics
4. Marry Anne K. K., Lloyd Yee Y., Brian K. A., Robbin L.C., Joseph G. B., Wayne A. K., Bradley R.W., Applied Therapeutics, The Clinical use of Drugs, The Point Lippincott Williams & Wilkins.
5. Mycek M.J, Gelnet S.B and Perper M.M. Lippincott‘s Illustrated Reviews-Pharmacology.
6. K.D.Tripathi. Essentials of Medical Pharmacology, , JAYPEE Brothers Medical Publishers (P) Ltd, New Delhi.
7. Sharma H. L., Sharma K. K., Principles of Pharmacology, Paras medical publisher
8. Modern Pharmacology with clinical Applications, by Charles R.Craig& Robert.
9. Ghosh MN. Fundamentals of Experimental Pharmacology. Hilton & Company, Kolkata.
10. Kulkarni SK. Handbook of experimental pharmacology. Vallabh Prakashan.

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH5004: Pharmacognosy-II (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Metabolic pathways in higher plants and their determination	7
2	General introduction, composition, chemistry & chemical classes, general methods of extraction & analysis, biosources, therapeutic uses and commercial applications of secondary metabolites	14
3	Isolation, Identification and Analysis of	6
4	Industrial production, estimation and utilization of the	10
5	Basics of Phytochemistry	8

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weight age (%)
1.	**Metabolic pathways in higher plants and their determination**	**7 Hours**	**15.55%**
	a) Brief study of basic metabolic pathways and formation of different secondary metabolites through these pathways- Shikimic acid pathway, Acetate pathways and Amino acid pathway. b) Study of utilization of radioactive isotopes in the investigation of Biogenetic studies.		
2.	General introduction, composition, chemistry & chemical classes, general methods of extraction & analysis, biosources, therapeutic uses and commercial applications of following secondary metabolites, Alkaloids, Phenylpropanoids and Flavonoids, Steroids, Cardiac Glycosides & Triterpenoids, Volatile oils,Tannins, Resins, Glycosides,Iridoids, Other terpenoids & Naphthaquinones	**14 Hours**	**31.11%**
	Vinca, Rauwolfia, Belladonna, Opium Lignans, Tea, Ruta Liquorice, Dioscorea, Mentha, Clove, Cinnamon, Fennel, Coriander Catechu, Pterocarpus, Myrobalan,Amla Benzoin, Guggul, Ginger, Asafoetida, Myrrh, Colophony Senna, Aloes, Bitter Almond Gentian, Artemisia, taxus, carotenoids		
	Isolation, Identification and Analysis of Phytoconstituents		

3.		**6 Hours**	**13.33%**
	a) Terpenoids: Menthol, Citral, Artemisin b) Glycosides: Glycyrhetinic acid & Rutin c) Alkaloids: Atropine,Quinine,Reserpine,Caffeine d) Resins: Podophyllotoxin, Curcumin		
4.	**Industrial production, estimation and utilization of the following phytoconstituents**	**10 Hours**	**22.22%**
	Forskolin, Sennoside, Artemisinin, Diosgenin, Digoxin, Atropine, Podophyllotoxin, Caffeine, Taxol, Vincristine and Vinblastine		
5.	**Basics of Phytochemistry**	**8 Hours**	**17.77%**
	Modern methods of extraction, application of latest techniques like Spectroscopy, chromatography and electrophoresis in the isolation, purification and identification of crude drugs.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe radio tracer technique and biogenesis of metabolites in higher plants.
CO2	describe pharmacognostic details for medicinal plants.
CO3	compare and contrast selected medicinal plants.
CO4	suggest, justify and describe appropriate extraction technique.
CO5	summarize the methods of industrial production, quality control and application of selected phyto-constituents.
CO6	suggest applications of chromatography and spectroscopy in phytochemical analysis.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	3	-	-	-	-	-	-	-	3
CO5	3	-	3	3	-	-	-	-	-	-	-
CO6	3	-	3	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH5004P: Pharmacognosy -II (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the practical
1	Morphology, histology and powder characteristics & extraction & detection of Cinchona, Cinnamon, Senna, Clove, Ephedra, Fennel and Coriander
2	**Exercise involving isolation & detection of active principles**
	a. Caffeine - from tea dust.
	b. Diosgenin from Dioscorea
	c. Atropine from Belladonna
	d. Sennosides from Senna
3	**Separation of sugars by Paper chromatography**
4	**TLC of herbal extract**
5	**Distillation of volatile oils and detection of phytoconstitutents by TLC**
6	**Analysis of crude drugs by chemical tests**
	(i) Asafoetida
	(ii) Benzoin
	(iii)Colophony
	(iv)Aloes
	(v)Myrrh

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	carry out morphological and microscopic evaluation of plant drugs
CO2	identify plant drugs powder through microscopic assessment
CO3	perform Thin Layer Chromatography and interprete the results.
CO4	separate phytoconstituents and ensure identity using TLC.
CO5	analyze the problem, communicate suggested solution and interprete the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	3	-	-	-	-	-	-	-
CO2	3	-	3	3	-	-	-	-	-	-	-
CO3	3	3	3	3	-	-	-	3	-	-	-
CO4	3	3	3	-	-	-	-	3	-	-	3
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1 W.C.Evans, Trease and Evans Pharmacognosy, 16th edition, W.B. Sounders & Co., London, 2009.
2 Mohammad Ali. Pharmacognosy and Phytochemistry, CBS Publishers & Distribution, New Delhi
3 Text book of Pharmacognosy by C.K. Kokate, Purohit, Gokhlae (2007), 37th Edition, Nirali Prakashan, New Delhi.
4 Herbal drug industry by R.D. Choudhary (1996), Ist Edn, Eastern Publisher, New Delhi.
5 Essentials of Pharmacognosy, Dr.SH.Ansari, IInd edition, Birla publications, New Delhi, 2007
6 Herbal Cosmetics by H.Pande, Asia Pacific Business press, Inc, New Delhi.
7 A.N. Kalia, Textbook of Industrial Pharmacognosy, CBS Publishers, New Delhi, 2005.
8 R Endress, Plant cell Biotechnology, Springer-Verlag, Berlin, 1994.
9 Pharmacognosy & Pharmacobiotechnology. James Bobbers, Marilyn KS, VE Tylor.
10 The formulation and preparation of cosmetic, fragrances and flavours.
11 Remington‘s Pharmaceutical sciences.
12 Textbook of Biotechnology by Vyas and Dixit.
13 Textbook of Biotechnology by R.C. Dubey.

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH5005: Pharmaceutical Jurisprudence (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Drugs and Cosmetics Act, 1940 and its rules 1945	10
2	Drugs and Cosmetics Act, 1940 and its rules 1945	10
3	Pharmacy Act –1948	10
	Medicinal and Toilet Preparation Act –1955	
	Narcotic Drugs and Psychotropic substances Act-1985	
4	Study of Salient Features of Drugs and magic remedies Act 1954 and its rules	8
	Prevention of Cruelty to animals Act-1960	
	National Pharmaceutical Pricing Authority	
5	Pharmaceutical Legislations	7
	Code of Pharmaceutical ethics	
	Medical Termination of pregnancy act	
	Right to information Act	
	Introduction to Intellectual Property Rights (IPR)	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weight age (%)
1.	**Drugs and Cosmetics Act, 1940 and its rules 1945**	**10 Hours**	**22.22%**
	Objectives, Definitions, Legal definitions of schedules to the act and rules. Import of drugs – Classes of drugs and cosmetics prohibited from import, Import under license or permit. Offences and penalties. Manufacture of drugs – Prohibition of manufacture and sale of certain drugs, Conditions for grant of license and conditions of license for manufacture of drugs, Manufacture of drugs for test, examination and analysis, manufacture of new drug, loan license and repacking license.		
2	**Drugs and Cosmetics Act, 1940 and its rules 1945**	**10 Hours**	**22.22%**
	Detailed study of Schedule G, H, M, N, P, T, U, V, X, Y, Part XII B, Sch F & DMR (OA). Sale of Drugs – Wholesale, Retail sale and Restricted license. Offences and penalties. Labeling & Packing of drugs- General labeling requirements and specimen labels for drugs and cosmetics, List of permitted colors. Offences and penalties.		

	Administration of the act and rules – Drugs Technical Advisory Board, Central drugs Laboratory, Drugs Consultative Committee, Government drug analysts, licensing authorities, controlling authorities, Drugs Inspectors.		
3.	**Pharmacy Act –1948, Medicinal and Toilet Preparation Act –1955, Narcotic Drugs and Psychotropic substances Act-1985 and Rules**	**10 Hours**	**22.22%**
	Objectives, Definitions, Pharmacy Council of India; its constitution and functions, Education Regulations, State and Joint state pharmacy councils; its constitution and functions, Registration of Pharmacists, Offences and Penalties. Objectives, Definitions, Licensing, Manufacture In bond and Outside bond, Export of alcoholic preparations, Manufacture of Ayurvedic, Homeopathic, and Patent & Proprietary Preparations. Offences and Penalties. Objectives, Definitions, Authorities and Officers, Constitution and Functions of narcotic & Psychotropic Consultative Committee, National Fund for Controlling the Drug Abuse, Prohibition, Control and Regulation, opium poppy cultivation and production of poppy straw, manufacture, sale and export of opium, Offences and Penalties.		
4.	**Study of Salient Features of Drugs and magic remedies Act 1954 and its rules, Prevention of Cruelty to animals Act-1960, National Pharmaceutical Pricing Authority**	**8 Hours**	**17.77%**
	Objectives, Definitions, Prohibition of certain advertisements, Classes of Exempted advertisements, Offences and Penalties. Objectives, Definitions, Institutional Animal Ethics Committee, Breeding and Stocking of Animals, Performance of Experiments, Transfer and acquisition of animals for experiment, Records, Power to suspend or revoke registration, Offences and Penalties. Drugs Price Control Order (DPCO) - 2013. Objectives, Definitions, Sale prices of bulk drugs, Retail price of formulations, Retail price and ceiling price of scheduled formulations, National List of Essential Medicines (NLEM).		
5.	**Pharmaceutical Legislations**, **Code of Pharmaceutical ethics, Medical Termination of pregnancy act, Right to information Act** **Introduction to Intellectual Property Rights (IPR)**	**7 Hours**	**15.55%**
	A brief review, Introduction, Study of drugs enquiry committee, Health survey and development committee, Hathi committee and Mudaliar committee. Definition, Pharmacist in relation to his job, trade, medical profession and his profession, Pharmacist‘s oath. Introduction, Termination of Pregnancies, Offences and Penalties.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	summarize and interprete Drugs and Cosmetics Act, 1940 and its rules 1945
CO2	analyse and infer pharmaceutical legislations and their implications in the development and marketing of pharmaceuticals.
CO3	describe the regulatory structure for governing the manufacture and sales of pharmaceuticals
CO4	appraise the code of ethics during the pharmaceutical practice

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	-	-	-	-	3	3	-	3	-	3
CO2	-	-	3	-	-	3	3	-	3	-	-
CO3	3	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	3	-	3	-	-

Recommended Study Material:

❖ **Textbook:**

1. Forensic Pharmacy by B. Suresh 124
2. Text book of Forensic Pharmacy by B.M. Mithal
3. Hand book of drug law-by M.L. Mehra
4. A text book of Forensic Pharmacy by N.K. Jain
5. Drugs and Cosmetics Act/Rules by Govt. of India publications.
6. Medicinal and Toilet preparations act 1955 by Govt. of India publications.
7. Narcotic drugs and psychotropic substances act by Govt. of India publications
8. Drugs and Magic Remedies act by Govt. of India publication
9. Bare Acts of the said laws published by Government. Reference books (Theory)

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
HS 124.01 B: PROFESSIONAL COMMUNICATION

Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**An Introduction Professional Communication** • *Concept & Applications of Professional Communication* • *Rhetoric in Professional Communication* • *Importance of Ethos, Logos, and Pathos in Professional Communication*	**03**
2	**Cross-cultural Communication and Globalization** • *Basic Concepts: Culture, Globalization andCross-cultural Communication* • *Social and People Skills* • *Communicating with People of Different Cultures* • *Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them*	**08**
3	**Group Discussion and Personal Interviews** • *Cover Letters and Resume* • *Styles, Formats and Content of Cover Letters* • *Types of Resume* • *Concept and Rationale of Group Discussion* • *Skills and Aspects assessed in Group Discussion* • *Concept and Rationale of Personal Interview* • *Types of Personal Interview*	**10**
4	**Group Dynamics and Leadership** • *An Introduction to Group Dynamics and Leadership* • *Groups and their Structures* • *Roles and Functions of Members in Groups* • *Leading a Group* • *Types of Leadership/Leaders* • *Roles and Functions of a Leader* • *Characteristics of an effective Leader*	**05**
5	**Statement of Purpose (SOP)** • Concept and Rationale of Statement of Purpose • Statement of Purpose as a part of Selection Process	**04**

	• Types, Format and Nature of Statement of Purpose • Content and Process of Statement of Purpose	
	Total	**30**

Course Outcome (COs):

After completion of the course, the student would be able to:

CO1	Gain basic conceptual understanding of communication skills in Professional settings
CO2	Develop awareness and competence in communicating across cultures in professional settings
CO3	Develop confidence and competence in speaking in formal interviews and to make formal presentations
CO4	Develop necessary soft skills to work collaboratively in a group and participate in a group Discussion for employment and in the workplace.
CO5	Develop team and group dynamics to work well in multi-disciplinary and cross-cultural work environment
CO6	Develop writing skills to prepare CV's,Resumes, Statement of Purpose and preparation of other formal documents.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	-	-	-	-	-	-	3	-	-	-
CO2	-	-	-	-	-	-	-	3	-	-	-
CO3	-	-	-	-	-	-	-	3	-	-	-
CO4	-	2	-	-	3	-	-	3	-	-	3
CO5	-	-	-	-	-	2	-	3	-	-	-
CO6	-	-	-	-	-	-	-	3	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

I. Reference Books

- Koneru, A. *Professional Communication*, Tata McGrow Hill Education Private Limited
- Disanza, J.R. &Legge, N. *Business and Professional Communication*, Pearson Education
- Anandamurugan, A. *Placement Interviews – Skills for Success,* Tata McGrow Hill Education Private Limited
- Raman, M & Singh, P. *Business Communication,* Oxford University Press
- Adair, J. *Adair on Leadership*, CREST Publishing House

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme

SYLLABI

(Semester – 6)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH6001 Medicinal Chemistry-II (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Antihistaminic agents	10
	H1–antagonists	
	H2-antagonists	
	Gastric Proton pump inhibitors	
	Anti-neoplastic agents	
	Alkylating agents	
	Antimetabolites	
	Antibiotics	
	Plant products	
	Miscellaneous	
2	Anti-anginal	10
	Vasodilators	
	Calcium channel blockers	
	Diuretics	
	Carbonic anhydrase inhibitors	
	Thiazides	
	Loop diuretics	
	Potassium sparing Diuretics	
	Osmotic Diuretics	
	Anti-hypertensive Agents	
3	Anti-arrhythmic Drugs	10
	Anti Hyperlipidaemic Agents	
	Coagulant and Anticoagulant	
	Drugs used in congestive heart Failure	
4	Drugs acting on Endocrine system	8
	Sex hormones	
	Drugs for erectile dysfunction	
	Oral contraceptives	
5	Antidiabetic agents	7
	Insulin and its preparations	
	Sulfonyl ureas	
	Thiazolidinediones	
	Meglitinides	
	Glucosidase inhibitors	
	Local Anesthetics	

	Amino Benzoic acid derivatives	
	Lidocaine/Anilide derivatives:	

Detailed Syllabus:

<table>
<tr><td colspan="4">Study of the development of the following classes of drugs, Classification, mechanism of action, uses of drugs mentioned in the course, Structure activity relationship of selective class of drugs as specified in the course and synthesis of drugs superscripted (*)</td></tr>
<tr><td>Sr. No.</td><td>UNIT</td><td>Hours</td><td>Weight age (%)</td></tr>
<tr><td rowspan="2">1.</td><td>Antihistaminic agents, H1–antagonists, H2-antagonists, Gastric Proton pump inhibitors, Anti-neoplastic agents, Alkylating agents, Antimetabolites, Antibiotics,Plant products, Miscellaneous</td><td rowspan="2">10 Hours</td><td rowspan="2">22.22%</td></tr>
<tr><td>
H1–antagonists: Diphenhydramine hydrochloride*, Dimenhydrinate, Doxylamines cuccinate, Clemastine fumarate, Diphenylphyraline hydrochloride, Tripelenamine hydrochloride, Chlorcyclizine hydrochloride, Meclizine hydrochloride, Buclizine hydrochloride, Chlorpheniramine maleate, Triprolidine hydrochloride*, Phenidamine tartarate, Promethazine hydrochloride*, Trimeprazine tartrate, Cyproheptadine hydrochloride, Azatidine maleate, Astemizole, Loratadine, Cetirizine, Levocetrazine Cromolyn sodium
H2-antagonists: Cimetidine*, Famotidine, Ranitidin.
Gastric Proton pump inhibitors: Omeprazole, Lansoprazole, Rabeprazole,
Pantoprazole
Anti-neoplastic agents:
Alkylating agents: Meclorethamine*, Cyclophosphamide, Melphalan, Chlorambucil, Busulfan, Thiotepa
Antimetabolites: Mercaptopurine*, Thioguanine, Fluorouracil, Floxuridine,
Cytarabine, Methotrexate*, Azathioprine
Antibiotics: Dactinomycin, Daunorubicin, Doxorubicin, Bleomycin
</td></tr>
</table>

	• **Plant products:** Etoposide, Vinblastin sulphate, Vincristin sulphate • **Miscellaneous:** Cisplatin, Mitotane		
2.	**Anti-anginal, Vasodilators, Calcium channel blockers, Diuretics, Carbonic anhydrase inhibitors, Thiazides, Loop diuretics, Potassium sparing Diuretics, Osmotic Diuretics, Anti-hypertensive Agents**	10 Hours	22.22%
	Anti-anginal: • **Vasodilators:** Amyl nitrite, Nitroglycerin*, Pentaerythritol tetranitrate, Isosorbide dinitrite*, Dipyridamole. • **Calcium channel blockers:** Verapamil, Bepridil hydrochloride, Diltiazem hydrochloride, Nifedipine, Amlodipine, Felodipine, Nicardipine, Nimodipine. **Diuretics:** • **Carbonic anhydrase inhibitors:** Acetazolamide*, Methazolamide, Dichlorphenamide. • **Thiazides:** Chlorthiazide*, Hydrochlorothiazide, Hydroflumethiazide, Cyclothiazide, • **Loop diuretics:** Furosemide*, Bumetanide, Ethacrynic acid. • **Potassium sparing Diuretics:** Spironolactone, Triamterene, Amiloride. • **Osmotic Diuretics:** Mannitol • **Anti-hypertensive Agents:** Timolol, Captopril, Lisinopril, Enalapril, Benazepril hydrochloride, Quinapril hydrochloride, Methyldopate hydrochloride,* Clonidine hydrochloride, Guanethidine monosulphate, Guanabenz acetate, Sodium nitroprusside, Diazoxide,Minoxidil, Reserpine, Hydralazine hydrochloride.		
3.	**Anti-arrhythmic Drugs, Anti Hyperlipidemic Agents, Coagulant and Anticoagulant, Drugs used in congestive heart Failure**	10 Hours	22.22%
	• **Anti-arrhythmic Drugs:** Quinidine sulphate, Procainamide hydrochloride, Disopyramide phosphate*, Phenytoin sodium,		

	Lidocaine hydrochloride, Tocainide hydrochloride, Mexiletine hydrochloride, Lorcainide hydrochloride, Amiodarone, Sotalol. • **Anti-hyperlipidemic agents:** Clofibrate, Lovastatin, Cholesteramine and Cholestipol • **Coagulant & Anticoagulants:** Menadione, Acetomenadione, Warfarin*, Anisindione, clopidogrel • **Drugs used in Congestive Heart Failure:** Digoxin, Digitoxin, Nesiritide, Bosentan, Tezosentan		
4.	**Drugs acting on Endocrine system, Sex hormones, Drugs for erectile dysfunction, Oral contraceptives, Corticosteroids, Thyroid and anti-thyroid drugs**	**8 Hours**	**17.77%**
	• **Drugs acting on Endocrine system** • Nomenclature, Stereochemistry and metabolism of steroids • **Sex hormones:** Testosterone, Nandralone, Progestrones, Oestriol, Oestradiol, Oestrione, Diethyl stilbestrol. • **Drugs for erectile dysfunction:** Sildenafil, Tadalafil. • **Oral contraceptives:** Mifepristone, Norgestril, Levonorgestrol • **Corticosteroids:** Cortisone, Hydrocortisone, Prednisolone, Betamethasone, Dexamethasone • **Thyroid and antithyroid drugs:** L-Thyroxine, L-Thyronine, Propylthiouracil, Methimazole.		
5.	**Antidiabetic agents, Insulin and its preparations, Sulfonyl ureas**,**Thiazolidinediones, Meglitinides, Glucosidase inhibitors, Local Anesthetics, Amino Benzoic acid derivatives, Lidocaine/Anilide derivatives:**	**7 Hours**	**15.55%**
	• **Antidiabetic agents:** • **Insulin and its preparations** • **Sulfonyl ureas:** Tolbutamide*, Chlorpropamide, Glipizide, Glimepiride, Biguanides: Metformin. • **Thiazolidinediones:** Pioglitazone, Rosiglitazone. • **Meglitinides:** Repaglinide, Nateglinide. • **Glucosidase inhibitors:** Acrabose, Voglibose. • **Local Anesthetics:** SAR of Local anesthetics		

	• Benzoic Acid derivatives; Cocaine, Hexylcaine, Meprylcaine, Cyclomethycaine, Piperocaine. • **Amino Benzoic acid derivatives:** Benzocaine*, Butamben, Procaine*, Butacaine, Propoxycaine, Tetracaine, Benoxinate. • **Lidocaine/Anilide derivatives:** Lignocaine, Mepivacaine, Prilocaine, Etidocaine. • Miscellaneous: Phenacaine, Diperodon, Dibucaine.*		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe chemistry of various category of drugs with respect to their pharmacological activity
CO2	summarize structural activity relationship of different class of drugs
CO3	illustrate the drug metabolic pathways and its impact on adverse effect and therapeutic effect of drugs
CO4	suggest, categorise and outline the chemical synthetic pathways of selected drugs

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	3	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH6002: Pharmacology-III (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Pharmacology of Drugs acting on Respiratory System	10
	Pharmacology of drugs acting on the Gastrointestinal Tract	
2	Chemotherapy	10
3	Chemotherapy	10
4	Chemotherapy	8
	Immunopharmacology	
5	Principles of toxicology	7
	Chronopharmacology	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Pharmacology of Drugs acting on Respiratory System, Pharmacology of drugs acting on the Gastrointestinal Tract**	**10 Hours**	**22.22%**
	Anti -asthmatic drugs Drugs used in the management of COPD Expectorants and antitussives Nasal decongestants Respiratory stimulants Antiulcer agents. Drugs for constipation and diarrhea. Appetite stimulants and suppressants. Digestants and carminatives. Emetics and anti-emetics		
2.	**Chemotherapy**	**10 Hours**	**22.22%**
	General principles of chemotherapy. Sulfonamides and cotrimoxazole. Antibiotics- Penicillins, cephalosporins, chloramphenicol, macrolides, quinolones and fluoroquinolins, tetracycline and aminoglycosides		
3.	**Chemotherapy**	**10 Hours**	**22.22%**
	Antitubercular agents Antileprotic agents		

	Antifungal agents Antiviral drugs Anthelmintics Antimalarial drugs Antiamoebic agents		
4.	**Chemotherapy of Neoplastic Diseases and Miscelleneous Drugs**	**8 Hours**	**17.77%**
	Urinary tract infections and sexually transmitted diseases. Chemotherapy of malignancy. Immunostimulants, Immunosuppressant, Protein drugs, monoclonal antibodies, target drugs to antigen, biosimilars		
5.	**Principles of toxicology,** Chronopharmacology	**7 Hours**	**15.55%**
	Definition and basic knowledge of acute, subacute and chronic toxicity. Definition and basic knowledge of genotoxicity, carcinogenicity, teratogenicity and mutagenicity General principles of treatment of poisoning Clinical symptoms and management of barbiturates, morphine, organophosphosphorus compound and lead, mercury and arsenic poisoning Definition of rhythm and cycles. B. Biological clock and their significance leading to chronotherapy.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	classify and provide specific examples for drugs affecting respiratory, GI system and chemotherapeutic agent
CO2	describe and compare mechanism of action, adverse effects and contraindications for drugs affecting respiratory, GI system and chemotherapeutic agent
CO3	evaluate clinical conditions and justify the appropriate use of drugs in the particular disease conditions
CO4	comprehend the principles of toxicology, toxicity studies and treatment of various poisonings.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme
BPH6002P: Pharmacology-III (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	List of Practical
1.	Dose calculation in pharmacological experiments
2.	Anti-allergic activity by mast cell stabilization assay
3.	Study of anti-ulcer activity of a drug using pylorus ligand (SHAY) rat model and NSAIDS induced ulcer model.
4.	Study of effect of drugs on gastrointestinal motility
5.	Effect of agonist and antagonists on guinea pig ileum
6.	Estimation of serum biochemical parameters by using semi- autoanalyser
7.	Effect of saline purgative on frog intestine
8.	Insulin hypoglycemic effect in rabbit
9.	Test for pyrogens (rabbit method)
10.	Determination of acute oral toxicity (LD50) of a drug from a given data
11	Determination of acute skin irritation / corrosion of a test substance
12	Determination of acute eye irritation / corrosion of a test substance
13	Calculation of pharmacokinetic parameters from a given data
14	Biostatistics methods in experimental pharmacology (student‘s t test, ANOVA)
15	Biostatistics methods in experimental pharmacology (Chi square test, Wilcoxon Signed Rank test)

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	demonstrate of screening model-based experiments
CO2	calculate various experimental pharmacological calculation
CO3	demonstrate toxicity testing in experimental Pharmacology
CO4	perform isolated tissues based experiments
CO5	analyze the problem, communicate suggested solution and interprete the results

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	3	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	3	-	-	-	-	-	-	-	3
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Rang H. P., Dale M. M., Ritter J. M., Flower R. J., Rang and Dale‘s Pharmacology, Churchil Livingstone Elsevier
2. Katzung B. G., Masters S. B., Trevor A. J., Basic and clinical pharmacology, Tata Mc Graw-Hill
3. Goodman and Gilman‘s, The Pharmacological Basis of Therapeutics
4. Marry Anne K. K., Lloyd Yee Y., Brian K. A., Robbin L.C., Joseph G. B., Wayne A. K., Bradley R.W., Applied Therapeutics, The Clinical use of Drugs. The Point Lippincott Williams & Wilkins
5. Mycek M.J, Gelnet S.B and Perper M.M. Lippincott‘s Illustrated Reviews-Pharmacology

6. K.D.Tripathi. Essentials of Medical Pharmacology, , JAYPEE Brothers Medical Publishers (P) Ltd, New Delhi
7. Sharma H. L., Sharma K. K., Principles of Pharmacology, Paras medical publisher Modern Pharmacology with clinical Applications, by Charles R.Craig& Robert,
8. Ghosh MN. Fundamentals of Experimental Pharmacology. Hilton & Company, Kolkata,
9. Kulkarni SK. Handbook of experimental pharmacology. VallabhPrakashan
10. N. Udupa and P.D. Gupta, Concepts in Chronopharmacology

❖ **Web Materials:**

1. http://heb-nic.in/ex-pharm
2. https://www.researchgate.net/publication/287995503_Guidelines_on_dosage_calculation_and_stock_solution_preparation_in_experimental_animals'_studies
3. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1637006/
4. https://msmedia.com.au/pharmacology-software

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH6003: Herbal Drug Technology (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Herbs as raw materials Biodynamic Agriculture Indian Systems of Medicine	11
2	Nutraceuticals Herbal-Drug and Herb-Food Interactions	7
3	Herbal Cosmetics Herbal excipients Herbal formulations	7
4	Evaluation of Drugs Stability testing of herbal drugs. Patenting and Regulatory requirements of natural Regulatory Issues	10
5	General Introduction to Herbal Industry Schedule T – Good Manufacturing Practice of Indian	10

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Herbs as raw materials, Biodynamic Agriculture, Indian Systems of Medicine**	**11 Hours**	**24.44%**
	Definition of herb, herbal medicine, herbal medicinal product, herbal drug preparation. Source of Herbs. Selection, identification and authentication of herbal materials. Processing of herbal raw material. Good agricultural practices in cultivation of medicinal plants including Organic farming. Pest and Pest management in medicinal plants: Biopesticides/Bioinsecticides. a) Basic principles involved in Ayurveda, Siddha, Unani and Homeopathy b) Preparation and standardization of Ayurvedic formulations viz Aristas and Asawas,Ghutika,Churna, Lehya, Taila, Ghrita and Bhasma.		
2.	**Nutraceuticals, Herbal-Drug and Herb-Food Interactions,**	**7 Hours**	**15.55%**
	General aspects, Market, growth, scope and types of products available in the market. Health benefits and role of Nutraceuticals in ailments like Diabetes,		

	CVS diseases, Cancer, Irritable bowel syndrome and various Gastrointestinal diseases. Study of following herbs as health food: Alfaalfa, Chicory, Ginger, Fenugreek, Garlic,Honey, Amla, Ginseng, Ashwagandha, Spirulina General introduction to interaction and classification. Study of following drugs and their possible side effects and interactions: Hypercium, kava-kava, Ginkobiloba, Ginseng, Garlic, Pepper & Ephedra.		
3.	**Herbal Cosmetics, Herbal excipients, Herbal formulations**	**7 Hours**	**15.55%**
	Sources and description of raw materials of herbal origin used via, fixed oils, waxes, gums colours, perfumes, protective agents, bleaching agents, antioxidants in products such as skin care, hair care and oral hygiene products. Herbal Excipients – Significance of substances of natural origin as excipients – colorants,sweeteners, binders, diluents, viscosity builders, disintegrants, flavors & perfumes. Conventional herbal formulations like syrups, mixtures and tablets and Novel dosage forms like phytosomes		
4.	**Evaluation of Drugs, Stability testing of herbal drugs, Patenting and Regulatory requirements of natural products, Regulatory Issues**	**10 Hours**	**22.22%**
	WHO & ICH guidelines for the assessment of herbal drugs. a) Definition of the terms: Patent, IPR, Farmers right, Breeder‘s right, Bioprospecting and Biopiracy b) Patenting aspects of Traditional Knowledge and Natural Products. Case study of Curcuma & Neem. Regulations in India (ASU DTAB, ASU DCC), Regulation of manufacture of ASU drugs - Schedule Z of Drugs & Cosmetics Act for ASU drugs.		
5.	**General Introduction to Herbal Industry**, S**chedule T – Good Manufacturing Practice of Indian systems of medicine**	**10 Hours**	**22.22%**
	Herbal drugs industry: Present scope and future prospects. A brief account of plant based industries and institutions involved in work on medicinal and aromatic plants in India. Components of GMP (Schedule – T) and its objectives. Infrastructural requirements, working space, storage area, machinery and equipments, standard operating procedures, health and hygiene, documentation and records.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	appreciate herbs as raw materials and as active component in cosmetics.
CO2	describe significance of good agricultural practices.
CO3	summarize the concept of nutraceuticals and suggest plant materials to be used as nutraceuticals.

CO4	describe methods for preparation and suggest the approach to standardize herbal drugs, herbal formulation and Ayurveda formulations.
CO5	summarize and interpret the intellectual property issues related to herbal drugs.
CO6	suggest the information to be supplied for registration of herbal/Ayurveda formulations.
CO7	summarize herbal drug industry and schedule T of Drug and Cosmetics act.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	3	-	-	-	-	-	-	-
CO5	3	-	-	3	-	-	-	-	-	-	3
CO6	3	-	-	3	-	-	-	-	-	-	-
CO7	3	-	-	-	-	-	-	-	-	-	-

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH6003P: Herbal Drug Technology (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	List of Practicals
1.	To perform preliminary phytochemical screening of crude drugs
2.	Determination of the alcohol content of Asava and Arista
3.	Evaluation of excipients of natural origin
4.	Incorporation of prepared and standardized extract in cosmetic formulations like creams, lotions and shampoos and their evaluation.
5.	Incorporation of prepared and standardized extract in formulations like syrups, mixtures and tablets and their evaluation as per Pharmacopoeial requirements.
6.	Monograph analysis of herbal drugs from recent Pharmacopoeias.
7.	Determination of Aldehyde content.
8.	Determination of Phenol content.
9.	Determination of total alkaloids

Course Outcome (Cos):

At the end of the course, the students would be able to

CO1	identify specific types of phytoconstituents from medicinal plants
CO2	evaluate (standardize) Ayurveda and herbal formulations
CO3	quantify the group of phytoconstituents on the basis of functional group present.
CO4	incorporate standardized extract in different formulations.
CO5	analyse the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	3	-	-	-	-	-	-	-
CO2	3	-	-	3	-	-	-	-	-	-	-
CO3	3	-	-	3	-	-	-	-	-	-	-
CO4	-	-	-	3	-	-	-	-	-	-	-
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Textbook of Pharmacognosy by Trease & Evans.
2. Textbook of Pharmacognosy by Tyler, Brady & Robber.
3. Pharmacognosy by Kokate, Purohit and Gokhale.
4. Essential of Pharmacognosy by Dr.S.H.Ansari.
5. Pharmacognosy & Phytochemistry by V.D.Rangari.
6. Pharmacopoeal standards for Ayurvedic Formulation (Council of Research in Indian Medicine & Homeopathy.
7. Mukherjee, P.W. Quality Control of Herbal Drugs: An Approach to evaluation of Botanicals. Business Horizons Publishers, New Delhi, India, 2002.

❖ **Web Materials:**

1.https://nitte.edu.in/learningoutcomes/faculty-of-pharmacy/bpharm-1/herbal-drug-technology.html

2.https://www.sciencedirect.com/science/article/pii/B9780128202845000277

3.https://media.neliti.com/media/publications/278981-standardization-and- evaluation-of-herbal-ce0e9f33.pdf

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH6004 Industrial Pharmacy-II

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Pilot plant scale up techniques	10
2	Technology development and transfer	10
3	Regulatory affairs Regulatory requirements for drug approval	10
4	Quality management systems	8
5	Indian Regulatory Requirements	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weight age (%)
1.	**Pilot plant scale up techniques** General considerations - including significance of personnel requirements, space requirements, raw materials, Pilot plant scale up considerations for solids, liquid orals, semi solids and relevant documentation, SUPAC guidelines and Introduction to platform technology	**10 Hours**	**22.22%**
2.	**Technology development and transfer** WHO guidelines for Technology Transfer (TT): Terminology, Technology transfer protocol, Quality risk management, Transfer from R & D to production (Process, packaging and cleaning), Granularity of TT Process (API, excipients, finished products, packaging materials) Documentation, Premises and equipment, qualification and validation, quality control, analytical method transfer, Approved regulatory bodies and agencies, Commercialization - practical aspects and problems (case studies), TT agencies in India - APCTD, NRDC, TIFAC, BCIL, TBSE / SIDBI; TT related documentation - confidentiality agreement, licensing, MoUs, legal issues	**10 Hours**	**22.22%**
3.	**Regulatory affairs** a) Introduction, Historical overview of Regulatory Affairs, b) Regulatory authorities, Role of Regulatory affairs department, Responsibility of Regulatory Affairs Professionals Drug Development Teams, Non-Clinical Drug Development, Pharmacology, Drug Metabolism and Toxicology,	**10 Hours**	**22.22%**

	General considerations of Investigational New Drug (IND) Application, Investigator‘s Brochure (IB) and New Drug Application (NDA), Clinical research / BE studies, Clinical Research Protocols, Biostatistics in Pharmaceutical Product Development, Data Presentation for FDA Submissions, Management of Clinical Studies.		
4.	**Quality management systems**	**8 Hours**	**17.77%**
	Quality management & Certifications: Concept of Quality, Total Quality Management, Quality by Design (QbD), Six Sigma concept, Out of Specifications (OOS), Change control, Introduction to ISO 9000 series of quality systems standards, ISO 14000, NABL, GLP		
5.	**Indian Regulatory Requirements**	**7 Hours**	**15.55%**
	a) Central Drug Standard Control Organization (CDSCO) and State Licensing Authority: Organization, Responsibilities, b) Certificate of Pharmaceutical Product (COPP), Regulatory requirements and approval procedures for New Drugs.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe the process of pilot plant and scale up for various pharmaceutical dosage forms.
CO2	comprehend the process of technology transfer from lab scale to commercial batch and commercialization.
CO3	describe regulatory requirements for drug approval approval.
CO4	narrate various Laws and Acts that regulate pharmaceutical industry.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	3	-	-
CO4	3	-	-	-	-	-	3	-	3	-	3

Recommended Books: (Latest Editions)

1. Regulatory Affairs from Wikipedia, the free encyclopedia modified on 7th April available at http,//en.wikipedia.org/wiki/Regulatory_Affairs

2. International Regulatory Affairs Updates, 2005. available at http://www.iraup.com/about.php
3. Douglas J Pisano and David S. Mantus. Textbook of FDA Regulatory Affairs A Guide for Prescription Drugs, Medical Devices, and Biologics'Second Edition.
4. Regulatory Affairs brought by learning plus, inc. available at http.//www.cgmp.com/ra.htm
5. Vyawahare, N., & Itkar, S. C. Drug Regulatory Affairs: Nirali Prakashan.
6. Patravale, V. B., Disouza, J. I., & Rustomjee, M. (2016). Pharmaceutical Product Development: Insights into Pharmaceutical Processes, Management and Regulatory Affairs: CRC Press.
7. Anjaneyulu, Y., & Marayya, R. (2005). Quality Assurance and Quality Management In Pharmaceutical Industry: Book Syndicate.
8. Levin, M. (2011). Pharmaceutical Process Scale-Up, Third Edition: Taylor & Francis**.**

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH6005: Pharmaceutical Biotechnology (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Brief introduction to Biotechnology	10
	Enzyme Biotechnology	
	Biosensors	
	Brief introduction to Protein Engineering	
	Use of microbes in industry	
	Basic principles of genetic engineering	
2	Study of cloning vectors, restriction endonucleases and	10
	Recombinant DNA technology	
	Application of r DNA technology and genetic	
	Brief introduction to PCR	
3	Types of immunity- humoral immunity, cellular	10
4	Immuno blotting techniques	8
	Genetic organization of Eukaryotes and Prokaryotes	
	Microbial genetics	
	Introduction to Microbial biotransformation	
	Mutation	
5	Fermentation methods	7
	Large scale production fermenter design	
	Study of the production	
	Blood Products	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**UNIT-I**	**10 Hours**	**22.22%**
	a) Brief introduction to Biotechnology with reference to Pharmaceutical Sciences. b) Enzyme Biotechnology- Methods of enzyme immobilization and applications.		

	c) Biosensors- Working and applications of biosensors in Pharmaceutical Industries. d) Brief introduction to Protein Engineering. e) Use of microbes in industry. Production of Enzymes- General consideration Amylase, Catalase, Peroxidase, Lipase, Protease, Penicillinase. f) Basic principles of genetic engineering.		
2.	**UNIT-II** a) Study of cloning vectors, restriction endonucleases and DNA ligase. b) Recombinant DNA technology. Application of genetic engineering in medicine. c) Application of r DNA technology and genetic engineering in the production of: i) Interferon ii) Vaccines- hepatitis- B iii) Hormones- Insulin. d) Brief introduction to PCR	**10 Hours**	**22.22%**
3.	**UNIT-III** Types of immunity- humoral immunity, cellular immunity a) Structure of Immunoglobulins b) Structure and Function of MHC c) Hypersensitivity reactions, Immune stimulation and Immune suppressions. d) General method of the preparation of bacterial vaccines, toxoids, viral vaccine, antitoxins, serum-immune blood derivatives and other products relative to immunity. e) Storage conditions and stability of official vaccines f) Hybridoma technology- Production, Purification and Applications g) Blood products and Plasma Substitutes	**10 Hours**	**22.22%**
4.	**UNIT-IV** a) Immuno blotting techniques- ELISA, Western blotting, Southern blotting. b) Genetic organization of Eukaryotes and Prokaryotes c) Microbial genetics including transformation, transduction, conjugation, plasmids and transposons. d) Introduction to Microbial biotransformation and applications. e) Mutation: Types of mutation/mutants.	**8 Hours**	**17.77%**
5.	**UNIT-V** a) Fermentation methods and general requirements, study of media, equipments, sterilization methods, aeration process, stirring. b) Large scale production fermenter design and its various controls. c) Study of the production of - penicillins, citric acid, Vitamin B12, Glutamic acid, Griseofulvin, d) Blood Products: Collection, Processing and Storage of whole human blood, dried human plasma, plasma Substitutes.	**7 Hours**	**15.55%**

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe principle and application of various Immobilized enzymes in Pharmaceutical Industries
CO2	explain recombinant DNA technology and Genetic engineering applications in relation to production of pharmaceutical
CO3	describe the use of microorganisms in fermentation technology
CO4	summarize principle, importance and testing of Monoclonal antibodies in Industries

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ **Textbook:**

1. B.R. Glick and J.J. Pasternak: Molecular Biotechnology: Principles and Applications of RecombinantDNA: ASM Press Washington D.C.
2. RA Goldshy et. al., : Kuby Immunology.
3. J.W. Goding: Monoclonal Antibodies.
4. J.M. Walker and E.B. Gingold: Molecular Biology and Biotechnology by Royal Society of Chemistry.
5. Zaborsky: Immobilized Enzymes, CRC Press, Degraland, Ohio.
6. S.B. Primrose: Molecular Biotechnology (Second Edition) Blackwell Scientific Publication.
7. Stanbury F., P., Whitakar A., and Hall J., S., Principles of fermentation technology, 2nd edition, Aditya books Ltd., New Delhi

- **Web Materials:**

1. https://www.khanacademy.org/science/ap-biology/gene-expression-and-regulation/biotechnology/v/dna-cloning-and-recombinant-dna
2. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7119977/
3. https://www.nacalai.co.jp/global/download/pdf/western_blot_protocols.pdf
4. https://www.creativebiomart.net/resource/principle-protocol-western-blot-protocol-351.htm
5. https://www.moleculardevices.com/applications/enzyme-linked-immunosorbent-assay-elisa#gref
6. https://www.mybiosource.com/learn/southern-blotting/
7. https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/fermentation-technique
8. https://www.britannica.com/science/mutation-genetics

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH6006: Instrumental Methods of Analysis (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	UV Visible spectroscopy	10
	Fluorimetry	
2	IR spectroscopy	10
	Flame Photometry	
	Atomic absorption spectroscopy	
	Nepheloturbidometry	
3	Introduction to chromatography	10
	Electrophoresis	
4	Gas chromatography	8
	High performance liquid chromatography (HPLC)	
5	Ion exchange chromatography	7
	Gel chromatography	
	Affinity chromatography	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**UV Visible spectroscopy, Fluorimetry**	10 Hours	22.22%
	UV Visible spectroscopy: Electronic transitions, chromophores, auxochromes, spectral shifts, solvent effect on absorption spectra, Beer and Lambert's law, Derivation and deviations. Instrumentation - Sources of radiation, wavelength selectors, sample cells, detectors- Photo tube, Photomultiplier tube, Photo voltaic cell, Silicon Photodiode. Applications - Spectrophotometric titrations, Single component and multi component analysis **Fluorimetry**: Theory, Concepts of singlet, doublet and triplet electronic states, internal and external conversions, factors affecting fluorescence, quenching, instrumentation and applications.		
2.	**IR spectroscopy, Flame Photometry, Atomic absorption spectroscopy and Nepheloturbidometry**	10 Hours	22.22%

	IR spectroscopy: Introduction, fundamental modes of vibrations in poly atomic molecules, sample handling, factors affecting vibrations. Instrumentation - Sources of radiation, wavelength selectors, detectors - Golay cell, Bolometer, Thermocouple, Thermister, Pyroelectric detector and applications **Flame Photometry:** Principle, interferences, instrumentation and applications. **Atomic absorption spectroscopy:** Principle, interferences, instrumentation and applications **Nepheloturbidometry:** Principle, instrumentation and applications.		
3.	**Introduction to chromatography, Electrophoresis**	**10 Hours**	**22.22%**
	Introduction to chromatography **Adsorption and partition column chromatography**: Methodology, advantages, disadvantages and applications. **Thin layer chromatography**: Introduction, Principle, Methodology, Rf values, advantages, disadvantages and applications. **Paper chromatography**: Introduction, methodology, development techniques, advantages, disadvantages and applications **Electrophoresis**: Introduction, factors affecting electrophoretic mobility, Techniques of paper, gel, capillary electrophoresis, applications.		
4.	**Gas chromatography, High performance liquid chromatography (HPLC)**	**8 Hours**	**17.77%**
	Gas chromatography: Introduction, theory, instrumentation, derivatization, temperature programming, advantages, disadvantages and applications **High performance liquid chromatography (HPLC):** Introduction, theory, instrumentation, advantages and applications.		
5.	**Ion exchange chromatography, Gel chromatography, and Affinity chromatography**	**7 Hours**	**15.55%**
	Ion exchange chromatography: Introduction, classification, ion exchange resins, properties, mechanism of ion exchange process, factors affecting ion exchange, methodology and applications **Gel chromatography:** Introduction, theory, instrumentation and applications **Affinity chromatography:** Introduction, theory, instrumentation and applications.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	explain theory and principle of various spectroscopic and chromatographic separation techniques.
CO2	describe instrumentation and application of various spectroscopic and chromatographic separation techniques
CO3	summarize quantitative & qualitative analysis of drugs using various analytical instruments.

CO4	elucidate structure of organic compounds by spectral analysis.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	3	-	-	-	-	-	-	-	3

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH6006P: Instrumental Methods of Analysis (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	List of Practicals
1.	Determination of absorption maxima and effect of solvents on absorption maxima of organic compounds
2.	Estimation of dextrose by colorimetry
3.	Estimation of sulfanilamide by colorimetry
4.	Simultaneous estimation of ibuprofen and paracetamol by UV spectroscopy
5.	Assay of paracetamol by UV- Spectrophotometry
6.	Estimation of quinine sulfate by fluorimetry
7.	Study of quenching of fluorescence
8.	Determination of sodium by flame photometry
9.	Determination of potassium by flame photometry
10.	Determination of chlorides and sulphates by nepheloturbidometry
11	Separation of amino acids by paper chromatography
12	Separation of sugars by thin layer chromatography
13	Separation of plant pigments by column chromatography
14	Demonstration experiment on HPLC
15	Demonstration experiment on Gas Chromatography

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	estimate pharmaceuticals by colorimetry, UV-Vis spectrophotometry and fluorimetry.
CO2	estimate pharmaceuticals by flame photometry and nephelo turbidometry
CO3	separate compoments of pharmaceutical mixture by various chromatographic techniques
CO4	demonstrate HPLC and GC.
CO5	analyse the problem, communicate suggested solution and interprete the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	3	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	3
CO5	-	-	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1. Instrumental Methods of Chemical Analysis by B.K Sharma
2. Organic spectroscopy by Y.R Sharma
3. Textbook of Pharmaceutical Analysis by Kenneth A. Connors
4. Vogel's Text book of Quantitative Chemical Analysis by A.I. Vogel
5. Practical Pharmaceutical Chemistry by A.H. Beckett and J.B. Stenlake
6. Organic Chemistry by I. L. Finar
7. Organic spectroscopy by William Kemp
8. Quantitative Analysis of Drugs by D. C. Garrett
9. Quantitative Analysis of Drugs in Pharmaceutical Formulations by P. D. Sethi
10. Spectrophotometric identification of Organic Compounds by Silverstein
11. Quantitative Analysis of Drugs in Pharmaceutical formulation - P D Sethi,3rd Edition, CBS Publishers, New Delhi.
12. Practical Pharmaceutical Chemistry – Beckett and Stenlake, Vol II, 4thedition, CBS Publishers, New Delhi.
13. Indian Pharmacopoeia 2014, Ministry of Health and Family Welfare Volume – I, II, III. The Indian Pharmacopoeia Commission, Ghaziabad.

❖ **Web Materials:**

1. https://freevideolectures.com/course/3029/modern-instrumental-methods-of-analysis

2. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5206469/
3. https://link.springer.com/chapter/10.1007/978-981-15-1547-7_2
4. https://www.sciencedirect.com/science/article/pii/S0149639503800247

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
HS134 B: CONTRIBUTOR PERSONALITY DEVELOPMENT

Total hours: 30 Hr

Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	**Concept of Personality:** • Meaning of Personality • Types of Personality • Factors contributing to Personality • Personality Traits	06
2	**Soft Skills and Personality Development:** • Critical, Creative and Positive Thinking • Leadership, Assertiveness and Negotiation Skills • Self-Management • People's Skills • Building Relationship Skills • Being a Team Player	08
3	**Developing Contributor Personality – Part I** • Concept of Contributor • Characteristics of a Contributor • The Contributor's Vision of Success & Career • The Scope of Contribution • Embarking on the Journey to Contributor-ship	06
4	**Developing Contributor Personality – Part II** • Focus on values • Engage deeply • Think in enlightened self-interest • Practice imaginative sympathy • Demonstrate trust behavior • Developing a sense of duty and morality	06
5	**Contemporary Issues in CPD** • Contemporary Practices & Trends in Contributor Personality Development • Case Study & Presentations	04
	Total	**30**

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Course Outcome (COs):
After completion of the course, the student would be able to:

CO1	develop conceptual understanding of the term Personality and understand one's own personality Traits
CO2	develop soft skills for holistic personality development for career and in personal life
CO3	develop assertiveness,self management and negotiation skills to navigate the personal and professional environment successfully
CO4	develop conceptual clarity of the term ' Contributor Personality' and develop understanding of the traits of a contributor
CO5	develop the qualities of a contributor such as trust, strong work ethic, responsibility and accountabililty at workplace
CO6	develop relevant skills required to be a contributor at workplace and achieve success in life

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	-	-	-	2
CO2	-	-	-	-	-	-	-	3	-	-	-
CO3	-	3	-	-	3	-	-	3	-	-	-
CO4	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	3	2	2	-	-
CO6	-	-	-	-	-	-	-	3	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Reference Books / Reading

- Contributor Personality Program Workbook (Volume 1,2),
- Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd.

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme

SYLLABI

(Semester – 7)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH7001: Medicinal Chemistry-III (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Antibiotics Anti-tubercular Agents	12
2	Antimalarials Sulphonamides and Sulfones Urinary tract anti-infective agents	10
3	Anti-tubercular agent Anti-Leprosy Agents Anti-viral agents	8
4	Anti- fungal agents Anti-protozoal agents Anthelmintics	7
5	Introduction to drug design Combinatorial Chemistry Prodrugs	8

Detailed Syllabus:

Course Content: **Study of the development of the following classes of drugs, Classification, mechanism of action, uses of drugs mentioned in the course, Structure activity relationship of selective class of drugs as specified in the course and synthesis of drugs superscripted by (*)**

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Antibiotics** Historical background, Nomenclature, Stereochemistry, Structure activity relationship, Chemical degradation classification and important products of the following classes. • **β-Lactam antibiotics:** Penicillin, Cepholosporins, β- Lactamase inhibitors, Monobactams • **Aminoglycosides:** Streptomycin, Neomycin, Kanamycin • **Tetracyclines:** Tetracycline, Oxytetracycline, Chlortetracycline, Minocycline, Doxycycline • **Macrolide:** Erythromycin Clarithromycin, Azithromycin • **Peptide Antibiotics:** Polymyxin, Bacitracin, Gramicidin • **Oxazolidinones:** Linezolid	**12 Hours**	**26.66%**

	• **Miscellaneous:** Chloramphenicol, Clindamycin		
2.	**Antimalarials, Sulphonamides and Sulfones and Urinary tract anti-infective agents** **Antimalarials:** Etiology of malaria • **Quinolines:** SAR, Quinine sulphate, Chloroquine, Amodiaquine, Primaquine phosphate, Pamaquine, Quinacrine hydrochloride, Mefloquine • **Biguanides and dihydro triazines:** Cycloguanil pamoate, Proguanil • **Miscellaneous:** Pyrimethamine, Artesunete, Artemether, Atovoquone **Sulphonamides and Sulfones** • Historical development, chemistry, classification and SAR of Sulfonamides: Sulphamethizole, Sulfisoxazole, Sulphamethizine, Sulfacetamide, Sulphapyridine, Sulfamethoxaole, Sulphadiazine, Mefenide acetate, Sulfasalazine. • **Folate reductase inhibitors:** Trimethoprim, Cotrimoxazole. • **Sulfones:** Dapsone. **Urinary tract anti-infective agents** • **Quinolones:** SAR of quinolones, Nalidixic Acid,Norfloxacin, Enoxacin, Ciprofloxacin, Ofloxacin, Lomefloxacin, Sparfloxacin, Gatifloxacin, Moxifloxacin • **Miscellaneous:** Furazolidine, Nitrofurantoin, Methanamine.	**10 Hours**	**22.22%**
3.	**Anti-tubercular Agents, Anti-Leprosy Agents and Antiviral agents** **Anti-tubercular Agents** • Synthetic anti tubercular agents: Isoniozid, Ethionamide, Ethambutol, Pyrazinamide, Para amino salicylic acid. • **Anti-tubercular antibiotics**: Rifampicin, Rifabutin, Cycloserine Streptomycine, Capreomycin sulphate. **Anti-Leprosy Agents**: Dapson, Clofazimine **Antiviral agents:** • Amantadine hydrochloride, Rimantadine hydrochloride, Idoxuridine trifluoride, Acyclovir, Gancyclovir, Zidovudine, Didanosine, Zalcitabine, Lamivudine, Loviride, Delavirding, Ribavirin, Saquinavir, Indinavir, Ritonavir.	**8 Hours**	**17.77%**
4.	**Antifungal agents, Anti-protozoal Agents and Anthelmintics** **Antifungal agents** • Antifungal antibiotics: Amphotericin-B, Nystatin, Natamycin, Griseofulvin. • Synthetic Antifungal agents: Clotrimazole, Econazole, Butoconazole, Oxiconazole Tioconozole, Miconazole,	**7 Hours**	**15.55%**

	Ketoconazole, Terconazole, Itraconazole, Fluconazole, Naftifine hydrochloride, Tolnaftate. **Anti-protozoal Agents** • Metronidazole, Tinidazole, Ornidazole, Diloxanide, Iodoquinol,Pentamidine Isethionate, Atovaquone, Eflornithine. **Anthelmintics** • Diethylcarbamazine citrate, Thiabendazole, Mebendazole, Albendazole, Niclosamide, Oxamniquine, Praziquantal, Ivermectin.		
5.	**Introduction to Drug Design, Combinatorial Chemistry and Prodrugs**	**8 Hours**	**17.77%**
	Introduction to Drug Design • Various approaches used in drug design. • Physicochemical parameters used in quantitative structure activity relationship (QSAR) such as partition coefficient, Hammet's electronic parameter, Taft's steric parameter and Hansch analysis. • Pharmacophore modeling and docking techniques. **Combinatorial Chemistry** • Concept and applications of combinatorial chemistry: solid phase and solution phase synthesis. **Prodrugs** Basic concepts and application of prodrugs design		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	understand the physicochemical properties of drugs, Structure Activity Relationship (SAR) and computational technique in relation to drug design and discovery.
CO2	describe the chemistry and structure activity relationship of drugs with respect to their pharmacological activity.
CO3	illustrate the drug metabolic pathways and its impact on adverse effect and therapeutic effect of drugs.
CO4	suggest, categorise and outline the chemical synthetic pathways of selected drugs.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	-

CO4	3	-	3	-	-	-	-	-	-	-	3

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH7001P: Medicinal Chemistry-III (Practical)

Total hours: 60 Hr

Outline of the course:

Sr. No.	List of Practicals
1	**Preparation of drugs and intermediates** • Sulphanilamide • 7-Hydroxy, 4-methyl coumarin • Chlorobutanol • Triphenyl imidazole • Tolbutamide
2	**Assay of drugs** • Isonicotinic acid hydrazide • Chloroquine • Metronidazole • Dapsone • Chlorpheniramine maleate • Benzyl penicillin
3	Preparation of medicinally important compounds or intermediates by Microwave irradiation technique
4	Drawing structures and reactions using chem draw®
5	Determination of physicochemical properties such as logP, clogP, MR, Molecular weight, Hydrogen bond donors and acceptors for class of drugs course content using drug design software Drug likeliness screening (Lipinski‘s Rule of Five)

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	perform the conventional and microwave synthesis techniques for the synthesis of drugs and its intermediates.
CO2	perform Assay of drugs and intermediates using appropriate analytical methods
CO3	determine various physicochemical properties of the drugs by drug design softwares.
CO4	analyze the problem, communicate suggested solution and interprete the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	3	3	-	-	-	-	3	-	-	-
CO2	3	3	3	3	-	-	-	3	-	-	-
CO3	3	3	3	3	-	-	-	3	-	-	-
CO4	3	3	3	-	-	-	-	3	-	-	-

Recommended Study Material:

❖ **Textbook:**

1.Wilson and Griswold‘s Organic medicinal and Pharmaceutical Chemistry.

2.Foye‘s Principles of Medicinal Chemistry.

3.Burger‘s Medicinal Chemistry, Vol. I to IV.

4.Introduction to principles of drug design- Smith and Williams.

5.Remington‘s Pharmaceutical Sciences.

6.Martindale‘s extra pharmacopoeia.

7.Graham L. Petrick‘s An introduction to Medicinal Chemistry

8.D. Sriram and P.Yogeshwari‘s Medicinal Chemistry

9.Camile G. Wermuth‘s The Practice of Medicinal Chemistry

10. Kadam, Mahadik and Bothra‘s Principle of Medicinal Chemistry (Vol I and II)

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH7002: Quality Assurance (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Quality Assurance and Quality Management concepts	10
	Total Quality Management (TQM)	
	ICH Guidelines	
	Quality by design (QbD)	
	ISO 9000 & ISO14000	
	NABL accreditation	
2	Organization and personnel	10
	Premises	
	Equipments and raw materials	
3	Quality Control	10
	Good Laboratory Practices	
4	Complaints	8
	Document maintenance in pharmaceutical industry	
5	Calibration and Validation	7
	Warehousing	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Quality Assurance and Quality Management concepts, Total Quality Management (TQM), ICH Guidelines, Quality by design (QbD), ISO 9000 & ISO14000 and NABL accreditation**	**10 Hours**	**22.22%**
	Quality Assurance and Quality Management concepts: Definition and concept of Quality control, Quality assurance and GMP Total Quality Management (TQM): Definition, elements, philosophies ICH Guidelines: purpose, participants, process of harmonization, Brief overview of QSEM, with special emphasis on Q-series guidelines, ICH stability testing guidelines Quality by design (QbD): Definition, overview, elements of QbD program, tools		

	ISO 9000 & ISO14000: Overview, Benefits, Elements, steps for registration NABL accreditation: Principles and procedures.		
2.	**Organization and personnel, Premises, and Equipment's and raw materials**	**10 Hours**	**22.22%**
	Organization and personnel: Personnel responsibilities, training, hygiene and personal records. Premises: Design, construction and plant layout, maintenance, sanitation, environmental control, utilities and maintenance of sterile areas, control of contamination. Equipment's and raw materials: Equipment selection, purchase specifications, maintenance, purchase specifications and maintenance of stores for raw materials		
3.	**Quality Control and Good Laboratory Practices**	**10 Hours**	**22.22%**
	Quality Control: Quality control test for containers, rubber closures and secondary packing materials. Good Laboratory Practices: General Provisions, Organization and Personnel, Facilities, Equipment, Testing Facilities Operation, Test and Control Articles, Protocol for Conduct of a Nonclinical Laboratory Study, Records and Reports, Disqualification of Testing Facilities.		
4.	**Complaints and Document maintenance in pharmaceutical industry**	**8 Hours**	**17.77%**
	Complaints: Complaints and evaluation of complaints, Handling of return good, recalling and waste disposal. Document maintenance in pharmaceutical industry: Batch Formula Record, Master Formula Record, SOP, Quality audit, Quality Review and Quality documentation, Reports and documents, distribution records.		
5.	**Calibration and Validation and Warehousing**	**7 Hours**	**15.55%**
	Calibration and Validation: Introduction, definition and general principles of calibration, qualification and validation, importance and scope of validation, types of validation, validation master plan. Calibration of pH meter, Qualification of UV-Visible spectrophotometer, General principles of Analytical method Validation. Warehousing: Good warehousing practice, materials management		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	explain the concept, philosophy and elements of Quality Control, Quality Assurance, and Quality Management System.
CO2	understand the importance of principles of Good Manufacturing Practices and scope of quality certifications in a pharmaceutical industry.
CO3	understand the importance of Good Laboratory Practices in Pharmaceutical Industry.

CO4	know the purpose and processes of the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use.
CO5	describe the importance of calibration, validation, and documentation in maintaining quality and safety standards of Pharmaceuticals.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3
CO5	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ **Textbook:**

1. Quality Assurance Guide by organization of Pharmaceutical Products of India.
2. Good Laboratory Practice Regulations, 2 nd Edition, Sandy Weinberg Vol. 69.
3. Quality Assurance of Pharmaceuticals- A compendium of Guide lines and Related materials Vol I WHO Publications.
4. A guide to Total Quality Management- Kushik Maitra and Sedhan K Ghosh
5. How to Practice GMP‘s – P P Sharma.
6. ISO 9000 and Total Quality Management – Sadhank G Ghosh
7. The International Pharmacopoeia – Vol I, II, III, IV- General Methods of Analysis and Quality specification for Pharmaceutical Substances, Excipients and Dosage forms
8. Good laboratory Practices – Marcel Deckker Series 9. ICH guidelines, ISO 9000 and 14000 guidelines

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH7003: Biopharmaceutics and Pharmacokinetic (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Biopharmaceutics	10
2	Elimination	10
	Bioavailability and Bioequivalence	
3	Pharmacokinetics	10
4	Multicompartment models	8
5	Nonlinear Pharmacokinetics	**7**

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Introduction to Biopharmaceutics**	**10 Hours**	**22.22%**
	Introduction to Biopharmaceutics **Absorption:** Mechanisms of drug absorption through GIT, factors influencing drug absorption though GIT, absorption of drug from Non per oral extra-vascular routes **Distribution**: Tissue permeability of drugs, binding of drugs, apparent, volume of drug distribution, plasma and tissue protein binding of drugs, factors affecting protein-drug binding, Kinetics of protein binding, Clinical significance of protein binding of drugs.		
2.	**Elimination and Bioavailability and Bioequivalence**	**10 Hours**	**22.22%**

	Elimination: Drug metabolism and basic understanding metabolic pathways, renal excretion of drugs, factors affecting renal excretion of drugs, renal clearance, Non renal routes of drug excretion **Bioavailability and Bioequivalence:** Definition and Objectives of bioavailability, absolute and relative bioavailability, measurement of bioavailability, in-vitro drug dissolution models, in-vitro-in-vivo correlations, bioequivalence studies, methods to enhance the dissolution rates and bioavailability of poorly soluble drugs.		
3.	**Pharmacokinetics**	**10 Hours**	**22.22%**
	Pharmacokinetics: Definition and introduction to Pharmacokinetics, Compartment models, Non compartment models, physiological models, One compartment open model. (a). Intravenous Injection (Bolus) (b). Intravenous infusion and (c) Extra vascular administrations. Pharmacokinetics parameters - KE, t1/2, Vd, AUC, Ka, Clt and CLR – definitions, methods of eliminations, understanding of their significance and Application		
4.	**Multicompartment model**	**8 Hours**	**17.77%**
	Multicompartment models: Two compartment open model. IV bolus Kinetics of multiple dosing, steady state drug levels, calculation of loading and maintenance doses and their significance in clinical settings.		
5.	**Nonlinear Pharmacokinetics**	**7 Hours**	**15.55%**
	Nonlinear Pharmacokinetics: (a) Introduction (b) Factors causing Non-linearity(c) Michaelis-menton method of estimating parameters, Explanation with example of drugs.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe use of plasma drug concentration-time data to calculate the pharmacokinetic parameters to describe the kinetics of drug absorption, distribution, metabolism, excretion, elimination
CO2	understand the concepts of bioavailability and bioequivalence of drug products and their significance.
CO3	understand various pharmacokinetic parameters and model with their significance and applications

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ **Textbook:**

1. Biopharmaceutics and Clinical Pharmacokinetics, Milo Gibaldi, Lea and Febrger, Pennsylvania, USA.
2. Biopharmaceutics and Pharmacokinetics: An introduction, Robert E Notari, Marcel Dekker Inc.
3. Applied biopharmaceutics and pharmacokinetics International edition, Leon Shargel and Andrew B. C. YU, The McGraw-Hill Companies Inc. USA
4. Biopharmaceutics and Pharmacokinetics-A Treatise, D. M. Brahmankar and Sunil B. Jaiswal, Vallabh Prakashan, New Delhi
5. Pharmacokinetics, Milo Glbaldi, Donald R., Marcel Dekker Inc.
6. Hand Book of Clinical Pharmacokinetics, By Milo Gibaldi and Laurie Prescott by ADIS Health Science Press.
7. Biopharmaceutics, James Swarbrick, Lea & Febiger, Philadelphia, USA
8. Clinical Pharmacokinetics, Concepts and Applications: By Malcolm Rowland and Thomas, N. Tozen, Lea and Febrger, Philadelphia, 1995.
9. Dissolution, Bioavailability and Bioequivalence, By Abdou H.M, Mack, Publishing Company, Pennsylvania 1989.
10. Biopharmaceutics and Clinical Pharmacokinetics-An introduction 4th edition Revised and expanded, Rebort E Notari Marcel Dekker Inc, New York and Basel, 1987.
11. Remington‘s Pharmaceutical Sciences, Mack Publishing Company Pennsylvnia

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH7004: Pharmacy Practice (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Hospital and it's organization	10
	Hospital pharmacy and its organization	
	Adverse drug reaction	
	Community Pharmacy	
2	Drug distribution system in a hospital	10
	Hospital formulary	
	Therapeutic drug monitoring	
	Medication adherence	
	Patient medication history interview	
	Community pharmacy management	
3	Pharmacy and therapeutic committee	10
	Drug information services	
	Patient counseling	
	Education and training program in the hospital	
	Prescribed medication order and communication skills	
4	Budget preparation and implementation	8
	Clinical Pharmacy	
	Over the counter (OTC) sales	
5	Drug store management and inventory control	7
	Investigational use of drugs	
	Interpretation of Clinical Laboratory Tests	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Hospital and it's organization, Hospital pharmacy and its organization, Adverse drug reaction, and Community Pharmacy**	10 Hours	22.22%
	Hospital and it's organization: Definition, Classification of hospital-Primary, Secondary and Tertiary hospitals, Classification based on clinical and non- clinical basis, Organization Structure of a Hospital, and Medical staffs involved in the hospital and their fun Hospital pharmacy and its organization: Definition, functions of hospital pharmacy, Organization structure, Location, Layout and staff requirements, and Responsibilities and functions of hospital pharmacists. Adverse drug reaction: Classifications - Excessive pharmacological effects, secondary pharmacological effects, idiosyncrasy, allergic drug reactions, genetically determined toxicity, toxicity following sudden withdrawal of drugs, Drug interaction- beneficial interactions, adverse interactions, and pharmacokinetic drug interactions, Methods for detecting drug interactions, spontaneous case reports and record linkage studies, and Adverse drug reaction reporting and management. Community Pharmacy: Organization and structure of retail and wholesale drug store, types and design, Legal requirements for establishment and maintenance of a drug store, Dispensing of proprietary products, maintenance of records of retail and wholesale drug store.		
2.	**Drug distribution system in a hospital, Hospital formulary, Therapeutic drug monitoring, Medication adherence, Patient medication history interview, and Community pharmacy management**	10 Hours	22.22%
	Drug distribution system in a hospital: Dispensing of drugs to inpatients, types of drug distribution systems, charging policy and labeling, Dispensing of drugs to ambulatory patients, and Dispensing of controlled drugs. Hospital formulary: Definition, contents of hospital formulary, Differentiation of hospital formulary and Drug list, preparation and revision, and addition and deletion of drug from hospital formulary. Therapeutic drug monitoring: Need for Therapeutic Drug Monitoring, Factors to be considered during the Therapeutic Drug Monitoring, and Indian scenario for Therapeutic Drug Monitoring. Medication adherence: Causes of medication non-adherence, pharmacist role in the medication adherence, and monitoring of patient medication adherence. Patient medication history interview: Need for the patient medication history interview, medication interview forms. Community pharmacy management: Financial, materials, staff, and infrastructure requirements		

3.	**Pharmacy and therapeutic committee, Drug information services, Patient counseling, Education and training program in the hospital, Education and training program in the hospital, and Prescribed medication order and communication skills**	**10 Hours**	**22.22%**
	Pharmacy and therapeutic committee: Organization, functions, Policies of the pharmacy and therapeutic committee in including drugs into formulary, inpatient and outpatient prescription, automatic stop order, and emergency drug list preparation. Drug information services: Drug and Poison information centre, Sources of drug information, Computerised services, and storage and retrieval of information. Patient counseling: Definition of patient counseling; steps involved in patient counseling, and Special cases that require the pharmacist Education and training program in the hospital: Role of pharmacist in the education and training program, Internal and external training program, Services to the nursing homes/clinics, Code of ethics for community pharmacy, and Role of pharmacist in the interdepartmental communication and community health education. Prescribed medication order and communication skills: Prescribed medication order- interpretation and legal requirements, and Communication skills- communication with prescribers and patients.		
4.	**Budget preparation and implementation, Clinical Pharmacy, and Over the counter (OTC) sales**	**8 Hours**	**17.77%**
	Budget preparation and implementation: Budget preparation and implementation Clinical Pharmacy: Introduction to Clinical Pharmacy, Concept of clinical pharmacy, functions and responsibilities of clinical pharmacist, Drug therapy monitoring - medication chart review, clinical review, pharmacist intervention, Ward round participation, Medication history and Pharmaceutical care. Over the counter (OTC) sales: Introduction and sale of over the counter, and Rational use of common over the counter medications.		
5.	**Drug store management and inventory control, Investigational use of drugs, and Interpretation of Clinical Laboratory Tests**	**7 Hours**	**15.55%**
	Drug store management and inventory control: Organization of drug store, types of materials stocked and storage conditions, Purchase and inventory control: principles, purchase procedure, purchase order, procurement and stocking, Economic order quantity, Reorder quantity level, and Methods used for the analysis of the drug expenditure Investigational use of drugs: Description, principles involved, classification, control, identification, role of hospital pharmacist, advisory committee. Interpretation of Clinical Laboratory **Tests:** Blood chemistry, hematology, and urinalysis		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	identify various drug distribution methods in a hospital, drug related problems, pharmacy stores management
CO2	understand the concept of rational use of drug therapy
CO3	interpret selected laboratory results (as monitoring parameters in therapeutics) of specific disease states
CO4	monitor medication chart review as well as clinical review and do Patient counseling.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	-
CO4	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ **Textbook:**

1. Merchant S.H. and Dr. J.S.Quadry. A textbook of hospital pharmacy, 4th ed. Ahmadabad: B.S. Shah Prakakshan; 2001.
2. Parthasarathi G, Karin Nyfort-Hansen, Milap C Nahata. A textbook of Clinical Pharmacy Practice- essential concepts and skills, 1st ed. Chennai: Orient Longman Private Limited; 2004.
3. William E. Hassan. Hospital pharmacy, 5th ed. Philadelphia: Lea & Febiger; 1986.
4. Tipnis Bajaj. Hospital Pharmacy, 1st ed. Maharashtra: Career Publications; 2008.
5. Scott LT. Basic skills in interpreting laboratory data, 4thed. American Society of Health System Pharmacists Inc; 2009.
6. Parmar N.S. Health Education and Community Pharmacy, 18th ed. India: CBS Publishers & Distributers; 2008.

❖ **Journals:**

1. Therapeutic drug monitoring. ISSN: 0163-4356
2. Journal of pharmacy practice. ISSN : 0974-8326
3. American journal of health system pharmacy. ISSN: 1535-2900 (online)
4. Pharmacy times (Monthly magazine)

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH7005: Novel Drug Delivery System (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Controlled drug delivery systems	10
	Polymers	
2	Microencapsulation	10
	Mucosal Drug Delivery system	
	Implantable Drug Delivery Systems	
3	Transdermal Drug Delivery Systems	10
	Gastroretentive drug delivery systems	
	Nasopulmonary drug delivery system	
4	Targeted drug Delivery	8
5	Ocular Drug Delivery Systems	7
	Intrauterine Drug Delivery Systems	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Controlled drug delivery systems, and Polymers**	**10 Hours**	**22.22%**
	Controlled drug delivery systems: Introduction, terminology/definitions and rationale, advantages, disadvantages, selection of drug candidates. Approaches to design controlled release formulations based on diffusion, dissolution and ion exchange principles. Physicochemical and biological properties of drugs relevant to controlled release formulations **Polymers:** Introduction, classification, properties, advantages and application of polymers in formulation of controlled release drug delivery systems.		
2.	**Microencapsulation, Mucosal Drug Delivery system, and Implantable Drug Delivery Systems**	**10 Hours**	**22.22%**

	Microencapsulation: Definition, advantages and disadvantages, microspheres /microcapsules, microparticles, methods of microencapsulation, applications **Mucosal Drug Delivery system:** Introduction, Principles of bioadhesion/mucoadhesion, concepts, advantages and disadvantages, transmucosal permeability and formulation considerations of buccal delivery systems **Implantable Drug Delivery Systems:** Introduction, advantages and disadvantages, concept of implants and osmotic pump.		
3.	**Transdermal Drug Delivery Systems, Gastroretentive drug delivery systems, and Nasopulmonary drug delivery system**	**10 Hours**	**22.22%**
	Transdermal Drug Delivery Systems: Introduction, Permeation through skin, factors affecting permeation, permeation enhancers, basic components of TDDS, formulation approaches **Gastroretentive drug delivery systems:** Introduction, advantages, disadvantages, approaches for GRDDS - Floating, high density systems, inflatable and gastroadhesive systems and their applications **Nasopulmonary drug delivery system:** Introduction to Nasal and Pulmonary routes of drug delivery, Formulation of Inhalers (dry powder and metered dose), nasal sprays, nebulizers.		
4.	**Targeted drug Delivery**	**8 Hours**	**17.77%**
	Targeted drug Delivery: Concepts and approaches advantages and disadvantages, introduction to liposomes, niosomes, nanoparticles, monoclonal antibodies and their applications		
5.	**Ocular Drug Delivery Systems and Intrauterine Drug Delivery Systems**	**7 Hours**	**15.55%**
	Ocular Drug Delivery Systems: Introduction, intraocular barriers and methods to overcome - Preliminary study, ocular formulations and ocuserts **Intrauterine Drug Delivery Systems:** Introduction, advantages and disadvantages, development of intra uterine devices (IUDs) and applications		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	understand fundamentals and methods of preparation, evaluation of controlled drug delivery systems.
CO2	determine novel transdermal , gastroretentive and nasopulmonary drug delivery system and microemulsion in the related research field.
CO3	understand fundamentals and methods of preparation, evaluation of mucosal, implantable drug delivery systems and microemulsion.
CO4	describe targeted drug delivery systems & its significance.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ **Textbook:**

1. Novel Drug Delivery Systems, Y W. Chien, Marcel Dekker, Inc., New York.
2. Controlled Drug Delivery Systems, Robinson, J. R., Lee V. H. L, Marcel Dekker, Inc., New York.
3. Encyclopedia of Controlled Delivery, Edith Mathiowitz, Published by Wiley Interscience Publication, John Wiley and Sons, Inc, New York.
4. Controlled and Novel Drug Delivery, N.K. Jain, CBS Publishers & Distributors, New Delhi.
5. Controlled Drug Delivery - Concepts and Advances, S.P. Vyas and R.K. Khar, Vallabh Prakashan, New Delhi.

❖ **Journals**

1. Indian Journal of Pharmaceutical Sciences (IPA)
2. Indian Drugs (IDMA)
3. Journal of Controlled Release (Elsevier Sciences)
4. Drug Development and Industrial Pharmacy (Marcel & Decker)
5. International Journal of Pharmaceutics (Elsevier Sciences)

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH7006: Practice School

Total hours: 180

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours	Percentage
1	Practice School	12 per week	100%

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	determine novel transdermal, gastroprotective and nasopulmonary drug delivery system and microemulsion in the related research field.
CO2	monitor medication chart review as well as clinical review and do Patient counselling.
CO3	perform Assay of drugs and intermediates using appropriate analytical methods
CO4	describe methods for preparation and suggest the approach to standardize herbal drugs ,herbal formulation and ayurveda formulations.
CO5	quantify drugs using various in-vivo and in-vitro experiments

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	3	3	3	-	-	-	3	-	-	-
CO4	3	-	-	3	-	-	-	-	-	-	-
CO5	3	3	-	-	-	-	-	-	-	-	3

Recommended Study Material:

- **Reference books:**

1. Anjaneyulu, Y., & Marayya, R. (2005). Quality Assurance And Quality Management in Pharmaceutical Industry: Book Syndicate
2. Pharmacopoeal standards for Ayurvedic Formulation (Council of Research in Indian Medicine & Homeopathy.
3. Mukherjee, P.W. Quality Control of Herbal Drugs: An Approach to evaluation of Botanicals. Business Horizons Publishers, New Delhi, India, 2002.
4. Drug discovery and Evaluation by Vogel H.G.
5. Introduction to biostatistics and research methods by PSS Sundar Rao and J Richard
6. Parthasarathi G, Karin Nyfort-Hansen, Milap C Nahata. A textbook of Clinical Pharmacy Practice- essential concepts and skills, 1st ed. Chennai: Orient Longman Private Limited; 2004.

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH7007: Entrepreneurship for Pharma Professionals (Theory)

Total hours: 15 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Entrepreneur-Entrepreneurship-Enterprise-Intrapreneurship	3
2	Start-Up process	3
3	Entrepreneurial Challenges in small business	4
4	Governments Role in Growth of MSMEs & SMEs	3
5	Case studies on successful and unsuccessful entrepreneurs and ventures	2

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Entrepreneur-Entrepreneurship-Enterprise- Intrapreneurship**	**3 Hours**	**20 %**
	• Conceptual issues. • Nature and Characteristics of Entrepreneurship • Factors affecting entrepreneurship • Entrepreneurship and economic development • Barriers to entrepreneurship • Entrepreneurial competencies. • Entrepreneurial process. • Entrepreneurship vs. Management. • Scope and Need of NGO's in Pharmaceuticals • Growth drivers in Indian Pharmaceutical Industry • Import and Export in Pharma Industry • Business ethics related to Pharmaceutical industry		
2.	**Start-Up process**	**3 Hours**	**20 %**

	• Role of creativity & innovation and business research. • Sources of business ideas • Small business as the seedbed of entrepreneurship. • Entrepreneurial motivation, performance and rewards. • Functions of entrepreneurs • The process of setting up a small business. (Legitimate compliances) • Business Plan- contents, importance, pre-requisites for an effective business plan. • Basic of market research, Introduction, different method of research, target group, etc. • Company Start up either Pharma marketing or Manufacturing • Current Government policies for Start-Up • Government support for new start-Up initiation		
3.	**Entrepreneurial Challenges in small business**	**4 Hours**	**26.66%**
	• Sources of venture capital, fixed capital, working capital and a basic awareness of financial services such as venture capital, private equity, leasing and factoring • The concept and application of product life cycle, advertising & publicity, sales & distribution management. • HR challenges- Availability of talent, retention, compensation, etc. • Legal and Government Challenges. • Licensing Process in Pharma Manufacturing and Marketing		
4.	**Governments Role in Growth of MSMEs & SMEs**	**3 Hours**	**20 %**
	• Government policies • Supporting system to promotes Micro, Small and Medium enterprise • Schemes from Central Governments and State Governments		
5.	**Case studies on successful and unsuccessful entrepreneurs and ventures**	**2 Hours**	**13.33%**

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	understand basic concepts in the area of entrepreneurship, the role and importance of entrepreneurship for economic development
CO2	identify the elements of successful start-ups
CO3	understand the challenges for starting a business venture
CO4	interpret their own business plan

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	3	3	-	3	-	-	-	-	-	3
CO3	3	3	-	-	-	3	-	-	-	-	3
CO4	3	3	-	-	-	3	-	-	-	-	3

Recommended Study Material:

❖ **Textbook:**

1. Brandt, Steven C., The 10 Commandments for Building a Growth Company, Third Edition, Macmillan Business Books, Delhi, 1977.
2. Dollinger, Mare J., Entrepreneurship: Strategies and Resources, Illinois, Irwin, 1955.
3. Entrepreneurial Business Law Journal
4. Entrepreneurship Theory and Practice
5. Holt, David H., Entrepreneurship: New Venture Creation, Prentice-Hall of India, latest Edition.
6. Kazmi, Azhar, ―What Young Entrepreneurs Think and Do: A Study of Second Generation Business Entrepreneurs,‖ The Journal of Entrepreneurship, 8, No. 1, 1999, pp. 67-78.
7. Small Business Economics
8. Vesper, KarlsH, New Venture Strategies, (Revised Edition), New Jersey, Prentice-Hall, 1990

❖ **Journals:**

1. Journal of Business Venturing
2. Journal of Developmental Entrepreneurship
3. Journal of International Entrepreneurship
4. Journal of Small Business Management
5. Journal: Entrepreneurship and Regional Development

❖ **Website:**

1. http://msmestartupkit.com
2. http://www.dcmsme.gov.in

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH7007P: Entrepreneurship for Pharma Professionals (Project)

Total hours: 30

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours/weeks	Percentage
1	Entrepreneurship for Pharma Professionals (Project)	2	100%

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	Understand basic concepts in the area of entrepreneurship, the role and importance of entrepreneurship for economic development
CO2	Identify the elements of successful start-ups
CO3	Understand the challenges for starting a business venture
CO4	Interpret their own business plan

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	3	-	-	-	-	3	-	-	3

FACULTY OF PHARMACY

Bachelor of Pharmacy Programme

SYLLABI

(Semester – 8)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8001: Biostatistics and Research Methodology (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction	10
	Measures of central tendency	
	Measures of dispersion	
	Correlation	
2	Regression	10
	Probability	
	Parametric test	
3	Non Parametric tests	10
	Introduction to Research	
	Designing the methodology	
	Graphs	
4	Regression modeling	8
	Clinical Trials Problems	
5	Design and Analysis of experiments	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weight age (%)
1.	**Introduction, Measures of central tendency, Measures of dispersion and Correlation**	**10 Hours**	**22.22%**
	Introduction: Statistics, Biostatistics, Frequency distribution **Measures of central tendency:** Mean, Median, Mode- Pharmaceutical examples **Measures of dispersion:** Dispersion, Range, standard deviation, Pharmaceutical problems **Correlation:** Definition, Karl Pearson‘s coefficient of correlation, Multiple correlation - Pharmaceuticals examples		
2.	**Regression, Probability and Parametric test**	**10 Hours**	**22.22%**
	Regression: Curve fitting by the method of least squares, fitting the lines y= a + bx and x = a + by, Multiple regression, standard error of regression– Pharmaceutical Examples		

	Probability: Definition of probability, Binomial distribution, Normal distribution, Poisson‘s distribution, properties – problems Sample, Population, large sample, small sample, Null hypothesis, alternative hypothesis, sampling, essence of sampling, types of sampling, Error-I type, Error-II type, Standard error of mean (SEM) - Pharmaceutical examples **Parametric test:** t-test(Sample, Pooled or Unpaired and Paired) , ANOVA, (One way and Two way), Least Significance difference		
3.	**Non Parametric tests, Introduction to Research, Graphs and Designing the methodology**	**10 Hours**	**22.22%**
	Non-Parametric tests: Wilcoxon Rank Sum Test, Mann-Whitney U test, Kruskal-Wallis test, Friedman Test 156 **Introduction to Research:** Need for research, Need for design of Experiments, Experiential Design Technique, plagiarism **Graphs:** Histogram, Pie Chart, Cubic Graph, response surface plot, Counter Plot graph **Designing the methodology:** Sample size determination and Power of a study, Report writing and presentation of data, Protocol, Cohorts studies, Observational studies, Experimental studies, Designing clinical trial, various phases.		
4.	**Regression modeling and Clinical Trials Problems**	**8 Hours**	**17.77%**
	Blocking and confounding system for Two-level factorials Regression modeling: Hypothesis testing in Simple and Multiple regressionmodels Introduction to Practical components of Industrial and Clinical Trials Problems: Statistical Analysis Using Excel, SPSS, MINITAB®, DESIGN OF EXPERIMENTS, R - Online Statistical Software‘s to Industrial and Clinical trial approach		
5.	**Design and Analysis of experiments**	**7 Hours**	**15.55%**
	Factorial Design: Definition, 2_2, 2_3design. Advantage of factorial design **Response Surface methodology**: Central composite design, Historical design, Optimization Techniques		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	understand the various statistical techniques to solve statistical problems
CO2	describe the operation of M.S. Excel, SPSS, R and MINITAB , DoE (Design of Experiment)
CO3	understand the concept of design of Experiment and perform optimization
CO4	interpret various statistical calculations using various statistical tools

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ **Textbook:**

1. Pharmaceutical statistics- Practical and clinical applications, Sanford Bolton, publisher Marcel Dekker Inc. NewYork.
2. Fundamental of Statistics – Himalaya Publishing House- S.C.Guptha
3. Design and Analysis of Experiments –PHI Learning Private Limited, R. Pannerselvam,
4. Design and Analysis of Experiments – Wiley Students Edition, Douglas and C. Montgomery

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8002: Social and Preventive (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Concept of health and disease	10
	Social and health education	
	Sociology and health	
	Hygiene and health	
2	Preventive medicine	10
3	National health programs, its objectives, functioning and outcome	10
4	National health intervention programme	8
5	Community services in rural, urban and school health	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weight age (%)
1.	**Concept of health and disease, Social and health education, Sociology and health, and Hygiene and health**	**10 Hours**	**22.22%**
	Concept of health and disease: Definition, concepts and evaluation of public health. Understanding the concept of prevention and control of disease, social causes of diseases and social problems of the sick. **Social and health education:** Food in relation to nutrition and health, Balanced diet, Nutritional deficiencies, Vitamin deficiencies, Malnutrition and its prevention. **Sociology and health:** Socio cultural factors related to health and disease, Impact of urbanization on health and disease, Poverty and health **Hygiene and health:** personal hygiene and health care; avoidable habits		
2.	**Preventive medicine**	**10 Hours**	**22.22%**
	Preventive medicine: General principles of prevention and control of diseases such as cholera, SARS, Ebola virus, influenza, acute respiratory infections, malaria, chicken guinea, dengue, lymphatic filariasis, pneumonia, hypertension, diabetes mellitus, cancer, drug addiction-drug substance abuse		

3.	**National health programs, its objectives, functioning and outcome of the following**	**10 Hours**	**22.22%**
	National health programs, its objectives, functioning and outcome of the following: HIV AND AIDS control programme, TB, Integrated disease surveillance program (IDSP), National leprosy control programme, National mental health program, National programme for prevention and control of deafness, Universal immunization programme, National programme for control of blindness, Pulse polio programme.		
4.	**National health intervention programme**	**8 Hours**	**17.77%**
	National health intervention programme for mother and child, National family welfare programme, National tobacco control programme, National Malaria Prevention Program, National programme for the health care for the elderly, Social health programme; role of WHO in Indian		
5.	**Community services**	**7 Hours**	**15.55%**
	Community services in rural, urban and school health: Functions of PHC, Improvement in rural sanitation, national urban health mission, Health promotion and education in school.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe high consciousness/realization of current issues related to health and pharmaceutical problems within the country and worldwide.
CO2	identify critical way of thinking based on current healthcare development.
CO3	interpret alternative ways of solving problems related to health and pharmaceutical issues

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

- ❖ **Textbook:**

1. Short Textbook of Preventive and Social Medicine, Prabhakara GN, 2nd Edition, 2010, ISBN: 9789380704104, JAYPEE Publications
2. Textbook of Preventive and Social Medicine (Mahajan and Gupta), Edited by Roy Rabindra Nath, Saha Indranil, 4th Edition, 2013, ISBN: 9789350901878, JAYPEE Publications
3. Review of Preventive and Social Medicine (Including Biostatistics), Jain Vivek, 6th Edition, 2014, ISBN: 9789351522331, JAYPEE Publications
4. Essentials of Community Medicine—A Practical Approach, Hiremath Lalita D, Hiremath Dhananjaya A, 2nd Edition, 2012, ISBN: 9789350250440, JAYPEE Publications
5. Park Textbook of Preventive and Social Medicine, K Park, 21st Edition, 2011, ISBN-14: 9788190128285, BANARSIDAS BHANOT PUBLISHERS.
6. Community Pharmacy Practice, Ramesh Adepu, BSP publishers, Hyderabad

❖ **Journals:**

1. Research in Social and Administrative Pharmacy
2. Elsevier
3. Ireland

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8003: Pharma Marketing Management (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Marketing	10
	Pharmaceutical market	
2	Product decision	10
3	Promotion	10
4	Pharmaceutical marketing channels	08
	Professional sales representative (PSR)	
5	Pricing	07
	Emerging concepts in marketing	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Marketing and Pharmaceutical market**	**10 Hours**	**22.22%**
	Marketing: Definition, general concepts and scope of marketing; Distinction between marketing & selling; Marketing environment; Industry and competitive analysis; Analyzing consumer buying behavior; industrial buying behavior. **Pharmaceutical market:** Quantitative and qualitative aspects; size and composition of the market; demographic descriptions and socio-psychological characteristics of the consumer; market segmentation& targeting.Consumer profile; Motivation and prescribing habits of the physician; patients' choice of physician and retail pharmacist.Analyzing the Market;Role of market research.		
2.	**Product decision**	**10 Hours**	**22.22%**
	Classification, product line and product mix decisions, product life cycle,product portfolio analysis; product positioning; New product decisions; Product branding, packaging and labeling decisions, Product management in pharmaceutical industry.		

3.	**Promotion**	**10 Hours**	**22.22%**
	Methods, determinants of promotional mix, promotional budget; An overview of personal selling, advertising, direct mail, journals, sampling, retailing, medical exhibition, public relations, online promotional techniques for OTC Products.		
4.	**Pharmaceutical marketing channels and Professional sales representative (PSR)**	**08 Hours**	**17.77%**
	Pharmaceutical marketing channels: Designing channel, channel members, selecting the appropriate channel, conflict in channels, physical distribution management: Strategic importance, tasks in physical distribution management. **Professional sales representative (PSR):** Duties of PSR, purpose of detailing, selection and training, supervising, norms for customer calls, motivating, evaluating, compensation and future prospects of the PSR.		
5.	**Design and Analysis of experiments**	**10 Hours**	**15.55%**
	Pricing: Meaning, importance, objectives, determinants of price; pricing methods and strategies, issues in price management in pharmaceutical industry. An overview of DPCO (Drug Price Control Order) and NPPA (National Pharmaceutical Pricing Authority). **Emerging concepts in marketing:** Vertical & Horizontal Marketing; RuralMarketing; Consumerism; Industrial Marketing; Global Marketing.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	describe marketing concepts and consumer behaviour
CO2	describe about product life cycle and pharmaceutical product management.
CO3	write various strategies for promotion of pharmaceutical products
CO4	explain various pricing strategies

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	3	3	-	3	-	3	3	-	-	3
CO2	3	3	-	-	-	-	-	-	-	-	3
CO3	3	3		-	3	-	-	3	-	-	3

CO4	3	3		-	-	-	-	-	-	-	3

Recommended Study Material:

- **Textbook:**
  1. Philip Kotler and Kevin Lane Keller: Marketing Management, Prentice Hall of India, New Delhi
  2. Walker, Boyd and Larreche : Marketing Strategy- Planning and Implementation, TataMC GrawHill, New Delhi.
  3. Dhruv Grewal and Michael Levy: Marketing, Tata MC Graw Hill
  4. Arun Kumar and N Menakshi: Marketing Management, Vikas Publishing, India
  5. Rajan Saxena: Marketing Management; Tata MC Graw-Hill (India Edition)
  6. Ramaswamy, U.S & Nanakamari, S: Marketing Managemnt:Global Perspective, IndianContext,Macmilan India, New Delhi.
  7. Shanker, Ravi: Service Marketing, Excell Books, New Delhi
  8. Subba Rao Changanti, Pharmaceutical Marketing in India (GIFT – Excel series) Excel

- **Reference book:**
  1. Philip Kotler and Kevin Lane Keller: Marketing Management, Prentice Hall of India, New Delhi
  2. Walker, Boyd and Larreche : Marketing Strategy- Planning and Implementation, TataMC GrawHill, New Delhi

- **Web Materials:**
  1. http://www.pharmabiz.com/
  2. http://www.expressbpd.com/pharma/
  3. http://www.cafepharma.com/
  4. https://www.fiercepharma.com/
  5. http://www.pharmalive.com/
  6. www.americanpharmaceuticalreview.com/1448-News/
  7. https://www.pharma-iq.com/
  8. https://www.pharmatutor.org/pharma-news
  9. http://www.pharmatimes.com/news
  10. https://www.worldpharmanews.com/
  11. https://www.businesstoday.in/sectors/pharma

12. www.reuters.com
13. http://globalpharmaupdate.com/
14. http://newsblur.com/
15. https://www.pharmatching.com/inforena/

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8004: Pharmaceutical Regulatory Science (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	New Drug Discovery and development	10
2	Regulatory Approval Process	10
3	Registration of Indian drug product in overseas market	10
4	Clinical trials	8
5	Regulatory Concepts	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**New Drug Discovery and development** Stages of drug discovery, Drug development process, pre-clinical studies, non-clinical activities, clinical studies, Innovator and generics, Concept of generics, Generic drug product development.	**10 Hours**	**22.22%**
2.	**Regulatory Approval Process** Approval processes and timelines involved in Investigational New Drug (IND), New Drug Application (NDA), Abbreviated New Drug Application (ANDA) in US. Changes to an approved NDA / ANDA. **Regulatory authorities and agencies** Overview of regulatory authorities of United States, European Union, Australia, Japan, Canada (Organization structure and types of applications)	**10 Hours**	**22.22%**
3.	**Registration of Indian drug product in overseas market** Procedure for export of pharmaceutical products, Technical documentation, Drug Master Files (DMF), Common Technical Document (CTD), electronic Common Technical Document (eCTD), ASEAN Common Technical Document (ACTD)research.	**10 Hours**	**22.22%**
4.	**Clinical trials** Developing clinical trial protocols, Institutional Review Board / Independent Ethics committee - formation and working procedures,	**8 Hours**	**17.77%**

	Informed consent process and procedures, GCP obligations of Investigators, sponsors & Monitors, Managing and Monitoring clinical trials, Pharmacovigilance - safety monitoring in clinical trials		
5.	**Regulatory Concepts**	**7 Hours**	**15.55%**
	Basic terminologies, guidance, guidelines, regulations, laws and acts, Orange book, Federal Register, Code of Federal Regulatory, Purple book		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	know the regulatory authorities and agencies governing the manufacture and sale of pharmaceuticals
CO2	describe the process of drug discovery and development
CO3	describe the regulatory approval process and their registration in Indian and international markets
CO4	narrate various guidelines related clinical trial regulations, managing and monitoring
CO5	describe the various regulatory terminologies, guidance, guidelines, regulations, laws and acts related to pharmaceuticals

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3
CO5	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

1. Drug Regulatory Affairs by Sachin Itkar, Dr. N.S. Vyawahare, Nirali Prakashan.

2. The Pharmaceutical Regulatory Process, Second Edition Edited by Ira R. Berry and Robert P. Martin, Drugs and the Pharmaceutical Sciences,Vol.185. Informa Health care Publishers.

3. New Drug Approval Process: Accelerating Global Registrations By Richard A Guarino, MD, 5th edition, Drugs and the Pharmaceutical Sciences,Vol.190.
4. Guidebook for drug regulatory submissions / Sandy Weinberg. By John Wiley & Sons. Inc.
5. FDA Regulatory Affairs: a guide for prescription drugs, medical devices, and biologics /edited by Douglas J. Pisano, David Mantus.
6. Generic Drug Product Development, Solid Oral Dosage forms, Leon Shargel and Isader Kaufer, Marcel Dekker series, Vol.143
7. Clinical Trials and Human Research: A Practical Guide to Regulatory Compliance By Fay A. Rozovsky and Rodney K. Adams
8. Principles and Practices of Clinical Research, Second Edition Edited by John I. Gallin and Frederick P. Ognibene
9. Drugs: From Discovery to Approval, Second Edition By Rick Ng

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8005: Pharmacovigilance (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Pharmacovigilance Introduction to adverse drug reactions Basic terminologies used in pharmacovigilance	10
2	Drug and disease classification Drug dictionaries and coding in pharmacovigilance Information resources in pharmacovigilance Establishing pharmacovigilance programme	10
3	Vaccine safety surveillance Pharmacovigilance methods Communication in pharmacovigilance	10
4	Statistical methods for evaluating medication safety data Safety data generation ICH Guidelines for Pharmacovigilance	8
5	Pharmacogenomics of adverse drug reactions Drug safety evaluation in special population CIOMS CDSCO (India) and Pharmacovigilance	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Unit I** **Introduction to Pharmacovigilance** • History and development of Pharmacovigilance • Importance of safety monitoring of Medicine • WHO international drug monitoring programme • Pharmacovigilance Program of India (PvPI) **Introduction to adverse drug reactions** • Definitions and classification of ADRs	**10 Hours**	**22.22%**

	• Detection and reporting • Methods in Causality assessment • Severity and seriousness assessment • Predictability and preventability assessment • Management of adverse drug reactions **Basic terminologies used in pharmacovigilance** • Terminologies of adverse medication related events • Regulatory terminologies		
2.	**Unit II**	**10 Hours**	**22.22%**
	Drug and disease classification • Anatomical, therapeutic and chemical classification of drugs • International classification of diseases • Daily defined doses • International Nonproprietary Names for drugs **Drug dictionaries and coding in pharmacovigilance** • WHO adverse reaction terminologies • MedDRA and Standardised MedDRA queries • WHO drug dictionary • Eudravigilance medicinal product dictionary **Information resources in pharmacovigilance** • Basic drug information resources • Specialized resources for ADRs **Establishing pharmacovigilance programme** • Establishing in a hospital • Establishment & operation of drug safety department in industry • Contract Research Organizations (CROs) • Establishing a national programme		
3.	**Unit III**	**10 Hours**	**22.22%**
	Vaccine safety surveillance • Vaccine Pharmacovigilance • Vaccination failure • Adverse events following immunization **Pharmacovigilance methods** • Passive surveillance – Spontaneous reports and case series • Stimulated reporting • Active surveillance – Sentinel sites, drug event monitoring and registries • Comparative observational studies – Cross sectional study, case control study and cohort study • Targeted clinical investigations **Communication in pharmacovigilance** • Effective communication in Pharmacovigilance • Communication in Drug Safety Crisis management		

	• Communicating with Regulatory Agencies, Business Partners, Healthcare facilities & Media		
4.	**Unit IV** **Statistical methods for evaluating medication safety data** **Safety data generation** • Pre-clinical phase • Clinical phase • Post approval phase **ICH Guidelines for Pharmacovigilance** • Organization and objectives of ICH • Expedited reporting • Individual case safety reports • Periodic safety update reports • Post approval expedited reporting • Pharmacovigilance planning • Good clinical practice in pharmacovigilance studies	**8 Hours**	**17.77%**
5.	**Unit V** **Pharmacogenomics of adverse drug reactions** **Drug safety evaluation in special population** • Paediatrics • Pregnancy and lactation • Geriatrics **CIOMS** • CIOMS Working Groups • CIOMS Form **CDSCO (India) and Pharmacovigilance** • D&C Act and Schedule Y • Differences in Indian and global pharmacovigilance requirements	**7 Hours**	**15.55%**

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	understand History and development of Pharmacovigilance
CO2	identify national and international scenario of Pharmacovigilance
CO3	know the Pharmacovigilance Program of India (PvPI) and Adverse drug reaction reporting systems and communication in Pharmacovigilance
CO4	describe the Statistical methods for evaluating medication safety data

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	3	-	-	-	3	3
CO4	3	-	-	-	-	-	-	-	-	3	3

Recommended Study Material:

1. Textbook of Pharmacovigilance: S K Gupta, Jaypee Brothers, Medical Publishers.

2. Practical Drug Safety from A to Z by Barton Cobert, Pierre Biron, Jones and Bartlett Publishers.

3. Mann's Pharmacovigilance:Elizabeth B. Andrews, Nicholas, Wiley Publishers.

4. Stephens' Detection of New Adverse Drug Reactions: John Talbot, Patrick Walle, Wiley Publishers.

5. An Introduction to Pharmacovigilance: Patrick Waller,Wiley Publishers.

6. Cobert's Manual of Drug Safety and Pharmacovigilance: Barton Cobert,Jones& Bartlett Publishers.

7. Textbook of Pharmacoepidemiolog edited by Brian L. Strom, Stephen E Kimmel, Sean Hennessy,Wiley Publishers.

8. A Textbook of Clinical Pharmacy Practice -Essential Concepts and Skills:G. Parthasarathi, Karin NyfortHansen,Milap C. Nahata

9. National Formulary of India

10. Text Book of Medicine by Yashpal Munjal

11. Text book of Pharmacovigilance: concept and practice by GP Mohanta and PK Manna

12. http://www.whoumc.org/DynPage.aspx?id=105825&mn1=7347&mn2=7259&mn3=72

13. http://www.ich.org/

14. http://www.cioms.ch/

15. http://cdsco.nic.in/

16. http://www.who.int/vaccine_safety/en/

17. http://www.ipc.gov.in/PvPI/pv_home.html

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8006: Quality Control and Standardization of Herbals (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Basic tests for drugs – Pharmaceutical substances, Medicinal plants materials and dosage forms. WHO guidelines for quality control of herbal drugs. Evaluation of commercial crude drugs intended for use	10
2	Quality assurance in herbal drug industry of cGMP, GAP, GMP and GLP in traditional system of medicine. WHO Guidelines on current good manufacturing Practices (cGMP) for Herbal Medicines WHO Guidelines on GACP for Medicinal Plants.	10
3	EU and ICH guidelines for quality control of herbal drugs. Research Guidelines for Evaluating the Safety and Efficacy of Herbal Medicines	10
4	Stability testing of herbal medicines. Application of various chromatographic techniques in standardization of herbal products. Preparation of documents for new drug application and export registration GMP requirements and Drugs & Cosmetics Act provisions.	8
5	Regulatory requirements for herbal medicines WHO guidelines on safety monitoring of herbal medicines in pharmacovigilance systems Comparison of various Herbal Pharmacopoeias. Role of chemical and biological markers in standardization of herbal products	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**UNIT-I** • Basic tests for drugs – Pharmaceutical substances, Medicinal plants materials and dosage forms. • WHO guidelines for quality control of herbal drugs. • Evaluation of commercial crude drugs intended for use	**10 Hours**	**22.22%**
2.	**UNIT-II** • Quality assurance in herbal drug industry of cGMP, GAP, GMP and GLP in traditional system of medicine. • WHO Guidelines on current good manufacturing Practices (cGMP) for Herbal Medicines • WHO Guidelines on GACP for Medicinal Plants.	**10 Hours**	**22.22%**
3.	**UNIT-III** • EU and ICH guidelines for quality control of herbal drugs. • Research Guidelines for Evaluating the Safety and Efficacy of Herbal Medicines	**10 Hours**	**22.22%**
4.	**UNIT-IV** • Stability testing of herbal medicines. Application of various chromatographic techniques in standardization of herbal products. • Preparation of documents for new drug application and export registration • GMP requirements and Drugs & Cosmetics Act provisions.	**8 Hours**	**17.77%**
5.	**UNIT-V** • Regulatory requirements for herbal medicines. • WHO guidelines on safety monitoring of herbal medicines in pharmacovigilance systems • Comparison of various Herbal Pharmacopoeias. • Role of chemical and biological markers in standardization of herbal products	**7 Hours**	**15.55%**

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	understand WHO guidelines for quality control of herbal drugs
CO2	describe Quality assurance in herbal drug industry
CO3	understand the regulatory approval process and their registration in Indian and international markets
CO4	describe EU and ICH guidelines for quality control of herbal drugs

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

1. Pharmacognosy by Trease and Evans
2. Pharmacognosy by Kokate, Purohit and Gokhale
3. Rangari, V.D., Text book of Pharmacognosy and Phytochemistry Vol. I , Carrier Pub., 2006.
4. Aggrawal, S.S., Herbal Drug Technology. Universities Press, 2002.
5. EMEA. Guidelines on Quality of Herbal Medicinal Products/Traditional Medicinal Products,
6. Mukherjee, P.W. Quality Control of Herbal Drugs: An Approach to Evaluation of Botanicals. Business Horizons Publishers, New Delhi, India, 2002
7. Shinde M.V., Dhalwal K., Potdar K., Mahadik K. Application of quality control principles to herbal drugs. International Journal of Phytomedicine 1(2009); p. 4-8
8. WHO. Quality Control Methods for Medicinal Plant Materials, World Health Organization, Geneva, 1998. WHO. Guidelines for the Appropriate Use of Herbal Medicines. WHO Regional Publications, Western Pacific Series No 3, WHO Regional office for the Western Pacific, Manila, 1998.
9. WHO. The International Pharmacopeia, Vol. 2: Quality Specifications, 3rd edn. World Health Organization, Geneva, 1981.
10. WHO. Quality Control Methods for Medicinal Plant Materials. World Health Organization, Geneva, 1999.
11. WHO. WHO Global Atlas of Traditional, Complementary and Alternative Medicine. 2 vol. set. Vol. 1 contains text and Vol. 2, maps. World Health Organization, Geneva, 2005.
12. WHO. Guidelines on Good Agricultural and Collection Practices (GACP) for Medicinal Plants. World Health Organization, Geneva, 2004.

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8007: Computer Aided Drug Design (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Drug Discovery and Development Lead discovery and Analog Based Drug Design	10
2	Quantitative Structure Activity Relationship (QSAR)	10
3	Molecular Modeling and virtual screening techniques Virtual Screening techniques	10
4	Informatics & Methods in drug design Introduction to Bioinformatics, chemo informatics. ADME databases, chemical, biochemical and pharmaceutical databases	8
5	Molecular Modeling: Introduction to molecular mechanics and quantum mechanics. Energy Minimization methods and Conformational Analysis, global conformational minima determination.	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**UNIT-I** **Introduction to Drug Discovery and Development** Stages of drug discovery and development **Lead discovery and Analog Based Drug Design** Rational approaches to lead discovery based on traditional medicine, Random screening, Non-random screening, serendipitous drug discovery, lead discovery based on drug metabolism, lead discovery based on clinical observation. Analog Based Drug Design: Bioisosterism, Classification, Bioisosteric replacement. Any three case studies	**10 Hours**	**22.22%**
2.	**UNIT-II** **Quantitative Structure Activity Relationship (QSAR)** SAR versus QSAR, History and development of QSAR, Types of physicochemical parameters, experimental and theoretical approaches for the determination of physicochemical parameters such as Partition	**10 Hours**	**22.22%**

	coefficient, Hammet's substituent constant and Tafts steric constant. Hansch analysis, Free Wilson analysis, 3D-QSAR approaches like COMFA and COMSIA.		
3.	**UNIT-III** Molecular Modeling and virtual screening techniques Virtual Screeningtechniques: Drug likeness screening, Concept of pharmacophore mapping and pharmacophore based Screening, Molecular docking: Rigid docking, flexible docking, manual docking, Docking based screening. De novo drug design.	**10 Hours**	**22.22%**
4.	**UNIT-IV** Informatics & Methods in drug design Introduction to Bioinformatics, chemoinformatics. ADME databases, chemical, biochemical and pharmaceutical databases	**8 Hours**	**17.77%**
5.	**UNIT-V** Molecular Modeling: Introduction to molecular mechanics and quantum mechanics. Energy Minimization methods and Conformational Analysis, global conformational minima determination.	**7 Hours**	**15.55%**

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	understand design and discovery of lead molecules
CO2	describe the role of drug design in drug discovery process
CO3	understand concept of QSAR and docking
CO4	describe various strategies to develop new drug like molecules.
CO5	interpret the design of new drug molecules using molecular modeling software

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	3
CO5	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

1. Robert GCK, ed., ―Drug Action at the Molecular Level‖ University Prak Press Baltimore.

2. Martin YC. ―Quantitative Drug Design‖ Dekker, New York.

3. Delgado JN, Remers WA eds ―Wilson & Gisvolds‘s Text Book of Organic Medicinal & Pharmaceutical Chemistry‖ Lippincott, New York.

4. Foye WO ―Principles of Medicinal chemistry ‗Lea & Febiger.

5. Koro lkovas A, Burckhalter JH. ―Essentials of Medicinal Chemistry‖ WileyInterscience.

6. Wolf ME, ed ―The Basis of Medicinal Chemistry, Burger‘s Medicinal Chemistry‖ JohnWiley& Sons, New York.

7. Patrick Graham, L., An Introduction to Medicinal Chemistry, Oxford UniversityPress.

8. Smith HJ, Williams H, eds, ―Introduction to the principles of Drug Design‖Wright Boston.

9. Silverman R.B. ―The organic Chemistry of Drug Design and Drug Action‖Academic Press New York. 172

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8008: Cell and Molecular Biology (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Cell and Molecular Biology	10
2	Genetic material of the Cell	10
3	Proteins and its regulation	10
4	Cell cycle and science of genetics	8
5	Cell signalling	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Cell and Molecular Biology** A. Cell and Molecular Biology: Definitions theory and basics and Applications. B. Cell and Molecular Biology: History and Summation. C. Theory of the Cell? Properties of cells and cell membrane. D. Prokaryotic versus Eukaryotic E. Cellular Reproduction F. Chemical Foundations – an Introduction and Reactions (Types)	**10 Hours**	**22.22%**
2.	**Genetic material of the Cell** A. DNA and the Flow of Molecular Structure B. DNA Functioning C. DNA and RNA D. Types of RNA E. Transcription and Translation	**10 Hours**	**22.22%**
3.	**Proteins and its regulation** A. Proteins: Defined and Amino Acids B. Protein Structure C. Regularities in Protein Pathways D. Cellular Processes E. Positive Control and significance of Protein Synthesis	**10 Hours**	**22.22%**

4.	**Cell cycle and science of genetics**	**08 Hours**	**17.77%**
	A. Science of Genetics B. Transgenics and Genomic Analysis C. Cell Cycle analysis D. Mitosis and Meiosis E. Cellular Activities and Checkpoints		
5.	**Cell signaling**	**07 Hours**	**15.55%**
	A. Cell Signals: Introduction B. Receptors for Cell Signals C. Signaling Pathways: Overview D. Misregulation of Signaling Pathways E. Protein-Kinases: Functioning		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	summarize history and applications of cell and molecular biology
CO2	describe cellular structure, functions, composition, reproduction and properties of cell membrane
CO3	understand structure and functions of DNA, RNA and proteins
CO4	understand basic molecular genetic mechanisms
CO5	describe cellular signaling pathways

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3
CO5	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

1. W.B. Hugo and A.D. Russel: Pharmaceutical Microbiology, Blackwell Scientific publications, Oxford London.

2. Prescott and Dunn., Industrial Microbiology, 4th edition, CBS Publishers & Distributors, Delhi.
3. Pelczar, Chan Kreig, Microbiology, Tata McGraw Hill edn.
4. Malcolm Harris, Balliere Tindall and Cox: Pharmaceutical Microbiology.
5. Rose: Industrial Microbiology.
6. Probisher, Hinsdill et al: Fundamentals of Microbiology, 9th ed. Japan
7. Cooper and Gunn's: Tutorial Pharmacy, CBS Publisher and Distribution.
8. Peppler: Microbial Technology.
9. Edward: Fundamentals of Microbiology.
10. N.K.Jain: Pharmaceutical Microbiology, Vallabh Prakashan, Delhi
11. Bergeys manual of systematic bacteriology, Williams and Wilkins- A Waverly company
12. B.R. Glick and J.J. Pasternak: Molecular Biotechnology: Principles and Applications of RecombinantDNA: ASM Press Washington D.C.
13. RA Goldshy et. al., : Kuby Immunology

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8009: Cosmetic Science (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Cosmetic excipients: Skin, Hair, Oral Cavity	10
2	Principles of formulation and building blocks of skin care products, Principles of formulation and building blocks of Hair care products, Principles of formulation and building blocks of oral care products.	10
3	Role of herbs in cosmetics: Skin Care, Hair care, Oral care, Analytical cosmetics	10
4	Principles of Cosmetic Evaluation	8
5	Cosmetic problems associated with Hair and scalp, with skin	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Cosmetic excipients: Skin, Hair, Oral Cavity** Classification of cosmetic and cosmeceutical products **Cosmetic excipients**: Surfactants, rheology modifiers, humectants, emollients, preservatives. Classification and application **Skin**: Basic structure and function of skin. **Hair:** Basic structure of hair. Hair growth cycle. **Oral Cavity:** Common problem associated with teeth and gu	**10 Hours**	**22.22%**
2.	**Principles of formulation and building blocks of skin care products** Face wash, Moisturizing cream, Cold Cream, Vanishing cream their relative skin sensory, advantages and disadvantages.Application of these products in formulation of cosmecuticals. **Principles of formulation and building blocks of Hair care products:** Conditioning shampoo, Hair conditioners,antidandruff shampoo. Hair oils. Chemistry and formulation of Para-phylene diamine based hair dye. Principles of formulation and building blocks of	**10 Hours**	**22.22%**

	oral care products: Toothpaste for bleeding gums, sensitive teeth. Teeth whitening, Mouthwash.		
3.	**Sun protection**	**10 Hours**	**22.22%**
	, Classification of Sunscreens and SPF. **Role of herbs in cosmetics:** Skin Care: Aloe and turmeric Hair care: Henna and amla. Oral care: Neem and clove **Analytical cosmetics**: BIS specification and analytical methods for shampoo, skincream and toothpaste.		
4.	**Principles of Cosmetic Evaluation**	**8 Hours**	**17.77%**
	Definition of cosmetics as per Indian and EU regulations, Evolution of cosmeceuticals from cosmetics, cosmetics as quasi and OTC drugs. Principles of Cosmetic Evaluation:Principles of sebumeter, corneometer. Measurement of TEWL, Skin Color, Hair tensile strength, Hair combing properties Soaps,and syndet bars. Evolution and skin benefits.		
5.	**Cosmetic problems associated with Hair and scalp, with skin**	**7 Hours**	**15.55%**
	Oily and dry skin, causes leading to dry skin, skin moisturisation. Basic understanding of the terms Comedogenic, dermatitis. Cosmetic problems associated with Hair and scalp: Dandruff, Hair fall causes Cosmetic problems associated with skin: blemishes, wrinkles, acne, prickly heat and body odor. Antiperspirants and Deodorants- Actives and mechanism of action		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	classify cosmetics, cosmeceuticals and describe cosmetics excipients, structure and function of skin and hair along with problem associated with oral cavity.
CO2	describe formulation and building blocks of skin care and hair care products.
CO3	describe role of herbs in skin care, hair care and oral care.
CO4	describe regulations for cosmetics and cosmeceuticals.
CO5	describe problems associated with skin, hair and scalp.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3
CO5	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

1) Harry‘s Cosmeticology, Wilkinson, Moore, Seventh Edition, George Godwin.
2) Cosmetics – Formulations, Manufacturing and Quality Control, P.P. Sharma, 4th Edition, Vandana Publications Pvt. Ltd., Delhi.
3) Text book of cosmeticology by Sanju Nanda & Roop K. Khar, Tata Publishers. 176

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8010: Pharmacological Screening Methods (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Laboratory Animals	08
2	Preclinical screening models	12
3	Preclinical screening models	10
4	Preclinical screening models	10
5	Research methodology and Bio-statistics	05

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Laboratory Animals:** Study of CPCSEA and OECD guidelines for maintenance, breeding and conduct of experiments on laboratory animals, Common lab animals: Description and applications of different species and strains of animals. Popular transgenic and mutant animals. Techniques for collection of blood and common routes of drug administration in laboratory animals, Techniques of blood collection and euthanasia.	**08 Hours**	**17.77%**
2.	**Preclinical screening models- CNS** A. Introduction: Dose selection, calculation and conversions, preparation of drug solution/suspensions, grouping of animals and importance of sham negative and positive control groups. Rationale for selection of animal species and sex for the study. B. Study of screening animal models for diuretics, nootropics, anti-Parkinson‘s, anti-asthmatics, **Preclinical screening models:** for CNS activity- analgesic, antipyretic, anti inflammatory, general anaesthetics, sedative and hypnotics, antipsychotic, antidepressant, antiepileptic, antiparkinsonism, alzheimer‘s disease	**12 Hours**	**26.66%**

3.	**Preclinical screening models- ANS**	**10 Hours**	**22.22%**
	for ANS activity, sympathomimetics, sympatholytics, parasympathomimetics, parasympatholytics, skeletal muscle relaxants, drugs acting on eye, local anaesthetics		
4.	**Preclinical screening models- CVS**	**10 Hours**	**22.22%**
	CVS activity: anti-hypertensives, diuretics, antiarrhythmic, anti-dyslipidemic activity, anti-platelet activity, coagulants, and anticoagulants Preclinical screening models for other important drugs like antiulcer, anti- diabetic, anticancer and anti-asthmatics.		
5	**Research methodology and Bio-statistics**	**05 Hours**	**11.11%**
	Selection of research topic, review of literature, research hypothesis and study design, Pre-clinical data analysis and interpretation using Students' test and One-way ANOVA. Graphical representation of data		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe the applications of various commonly used laboratory animals.
CO2	understand the various screening methods used in preclinical research
CO3	understand the importance of biostatistics and research methodology

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

1. Fundamentals of experimental Pharmacology-by M.N.Ghosh
2. Hand book of Experimental Pharmacology-S.K.Kulakarni
3. CPCSEA guidelines for laboratory animal facility.

4. Drug discovery and Evaluation by Vogel H.G.
5. Drug Screening Methods by Suresh Kumar Gupta and S. K. Gupta
6. Introduction to biostatistics and research methods by PSS Sundar Rao and J Richard

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8011: Advanced Instrumentation Techniques (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Nuclear Magnetic Resonance Spectroscopy	10
	Mass Spectrometry	
2	Thermal Methods of Analysis	10
	X-Ray Diffraction Methods	
3	Calibration and validation:	10
4	Radio immune assay	08
	Extraction techniques	
5	Hyphenated techniques	07

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**UNIT-I**	**10 Hours**	**22.22%**
	Nuclear Magnetic Resonance Spectroscopy: Principles of H-NMR and C-NMR, chemical shift, factors affecting chemical shift, coupling constant, Spin - spin coupling, relaxation, instrumentation and applications. **Mass Spectrometry:** Principles, Fragmentation, Ionization techniques – Electron impact, chemical ionization, MALDI, FAB, Analyzers-Time of flight and Quadrupole, instrumentation, applications.		
2.	**UNIT-II**	**10 Hours**	**22.22%**
	Thermal Methods of Analysis: Principles, instrumentation and applications of Thermogravimetric Analysis (TGA), Differential Thermal Analysis (DTA), Differential Scanning Calorimetry (DSC)		

	X-Ray Diffraction Methods: Origin of X-rays, basic aspects of crystals, X-ray Crystallography, rotating crystal technique, single crystal diffraction, powder diffraction, structural elucidation and applications.		
3.	**UNIT-III**	**10 Hours**	**22.22%**
	Calibration and validation: As per ICH and USFDA guidelines **Calibration of following Instruments**: Electronic balance, UV-Visible spectrophotometer, IR spectrophotometer, Fluorimeter, Flame Photometer, HPLC and GC.		
4.	**UNIT-IV**	**08 Hours**	**17.77%**
	Radio immune assay: Importance, various components, Principle, different methods, Limitation and Applications of Radio immuno assay. **Extraction techniques**: General principle and procedure involved in the solid phase extraction and liquid-liquid extraction.		
5	**UNIT-V**	**07 Hours**	**15.55%**
	Hyphenated techniques: LC-MS/MS, GC-MS/MS, HPTLC-MS.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	understand the advanced instruments used and its applications in drug analysis.
CO2	interpret the chromatographic separation and analysis of drugs.
CO3	describe the calibration of various analytical instruments.
CO4	describe analysis of drugs using various analytical instruments

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

1. Instrumental Methods of Chemical Analysis by B.K Sharma

2. Organic spectroscopy by Y.R Sharma
3. Text book of Pharmaceutical Analysis by Kenneth A. Connors
4. Vogel's Text book of Quantitative Chemical Analysis by A.I. Vogel
5. Practical Pharmaceutical Chemistry by A.H. Beckett and J.B. Stenlake
6. Organic Chemistry by I. L. Finar
7. Organic spectroscopy by William Kemp
8. Quantitative Analysis of Drugs by D. C. Garrett
9. Quantitative Analysis of Drugs in Pharmaceutical Formulations by P. D. Sethi
10. Spectrophotometric identification of Organic Compounds by Silverstein

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8012: Dietary Supplements and Nutraceuticals (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Definitions of Functional foods, Nutraceuticals and Dietary supplements	07
2	Phytochemicals as nutraceuticals: Occurrence and characteristic features (chemical nature medicinal benefits)	15
3	a) Introduction to free radicals: b) Dietary fibres and complex carbohydrates as functional food ingredients.	07
4	a) Role of free radicals in various diseases and free radicals theory of ageing. b) Antioxidants: c) Functional foods for chronic disease prevention	10
5	a) Effect of processing, storage and interactions of various environmental factors on the potential of Nutraceuticals. b) Regulatory Aspects on Food Safety. Adulteration of foods. c) Pharmacopoeial Specifications for dietary supplements and Nutraceuticals.	06

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**UNIT-I**	**7 Hours**	**15.55%**
	a. Definitions of Functional foods, Nutraceuticals and Dietary supplements. Classification of Nutraceuticals, Health problems and diseases that can be prevented or cured by Nutraceuticals i.e. weight control, diabetes, cancer, heart disease, stress, osteoarthritis, hypertension etc.		

	b. Public health nutrition, maternal and child nutrition, nutrition and ageing, nutrition education in community. c. Source, Name of marker compounds and their chemical nature, Medicinal uses and health benefits of following used as nutraceuticals/functional foods: Spirulina, Soyabean, Ginseng, Garlic, Broccoli, Gingko, Flaxseeds		
2.	**UNIT-II**	**15 Hours**	**33.33%**
	Phytochemicals as nutraceuticals: Occurrence and characteristic features(chemical nature medicinal benefits) of following a) Carotenoids- α and β-Carotene, Lycopene, Xanthophylls, leutin b) Sulfides: Diallyl sulfides, Allyl trisulfide. c) Polyphenolics: Reservetrol d) Flavonoids- Rutin , Naringin, Quercitin, Anthocyanidins, catechins, Flavones e) Prebiotics / Probiotics.: Fructo oligosaccharides, Lacto bacillum f) Phyto estrogens : Isoflavones, daidzein, Geebustin, lignans g) Tocopherols h) Proteins, vitamins, minerals, cereal, vegetables and beverages as functional foods: oats, wheat bran, rice bran, sea foods, coffee, tea and the like.		
3.	**UNIT-III**	**7 Hours**	**15.55%**
	a) Introduction to free radicals: Free radicals, reactive oxygen species, production of free radicals in cells, damaging reactions of free radicals on lipids, proteins, Carbohydrates, nucleic acids. b) Dietary fibres and complex carbohydrates as functional food ingredients..		
4.	**UNIT-IV**	**10 Hours**	**22.22%**
	a) Free radicals in Diabetes mellitus, Inflammation, Ischemic reperfusion injury, Cancer,Atherosclerosis, Free radicals in brain metabolism and pathology, kidney damage,muscle damage. Free radicals involvement in other disorders. Free radicals theory of ageing. b) Antioxidants: Endogenous antioxidants – enzymatic and nonenzymatic antioxidant defence, Superoxide dismutase, catalase, Glutathione peroxidase, Glutathione Vitamin C, Vitamin E, α- Lipoic acid, melatonin Synthetic antioxidants: Butylated hydroxy Toluene, Butylated hydroxy Anisole. c) Functional foods for chronic disease prevention		
5	**UNIT-V**	**06 Hours**	**13.33%**
	a) Effect of processing, storage and interactions of various environmental factors on the potential of Nutraceuticals.		

	b) Regulatory Aspects; FSSAI, FDA, FPO, MPO, AGMARK. HACCP and GMPs on Food Safety. Adulteration of foods. c) Pharmacopoeial Specifications for dietary supplements and Nutraceuticals		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	understand the need of supplements by the different group of people to maintain healthy life.
CO2	understand the role of dietary supplements and nutraceuticals in prevention of lifestyle diseases.
CO3	appreciate the components in dietary supplements and nutraceuticals and their application.
CO4	appreciate the regulatory and commercial aspects of dietary supplements including health claims.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

1. Dietetics by Sri Lakshmi
2. Role of dietary fibres and neutraceuticals in preventing diseases by K.T Agusti and P.Faizal: BSPunblication.
3. Advanced Nutritional Therapies by Cooper. K.A., (1996).
4. The Food Pharmacy by Jean Carper, Simon & Schuster, UK Ltd., (1988).
5. Prescription for Nutritional Healing by James F.Balch and Phyllis A.Balch 2nd Edn., Avery Publishing Group, NY (1997).
6. G. Gibson and C.williams Editors *2000 Functional foods* Woodhead Publ.Co.London.
7. Goldberg, I. *Functional Foods*. 1994. Chapman and Hall, New York.

8. Labuza, T.P. 2000 Functional Foods and Dietary Supplements: Safety, Good Manufacturing Practice (GMPs) and Shelf Life Testing in *Essentials of Functional Foods* M.K. Sachmidl and T.P. Labuza eds. Aspen Press.
9. Handbook of Nutraceuticals and Functional Foods, Third Edition (Modern Nutrition)
10. Shils, ME, Olson, JA, Shike, M. 1994 *Modern Nutrition in Health and Disease*. Eighth edition. Lea and Febiger.

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8013: Pharmaceutical Product Development (Theory)

Total hours: 45 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to pharmaceutical product development	10
2	An advanced study of Pharmaceutical Excipients in pharmaceutical product development	10
3	An advanced study of Pharmaceutical Excipients in pharmaceutical product development	10
4	Optimization techniques in pharmaceutical product development. A study of various	8
5	Selection and quality control testing of packaging materials for pharmaceutical product development- regulatory considerations.	7

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**UNIT-I**	**10 Hours**	**22.22%**
	Introduction to pharmaceutical product development, objectives, regulations related to preformulation, formulation development, stability assessment, manufacturing and quality control testing of different types of dosage forms		
2.	**UNIT-II**	**10 Hours**	**22.22%**
	An advanced study of Pharmaceutical Excipients in pharmaceutical product development with a special reference to the following categories i. Solvents and solubilizers ii. Cyclodextrins and their applications iii. Non - ionic surfactants and their applications iv. Polyethylene glycols and sorbitols v. Suspending and emulsifying agents vi. Semi solid excipients		
3.	**UNIT-III**		**22.22%**

	An advanced study of Pharmaceutical Excipients in pharmaceutical product development with a special reference to the following categories i. Tablet and capsule excipients ii. Directly compressible vehicles iii. Coat materials iv. Excipients in parenteral and aerosols products v. Excipients for formulation of NDDS Selection and application of excipients in pharmaceutical formulations with specific industrial applications	**10 Hours**	
4.	**UNIT-IV**	**08 Hours**	**17.77%**
	Optimization techniques in pharmaceutical product development. A study of various optimization techniques for pharmaceutical product development with specific examples. Optimization by factorial designs and their applications. A study of QbD and its application in pharmaceutical product development		
5	**UNIT-V**	**07 Hours**	**15.55%**
	Selection and quality control testing of packaging materials for pharmaceutical product development- regulatory considerations.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe the fundamental aspects of Pharmaceutical product development
CO2	understand the different class and applications of excipients used in development of pharmaceutical formulations.
CO3	summarize the different optimization techniques for development of products.
CO4	describe the regulatory and commercial aspects of packaging materials for pharmaceutical products.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

1. Pharmaceutical Statistics Practical and Clinical Applications by Stanford Bolton, CharlesBon; Marcel Dekker Inc.
2. Encyclopedia of Pharmaceutical Technology, edited by James swarbrick, Third Edition,Informa Healthcare publishers.
3. Pharmaceutical Dosage Forms, Tablets, Volume II, edited by Herbert A. Lieberman and Leon Lachman; Marcel Dekker, Inc.
4. The Theory and Practice of Industrial Pharmacy, Fourth Edition, edited by Roop K. Khar, S P Vyas, Farhan J Ahmad, Gaurav K Jain; CBS Publishers and Distributors Pvt.Ltd. 2013.
5. Martin‘s Physical Pharmacy and Pharmaceutical Sciences, Fifth Edition, edited by Patrick J. Sinko, BI Publications Pvt. Ltd.
6. Targeted and Controlled Drug Delivery, Novel Carrier Systems by S. P. Vyas and R. K.Khar, CBS Publishers and Distributors Pvt. Ltd, First Edition 2012.
7. Pharmaceutical Dosage Forms and Drug Delivery Systems, Loyd V. Allen Jr., Nicholas B.Popovich, Howard C. Ansel, 9th Ed. 40
8. Aulton‘s Pharmaceutics – The Design and Manufacture of Medicines, Michael E. Aulton,3rd Ed.
9. Remington – The Science and Practice of Pharmacy, 20th Ed.
10. Pharmaceutical Dosage Forms – Tablets Vol 1 to 3, A. Liberman, Leon Lachman and Joseph B. Schwartz
11. Pharmaceutical Dosage Forms – Disperse Systems Vol 1 to 3, H.A. Liberman, Martin, M.R and Gilbert S. Banker.
12. Pharmaceutical Dosage Forms – Parenteral Medication Vol 1 & 2, Kenneth E. Avis and H.A. Libermann.
13. Advanced Review Articles related to the topics.

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
BPH8014: Project work

Course Outcome (COs):

At the end of the course, the students would be able to

	CO
CO1	analyze the literature and identify the gap.
CO2	propose the solution and suggest the planning of the work
CO3	implement the strategy and carry out the studies
CO4	compile the results and rationalize the conclusion.
CO5	present the findings.
CO6	prepare report.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	3		-	-	-	-	-	-	-	3
CO3	3	-	3	3	-	-	-	-	-	-	3
CO4	3	-	-	3	-	3	-	3	-	-	3
CO5	3	-	-	-	-	-	-	-	-	-	3
CO6	3	-	-	-	-	-	3	3	-	-	3

UNIVERSITY ELECTIVE LEVEL-UNDERGRADUATE

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
List of University Elective Courses
3rd Semester Undergraduate Programme
(Academic year 2019-20)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	EC281.01: Introduction to MATLAB Programming	EC / FTE
2	CE281.01: Art of Programming	CE / FTE
3	CL281.01: Environmental Sustainability and Climate Change	CL / FTE
4	EE283: Python for Electrical Engineers	EE/FTE
5	IT281.01: ICT Resources and Multimedia	IT/FTE
6	ME281.01: Engineering Drawing	ME/FTE
7	PH233.01: Fundamentals of Packaging	RPCP/FPH
8	PD260.01: Basic Laboratory Techniques	PDPIAS/FAS
9	NR251.01: First Aid & Life Support	NURSING / FMD
10	PT191.01: Health Promotion and Fitness	ARIP / FMD
11	CA224: Introduction to Web Designing	CMPICA / FCA
12	BM231: Banking and Insurance	I2IM / FMS
13	PD261: Astrophysics, Space and Cosmos-1 (ASC–1)	PDPIAS/FAS

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

EC281.01: INTRODUCTION TO MATLAB PROGRAMMING
B. TECH SEMESTER-III (UNIVERSITY ELECTIVE)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Objective of the Course:

The course, intended for students with no programming experience, provides the foundations of programming in MATLAB. Variables, arrays, conditional statements, loops, functions, and plots are covered. At the end of the course, students should be able to use MATLAB in their own work and be prepared to deepen their MATLAB programming skills and tackle other programming languages.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to MATLAB Basics	3
2.	Basic MATLAB Functions	8
3.	Interactive computation	8
4.	Scripts and Functions in MATLAB	8
5.	Applications	3

Total Hours (Theory): 0

Total Hours (Lab): 30

Total Hours: 30

C. Detailed Syllabus:

1.	**Introduction to MATLAB Basics 3 Hours**	**10%**
1.1	MATLAB windows, On-line help	1Hr
1.2	Input- Output, File Types	1 Hr
1.3	General commands to remember	1 Hr
2.	**Basic MATLAB Functions 8 Hours**	**25%**
2.1	Working with Arrays of Number	1Hr
2.2	Creating and Printing Simple Plots	1Hr
2.3	Creating, Saving, and Executing a Script File	2Hrs
2.4	Creating and Executing a Function File	2Hrs
2.5	Working with Files and Directories	2Hrs
3.	**Interactive computation 8 Hours**	**25%**
3.1	Matrix and Vectors	1Hr
3.2	Matrix and Array Operations	3Hrs
3.3	Creating and Using Inline Functions	2Hrs
3.4	Using Built in Functions	2Hrs
4.	**Scripts and Functions in MATLAB 8 Hours**	**25%**
4.1	**Scripts Files**	2Hrs
4.2	**Function Files**	3Hrs
4.3	**Language-Specific Features**	3Hrs
5.	**Applications 3 Hours**	**15%**
5.1	Solving a Linear system	1Hr
5.2	Finding eigenvalues and eigenvectors	1 Hr
5.3	Matrix Factorizations	1 Hr

D. Instructional Methods and Pedagogy

- Chapter wise Assignments
- Quiz
- Audio Visual Presentations
- Chalk + Board
- White Board
- Online Demo

E. Student Learning Outcomes:

Students will acquire the following skills:

- Be fluent in the use of procedural statements--assignments, conditional statements, loops, function calls--and arrays.
- Be able to design, code, and test small MATLAB programs that meet requirements expressed in English. This includes a basic understanding of top-down design.

F. Recommended Study Materials

❖ **Reference Book & Text Book:**
1. Getting Started with MATLAB, Rudrapratap (IISc, Banglore), Oxford University Press. 2. A Guide to MATLAB, Brian R Hunt, Ronald L Lipsman, Cambridge University Press.
❖ **Web Materials / Reading Materials:**
1. Lecture notes 2. Handouts 3. Chapter wise Assignment

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
U & P U. PATEL DEPARTMENT OF COMPUTER ENGINEERING
CE281.01: ART OF PROGRAMMING
B TECH SEMESTER-III (UNIVERSITY ELECTIVE)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Objective of the Course:

- To be able to understand the various data structures available in programming language and apply them in solving computational problems.
- To be able to do testing and debugging of code written programming language.
- To create students' interest for programming related subjects and to make them aware of how to communicate with computers by writing a program.
- To foster the ability of solving various analytical and mathematical problems with algorithms within students.
- To make them learn regarding different data structures and memory management in the programming language.
- To promote skills like Development of logic and implementation of basic mathematical and other problems at individual level.
- To make them learn and understand coding standards, norms, variable naming conventions, commenting adequately and how to form layout of efficient program.
- To explain them concepts of pointer & file management concepts.

B. Outline of the Course:

Sr.No	Title of the Unit	Minimum number of Hours
1.	Introduction to Computer Systems	02
2.	Data Storage and Operations	03
3.	Algorithms and Flow charting	04
4	Algorithm to Program	06
5	Loops and Controls Construct	04

6	**Errors and Debugging**	**04**
7	**Structured Programming**	**04**
8	**Coding Conventions**	**03**

Total Hours (Theory):30

Total Hours (Lab): 00

Total Hours: 30

C. Detailed Syllabus:

1.	**Introduction to Computer Systems 02 Hours**	**7%**
1.1	Basic computer organisation, operating system, editor, compiler, interpreter, loader, linker, program development.	
2.	**Data Storage and Operations 03 Hours**	**10%**
	Various data representation techniques, data types, constants, variables (local and global), arrays, various arithmetic and logical operations in a typical programming environment, working with numbers, String and operators, finding pattern in string.	
3.	**Algorithms and Flow charting 04 Hours**	**13%**
	Introduction to computer problem solving, concepts and algorithms and flow chart, tracing of an algorithms, writing conditional code,	
4	**Algorithm to Program 06 Hours**	**21%**
	Specifications, top down development and stepwise refinement as per programming environment needs. Imperative style of correct and efficient programming, introductory concepts of time and space complexities.	
5	**Loops and Controls Construct 04 Hours**	**13%**
	Conditional and unconditional execution.Simple versus nested controls.Various aspects of repetitive executions, iterative versus recursive programming styles, assertions and loop invariants.	
6	**Errors and Debugging 04 Hours**	**13%**
	Types of errors, error handling, debugging, tracing/stepwise execution of program, watching variables values in memory	
7	**Programming 04 Hours**	**13%**
	Introduction to modular approach of problem solving, concepts of procedure and functions for effective programming,Making code modular	
8	**Coding Conventions 03 Hours**	**10%**

	Variable naming, function naming, indentation, usage and significance of comments for readability and program maintainability.	

D. Instructional Methods and Pedagogy

- Quiz
- Audio Visual Presentations
- White Board
- Online Demo

E. Student Learning Outcomes

- Students will get acquainted with basic components and capabilities of a typical computing system.
- Students will be able to critically think about basic problems and develop algorithms to solve, validate and verify with computing systems.
- Students will be able to identify appropriate language constructs and approach to computational problems.
- Students will be acquainted with coding standards including documentation which are required to be used for the development of effective, efficient and maintainable programs.

F. Recommended Study Materials

❖ **Reference Book & Text Book:**
3. Joyce Farrell, Programming Logic and Design Comprehensive, Cenage Learning 4. Sedgewick R., Algorithms in C, Addison Wesley 5. V. Rajaraman, Fundamentals of Computers, Prentice Hall of India
❖ **Web Materials / Reading Materials:**
4. Lecture notes 5. Handouts 6. Chapter wise Assignment

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
MANUBHAI S. PATEL DEPARTMENT OF CIVIL ENGINEERING

CL281.01: ENVIRONMENTAL SUSTAINABILITY AND CLIMATE CHANGE

SEMESTER-III (UNIVERSITY ELECTIVE)

Credits and Hours:

Teaching Scheme	Theory	Tutorial	Practical	Total	Credit
Hours/week	-	-	2	2	2
Marks	-	-	100	100	

A. Objectives of the Course:

The main objectives of the course are:

- To provide a basic understanding of the major environmental problems that need to be addressed to ensure sustainable development
- To provide a basic understanding about various management approaches towards a sustainable development
- To introduce students to the environmental aspects of specific industrial sectors, such as energy, transport, land and water use, and the built environment
- To provide basic understanding about climate changes, their causative factors and the possible mitigation

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introducing Sustainability Basics and Environmental Management	03
2	Environmental Challenges	03
3	Principles of Environmental Management	08
4	Environmental Sustainability	04
5	Introduction to Climate Change	07
6	Climate Change-Mitigation	05

Total Hours (Theory): 30

Total Hours (Lab): 00

Total Hours: 30

C. Detailed Syllabus:

1	**Introducing Sustainability Basics and Environmental Management**	**03 Hours**	**10%**
1.1 1.2 1.3 1.4 1.5	What is Unsustainable? What is Sustainability? Defining the Terms Development & Environment Environmental Strategy: The New Business Playing Field Environmental Management		
2	**Environmental Challenges**	**03 Hours**	**10%**
2.1 2.2 2.3 2.4 2.5	Depletion of Water Resources Population Agriculture Land Degradation Energy Security		
3	**Principles of Environmental Management**	**08 Hours**	**26%**
3.1 3.2 3.3 3.4 3.5 3.6 3.7	Environmental Concerns in India International Environmental Movement Definition, Goals, Need, Tools of Environmental Management Participants in EM Ethics and the Environment Ecology and the Environment Environmental Management Systems & Standards		
4	**Environmental Sustainability**	**04 Hours**	**14%**
4.1 4.2 4.3 4.4 4.5	Strategies for Sustainability Land Use and Urban Planning Energy and Climate Change Transportation Balancing Population with Food and Water Resources		
5	**Introduction to Climate Change**	**07 Hours**	**23%**
5.1 5.2 5.3 5.4 5.5 5.6 5.7	Climate Change-Way &Means What Do We Know and Don’t Know? The Physical Science of Climate Change Causes of Climate Change Global Atmospheric Composition Greenhouse Gases and Aerosols Extreme Weather Events & Sea Level Rise		

5.8	Climate Projections and their Uncertainties		
6	**Climate Change-Mitigation**	**05 Hours**	**17%**
6.1 6.2 6.3 6.4	Global Carbon Cycle Concept of Carbon Sequestration Carbon Credits and Carbon Footprints Policy Perspective: UNFCC, IPCC, Kyoto Protocol, MoEFCC		

D. Instructional Method and Pedagogy:

At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams will be conducted as per pedagogy as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar will be conducted as per pedagogy as a part of internal theory evaluation.

E. Students Learning Outcomes:

On the completion of the course the students will be able to:

- Understand & appreciate for the value of quantitative, systems and transdisciplinary thinking of Environmental Sustainability
- Expand their awareness about the environment as an increasing part of the core business model and day-to-day operations of many organizations
- Develop an environmental blueprint for action
- Think strategically and act entrepreneurially to create sustainable future
- Review on Climate Change and related strategies

F. Recommended Study Materials:

Text Books:

1. Environmental Management, T. V. Ramchandra & Vijay Kulkarni, Teri Press, New Delhi, 2009.
2. Handbook of Environmental Laws, Acts, Guidelines, Compliances &Standard Policy, R. K. Trivedy, B.S. Publishers, 2010.

3. Climate Change & India, Vulnerability Assessment and Adaption, P. R. Shukla, University Press, Hyderabad, 2003.

Reference Books:

1. Environmental Management, Principles and Practice, C. J. Barrow, Psychology Press, 1999.
2. Environmental Management in Practice, Nath B., Hens, L., Compton, P. and Devuyst, D, Vol I, Routledge, London and New York, 1998.
3. Handbook of Environmental Management and Technology: Gwendolyn Holmes, Ben Ramnarine Singh, and Louis Theodore, Wiley, 2004.
4. Corporate Environmental Management: Welford R, University Press, Hyderabad, 1999.

Web Materials:

1. http://nptel.ac.in/courses/122102006/7
2. http://envfor.nic.in/
3. http://cpcb.nic.in/
4. http://gpcb.gov.in/
5. http://nptel.ac.in/courses/119106008/40
6. https://unccelearn.org/course/
7. http://www.open.edu/openlearn/nature-environment/the-environment/climate-change/content-section-0
8. http://www.openlearningworld.com/

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF ELECTRICAL ENGINEERING
EE283: PYTHON FOR ELECTRICAL ENGINEERING
Semester- III (Level II)

Credit Hours:

Teaching Scheme	**Theory**	**Practical**	**Total**
Hours/week	-	2	2
Marks	-	100	100

A. Objective of the Course:

In this world of digitization, as an electrical engineer it is very important to opt for a precise computational alternative. The objectives of the course are:

- To introduce the students with the fundamentals and detail knowledge of Python. (1, 4, 5)
- To learn how to develop the program using basic commands of Python. (2, 4,6)
- To develop programs and user defined algorithms to provide optimized output. (1,5,6,7)
- Ability to perform and develop different computational skills using software recognized worldwide. (3,6)
- To Learn how to implement this computational techniques in solving problems of power system. (9)
- To be able to participate in Python based coding competition. (11)

B. Examination Scheme:

Theory Marks		**Practical Marks**		**Total Marks**
Internal	**External**	**Internal**	**External**	
-	**-**	**50**	**50**	**100**

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Variables and type	05
2	Function, Basic recursion	05
3	Control flow: Branching and repetition	05
4	Introduction to objects: Strings and lists	05
5	Python modules, debugging programs	05
6	Introduction to data structures : Dictionaries	05

Total Hours: 30

D. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
05	15	20	20	25	15

E. Course Outcomes (Learning Outcomes):

Upon successful completion of this course, a student will be able to

- Students will be aware about the use of Python for various problem solving.
- To strengthen the fundamentals of mathematics
- To enhance the programming skills in Python for electrical engineering.

F. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Laboratories will be conducted with the aid of multimedia projector.
- A student has to prepare a laboratory term work as per instruction given by lab instructor.
- A student has to prepare a laboratory term work as per instruction given by lab instructor.

- Attendance is compulsory in laboratory, which carries five marks of the overall evaluation.
- Two viva voce will be conducted during the semester and average of two will be considered as a part of overall evaluation.

G. Recommended Study Material:

Text Books:

1. Think Python – How to Think like a Computer Scientist, Allen Downey, Green Tea Press.

Reference Books:

1. Python Programming: A Complete Guide for Beginners to Master, PythonProgramming Language , Brian Draper
2. Python: The Complete Reference, Martin C. Brown, Tata McGraw Hill

Web Material:

1. https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-0001-introduction-to-computer-science-and-programming-in-python-fall-2016
2. https://www.python.org/about/gettingstarted
3. https://onlinecourses.nptel.ac.in/noc18_cs21/preview

FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF INFORMATION TECHNOLOGY
IT281.01: ICT RESOURCES AND MULTIMEDIA
SEMESTER-III (UNIVERSITY ELECTIVE)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	2
Marks	-	100	-	100	

A. Objective of the Course:

The main objectives for offering the course ICT RESOURCES AND MULTIMEDIA are:

1. To provide you the conceptual and technological developments in the field of information technology with the emphasis on comprehensive knowledge of Internet.
2. To introduce the fundamental elements of multimedia.
3. Understand to use the packages of presentation in detail.
4. Understand the impact of Information Technology on the Society and Various applications.

B. Outline of the Course

Sr. No.	Title of the Unit	Minimum Number of Hours
1	ICT Utilities and Tools	8
2	Introduction to Multimedia	4
3	Presentation Package	6
4	Networking Concepts	6
5	Information Technology and Society	6

Total hours (Theory): 30
Total hours (Lab): 00
Total hours: 30

C. Detailed Syllabus:

Following contents will be delivered to the students during laboratory sessions.

Sr. No.	UNIT	Hours	Weightage (%)
1.	**ICT Utilities and Tools**	**08 Hours**	**30%**
	Compression Utilities: WinZip, PKZIP, Concept of compression, Defragmenting Hard, disk using defrag, Scan Disk for checking disk space, lost files and recovery, Formatting Hard disk, Setting System Date and Time, Antivirus. Tools: Prezi, Macromedia Director & Flash, Gliffy, Edmodo, Google Classroom, Glogster.		
2.	**Introduction to Multimedia**	**04 Hours**	**10 %**
	What is multimedia, Components of multimedia, Web and Internet multimedia applications, Transition from conventional media to digital media.		
3.	**Presentation Package**	**06 Hours**	**20%**
	Microsoft PowerPoint: Slide layout, Slide design (Proper selection based on audience), Header and Footer in slides, Slide transition, Slide Master, Insert Picture-Smart Art, Insert animations to different objects, Hide Slide, Rehearse Timings, Record slide show. How to prepare professional presentation.		
4.	**Networking Concepts**	**06 Hours**	**20%**
	What is Networking, Local Area Networking (LANs), Metropolitan Area Network , MAN), Wide Area Network (WAN), Networking Topologies, Transmission media & method of communication.		
5.	**Information Technology and Society**	**06 Hours**	**20%**
	Indian IT Act, Intellectual Property Rights – issues. Application of information Technology in Railways, Airlines, Banking, Insurance, Inventory Control, Financial systems, Hotel management, Education, Video games, Telephone exchanges, Mobile phones, Information kiosks, special effects in Movies.		

D. Instructional Method and Pedagogy:

- In the very beginning, the course delivery pattern, prerequisites of the subject will be discussed.

- Multimedia and overhead Projectors, Chalk - Board and White - Board will be used for Class room teaching.
- Quiz / Q-A session will be conducted by the concerned faculty / for the theory.
- Audio Visual Presentations through electronic means and related software, and on-line demonstrations from the authentic web sites of the other premium institutes.
- Internal tests (as per the directions from the head and dean) will be conducted as a part of the regular curriculum.
- Seminars on advanced topics related to this subject will be key features.
- Academic counselling will reduce the formal distance between / amongst the students and faculty every fortnight.
- Students will be provided the latest updates such as technical articles, e-resources, printed materials and projects from magazines and journals.
- Attendance is compulsory in lectures which is part of overall evaluation.

E. Student Learning Outcomes:

At the end of the course the students will be able to:

- A student will be having the basic knowledge of the Information Technology.
- A student will be able to understand effectively use miscellaneous utilities such as: Compression, CD writing, Antivirus etc.
- A student will be able to use different tools for analysis and presentation.
- A student will be able to know the impact of ICT tools and Information Technology in Society.

F. Recommended Study Material:

❖ **Text Books:**

- ITL Educational Society, "Introduction to IT", Pearson Education, 2009.
- William Stallings, "Data and Computer Communication", Prentice, Hall of India Private Limited.
- Mavis Beacon, "All-in-one MS Office" CD based views for self-learning, BPB Publication, 2008
- Li & Drew, "Fundamentals of Multimedia", Pearson Education, 2009.

❖ **Reference Books:**

- Tay Vaughan, "Multimedia making it work", Tata McGraw-Hill, 2008.
- Parekh Ranjan, "Principles of Multimedia", Tata McGraw-Hill, 2007.

❖ **Web Materials:**

- https://www.tutorialspoint.com
- http://computingcareers.acm.org

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY &ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING
ME 281.01: ENGINEERING DRAWING
B. TECH SEMESTER-III (UNIVERSITY ELECTIVE)

Credits Hours:

Teaching Scheme	**Theory**	**Practical**	**Total**	**Credit**
Hours/week	-	2	2	2
Marks	-	100	100	

A. Objective of the Course:

- To introduce the student to the universal language and tool ofc communication of engineers.
- To make them thorough in understanding and using the various concepts of Engineering Drawing**.**

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1	Introduction to Engineering Drawing	05
2.	Visualization of Objects	10
3.	Sectional View of Objects	08
4.	Assembly and Detailed Drawing	04
5.	Introduction to Computer Aided Drawing	03

Total hours (Theory): 30

Total hours (Lab): 00

Total: 30

C. Detailed Syllabus:

1 Introduction to Engineering Drawing 05 Hours 17%

1.1 Basic of engineering graphics.

1.2 Lines, types of lines, use of different line in engineering drawing.

1.3 Dimensioning, placing of dimensions, general rules of dimensions.

2 Visualization of Objects **10 Hours 34%**

2.1 Principle of visualization of objects, need of visualization, methods of visualization of three dimensional objects.

2.2 Interpretation of line and area in orthographic drawing.

2.3 Visualization of objects -pictorial view and orthographic view.

2.4 Visualization of objects- Isometric view.

3 Sectional view of Objects **08 Hours 27%**

3.1 Principle of sectional view of objects,

3.2 Classification of sectional views,

3.3 Full sectional views, half sectional views, convection for sectioning.

4 Assembly and Detailed Drawing **04 Hours 12%**

4.1 Assembly drawing with sectioning from given detailed

4.2 Detail drawing from given assembly

5. Introduction to Computer Aided Drawing **03 Hours 10%**

5.1 Introduction to 2D Drawing facilities in CAD software

5.2 Introduction to 3D modeling & its relationship with 2D drawing views

D. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi–media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

E. Student Learning Outcomes / objectives:

Upon successful completion of this course, the student will be able to:

1. Understand the basics of drawing which is used in industries.
2. To convert sketches to engineered drawings will increase.
3. Improve their visualization skills so that they can apply these skills in developing new products.
4. KnowthefundamentalofComputerAidedDrawing&3D Modeling.

F. Recommended Study Material:

Text Books:

1. Shah,P.J.,EngineeringDrawingVol.I&II,S.Chand&Co.
2. Bhatt,N.D.,EngineeringDrawing,CharotarPublishingHouse

Reference books:

1. GopalKrishnaK.L.,EngineeringDrawing,SubhasPublications
2. Venugopal,K.,EngineeringDrawingmadeEasy,WileyEasternLtd.
3. Agrawal,M.L.&Garg,R.K.,EngineeringDrawingVol-I,Dhanpatrai&Co.
4. French,T.E.,Vierck,C.J.&Foster,R.J.,GraphicScienceandDesign,McGrawHill
5. Luzadder,W.J.&Duff,J.M.,FundamentalsofEngg.Drawing,PrenticeHall
6. Venugopal,K.,Engg.DrawingandGraphics,NewAgeinternationalPry.Ltd.

Web Materials:

1. users.rowan.edul~eyerettlcourseslfrcliilLectureslDraw.ppt
2. mechanical-engineering-drawing.ppt.fyxm.net

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF PHARMACY
Ramanbhai Patel College of Pharmacy
University Level Elective for Undergraduate Students of CHARUSAT
SEMESTER III
Fundamentals of Packaging [PH233.01]

Credit and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hr/Week	Theory		Practical		Total
					Internal	External	Internal	External	
III	PH233	Fundamen-tals of Packaging	02	02	-	-	30	70	100

Course Objectives

The course is structured to introduce the students, with diversified background, to different types of packaging materials generally employed and evaluation methods to be adopted for those packages.

Pre requisite: None

Methodology and Pedagogy:

During the sessions the students will be exposed to the concept with suitable examples, and are expected to learn through active learning, by preparing, e.g case studies, seminars, theoretical projects etc. The exercises will be given in groups of 2-3 students. Audio – Visual and IT resources will be used to transact the components of studies.

Learning Outcome:

Up on completion of the course, students would be able to,

1. Describe and indentify different types of packages
2. Be able to describe significance of tests to be performed to evaluate packages as well to suggest the type of tests to be performed for different type of packages
3. Suggest the type of material likely to be adopted for particular product
4. Describe the steps involved in aseptic manufacturing and packing of products
5. Able to identify different component of aerosol packages and be able to describe role of each of those component **Outline of the Course**

Sr No.	Unit
1	Introduction to Packaging
2	Packaging of oral solid formulation
3	Sterilization and Sterile Products Packaging including Aerosol Packaging

Detailed Syllabus

Sr.No	Units
1	**Introduction to Pharmaceutical Packaging**

	Definition, introduction to packaging, role of packaging, components of packaging, Overview of the Packaging Development and various aspects of it, Evaluation of packages and Physicochemical characteristics
2	Packaging of Oral Solids
	Flexible Packaging Materials – Introduction to Plastic, Cellulose and Semi Synthetic Polyemeric Materials , Overview of Packaging of Tablets, Capsules and Powders
3	**Sterilization and Sterile Products Packaging** Over view of Sterilization Processes, Asepctic Packaging, Packaging Systems, Assessment of Sterility
	Aerosol Packaging

Core Books

1. Encyclopedia of Pharmaceutical Technology Vol.1-3, Swarbric, J and Bolyln, J. C., Marcel Dekker, Inc., New York.
2. Smart Packaging Technologies for fast moving consumer goods, Editor Joseph Kerry and Paul Butler, Wiley
3. Pharmaceutical Packaging Technology, Dean, D. A. Evans, E. R. and Hall, j. H., Taylor and Francis, London.
4. Handbook of Packaging Technology, by Eiri Board (Engineers India Research Institute).

Reference Books

1. Packaging of Pharmaceutical & Healthcare products, H. Lockhart, F. A. Paine, Champman and Hall, London.
2. Packaging of Pharmaceuticals, C.F. Ross, Newnes-Butterworth.
3. The Theory and Practice of Industrial pharmacy, Lachmann, L., Lieberman, H.A. &Kanig, J.I., Lea and Fibiger, CBS Publishers and Distributers, New Delhi.

Web References

1. http://www.creativebloq.com/branding/packaging-design-resources-4131480
2. https://www.greenerpackage.com/newsletter

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

University Elective (UG)
PD260.01 BASIC LABORATORY TECHNIQUES
Semester-III

Credit: 2

A. Objective of the course.

The objectives of this course is to introduce students to the use of various electrical/electronic instruments, theirconstruction, applications, principles of operation, standards and units of measurements; and provide students with opportunities to develop basic skills in the design of electronic equipment.

B. Outline of the Course

1.	Error and uncertainty in measurements: Accuracy and precision, Significant figures, Error and uncertainty analysis, Types of errors: Gross error, systematic error, random error. Statistical analysis of data, (Arithmetic mean, deviation from mean, average deviation, standard deviation, chi-square), and curve fitting,Guassian distribution.	(10 Lectures)
2.	Basic of Measurement: Instruments accuracy, precision, sensitivity, resolution range etc. Errors in measurements and loading effects. Multimeter: Principles of measurement of dc voltage and dc current, ac voltage, ac current and resistance. Specifications of a multimeter and their significance.	(5 Lectures)
3.	DC and AC indicating Instruments: Ideal Constant-voltage and Constant-current Sources, Voltmeter, Ammeters, Cathode Ray Oscilloscope	(5 Lectures)
4.	Signal Generators and Analysis Instruments: Block diagram, explanation and specifications of low frequency signal generators. pulse generator, and function generator. Brief idea for testing, specifications. Distortion factor meter, wave analysis.	(5Lectures)
5.	Digital Instruments: Principle and working of digital meters. Comparison of analog& digital instruments. Characteristics of a digital meter. Working principles of digital voltmeter, working of a digital multimeter	(5Lectures)

C. Instructional Methods and Pedagogy:

The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Unit tests will be conducted regularly as a part of continuous evaluation and suggestions will be given to student in order to improve their performance.

D. Student Learning Outcomes / objectives:

Upon successful completion of this course, the student will be able to (Knowledge based) identify electronics/ electrical instruments, their use, peculiar errors associated with the instruments and how to minimise such errors.

E. Recommended Study Material:

1. Gupta B. R. (2003). Electronics and Instrumentation.Published by S. Chand and Company Ltd, New Delhi, India.
2. Measurement, Instrumentation and Experiment Design in Physics and Engineering, M. Sayer and A. Mansingh, PHI Learning Pvt. Ltd.
3. Experimental Methods for Engineers, J.P. Holman, McGraw Hill
4. Introduction to Measurements and Instrumentation, A.K. Ghosh, 3rd Edition, PHI Learning Pvt. Ltd.
5. Principles of Electronic Instrumentation, D. Patranabis, PHI Learning Pvt. Ltd.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ManikakaTopawala Institute of Nursing

University Level Elective for Undergraduate students:
NR 251.01- First Aid & Life Support (FALS)
Semester- III

I. Credits & Scheme:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
III	NR 251	First Aid & Life Support (FALS)	02	02	--	--	30	70	100

II. Course Objectives: Upon completing the course, students will be able to

- Demonstrate basic first aid skills needed to control bleeding and immobilize injuries.
- Demonstrate the skill needed to assess the ill or injured person.
- Demonstrate skills to assess and manage foreign body airway obstruction in infants, children and adults.
- Demonstrate skills to provide one- and two- person cardiopulmonary resuscitation to infants, children and adults

Course Outline:

Unit No.	Title of Unit	Hours
I.	**Introduction and Basics of First Aid:** • Rescuer Duties, Victim and Rescuer Safety • Looking for Help • After the emergency	2
II.	**Medical emergencies and their first aid**: • Breathing Problems • Choking in an Adult • Allergic Reactions • Heart Attack • Fainting • Diabetes and Low Blood Sugar • Stroke	12

	• Shock Abdominal maneuver, CPR, ventilation etc.	
III.	Injuries emergencies and their first aid: • Bleeding: You Can See/ You Can't See • Wounds • Burns and Electrical Injuries • Fractures Bandaging, immobilization, transferring etc.	8
IV.	Environmental Emergencies and first aid: • Bites and Stings • Poison Emergencies • Heat-Related Emergencies • Cold-Related Emergencies	7
V.	Preparation of First Aid Kit	1
Total		30 Hours

IV. Instruction Method and Pedagogy

The course is based on theory &practical learning. Teaching will be facilitated by reading material, discussion, microteaching, task-based learning, assignments, field visit and various interpersonal activities like group work, independent and collaborative research, presentations etc. Practical will be facilitated by demonstrations and preparing check-lists etc.

V. Evaluation: The students will be evaluated continuously in the form of internal as well as external examinations. The evaluation (Theory) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignments	1	8	8
2	Internal Test/ Model Exam	1	12	12
3	Attendance and Class Participation	Minimum 80% attendance		10
			Total	**30**

External Evaluation

The University Theory examination will be of 70 marks and will test the logic and critical thinking skills of the students by asking them theoretical as well as application based questions. The examination will avoid, as far as possible, grammatical errors and will focus on applications.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical Exam	01	70	70
			Total	70

VI. Learning Outcomes: At the end of the course, learners will be able to:

- Demonstrate bandaging and immobilization of patient
- Demonstrate the skill for transferring injured person
- Demonstrate skills to clear airway & ventilate the patient
- Demonstrate skills to perform CPR

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
Semester III (University Elective)
PT191.01HEALTH PROMOTION&FITNESS

CREDIT HOURS:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
-	2	2	-	2	2	-	100	100

A. OBJECTIVES OF THE COURSE:

This course will introduce students to the basic concept of health promotion, fitness and screening and basic assessment of fitness.

B. OUTLINE OF THE COURSE:

Sr. No.	Title of the unit	Minimum number of hours
1.	BASIC CONCEPT OF HEALTH PROMOTION	12
2.	EPIDEMIOLOGY AND HEALTH PROMOTION IN DIFFERENT SETTING	12
3.	BASIC CONCEPT OF FITNESS	12
4.	FITNESS ASSESMENT	12

Total hours (Theory) : 48

Total hours (Practical): 00

Total hours:48

C. **DETAILED SYLLABUS:**

1	**BASIC CONCEPT OF HEALTH PROMOTION**	**12hrs**
1.1 1.2 1.3 1.4 1.5 1.6	Meaning of health and Wellness Cultural &Social determinants of Health Physical, Environmental, Emotional & Psychological health Promotion of Healthy Lifestyles through Physical Activity, Diet, Stress Management, Avoiding Tobacco – Alcohol Promotion of Personal Hygiene, Treatment Seeking Behavior, Treatment Compliance and Reducing Stigma Need of health promotion in India	
2	**EPIDEMIOLOGY AND HEALTH PROMOTION IN DIFFERENT SETTING**	**12hrs**
2.1 2.2 2.3 2.4	Health Statistics: Analysis and Interpretation of Data Related to Health Promotion Use of Health Management Information System and Information Technologies in Health Promotion Health promotion in different settings - emergency and disaster Different areas of health promotion in India as compared to developed countries	
3	**BASIC CONCEPT OF FITNESS**	**12 Hrs**
3.1 3.2 3.3 3.4	Introduction definition of term :Fitness Basic Concepts Of Fitness Mental and physical fitness Health benefits of activity and Fitness	
4	**FITNESS ASSESSMENT**	**12Hrs**
4.1 4.2 4.3 4.4	Multifactorial fitness assessment and screening : Physical activity screening :Identify risk factors,height,weight,BMI,Physically active hours Aerobic fitness,Muscular Fitness,Activity and Weight control Vitality and Longevity Clinical preventive screening for infants Nutritional screening	

D. INSTRUCTIONAL METHOD AND PEDAGOGY:

- ❖ Interactive class room sessions using black-board and audio-visual aids.
- ❖ Using the available technology and resources for E- learning.
- ❖ Students will be focused on self-learning, practical learning which will be guided and facilitated by the faculty.
- ❖ Students will be enabled for continuous evaluation.
- ❖ Case study, group discussions, role-plays and simulation exercises.

E. STUDENT LEARNING OUTCOMES/OBJECTIVES:

At the end of the semester the student will be able:

- ❖ To review the basics health, Promotion& Fitness
- ❖ To understand the need of Health Promotion
- ❖ To understand basic concept of fitness
- ❖ To develop necessary skill to screen and assess basics of fitness

F. RECOMMENDED STUDY MATERIAL:

TEXTBOOKS:

1. Textbook of Preventive & Social Medicine- Dr. K. Park
2. Textbook of community medicine: V. K. Mahajan
3. Chiropractic, Health,Promotion and Wellness –MeridelI.Gatterman MA, DC,Med
4. Health ,Promotion and Wellness :evidence based guide to clinical preventive services— Cheryl Hawk & Will Evas
5. Fitness and Health – 6th edition – Brian J Sharkey,PhD

G. REFERENCE BOOKS:

- ❖ Principles Of Health Education And Health Promotion, (2^{nd}edition), J. Thomas Butler, Morton Publishing Company, Englewood, Colorado
- ❖ Foundations Of Health Education, R. M. Eberst, Editor, Coyote Press, San Bernardino: 1998-99
- ❖ Evaluation in health promotion – principles and perspective- WHO Regional Publications, European Series, No. 92
- ❖ Principles and foundation of health promotion and education(5th edition) by Randall R. Cottrell, James T. Girvan, James F. McKenzie

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

CA 224 Introduction to Web Designing
Semester-III

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Objective: The objective of the course is to provide basic understanding of designing professional web page templates with Markup Language Tags.

Pre-requisite: None.

Methodology & Pedagogy: During the sessions, topics related to design the pages of a website will be covered with suitable examples and students will be required to design and develop entire web sites using several Markup Language Tags and suitable editors.

Learning Outcome: Upon successful completion of the course, students will understand basic concepts of internet and web page designing and design and develop entire web sites using several web designing editors and HTML scripting language.

B. Outline of the Course:

Week No.	Content
1	Overview of Internet and WWW, Basic elements of the Internet, Internet services, Internet Browsers and Servers, Hardware and Software requirements to connect to the internet, Internet Service Provider (ISP), Introduction to Internet Protocols
2	Introduction to Web Page, Web Site, Web Browser, Overview of HTML, Structure of HTML Documents, HTML comments
3	HTML Basics Tags :Paragraph Tags, Horizontal Rule Tag, Heading Tags, Block quote Tags, Address Tags, PRE Tag,
4	Other HTML tags:- Formatting tags , Marquee tag, DIV tag, and SPAN tag
5	HTML List & Hyperlink in HTML (text link, image link, email link and many more)
6	HTML Images:- <img> tag and <map> tag
7-8	HTML Table
9-10	HTML Form
11-12	HTML Frames

Total Hours (Practical): 24

❖ **Core Books:**

1. Harley Hahn: The Internet Complete Reference, 2 nd Edition, Tata McGraw-HILL Edition.
2. Thomas a Powell: The Complete reference HTML, 3rd Edition, McGraw Hill,2001.
3. A. Whyte: Basic HTML, 2nd Edition, Payne-Gallway, Oxford, 2003.

4. Farrar : HTML Example book, BPB,2007.

❖ **Reference Books**:

1. Ivan Bayross: Web Enabled Commercial Application Development using HTML, JavaScript, DHTML and PHP, 4th revised edition, BPB Publication.

2. Jeremy Keith : HTML5 for Web Designers, A Book Apart Jeffrey Zeldmann, 2010

3. Peter Morville & Louis Rosenfeld, Information Architecture for WWW , 3rd Edition, O'Reilly Publication, 2006.

❖ **Web References**:

1. http://www.w3schools.com/html/[HTML notes]
2. http://www.tutorialspoint.com/html/ [HTML Materials]
3. http://www.tutorialspoint.com/html5/[HTML5 notes]

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

BM231: BANKING AND INSURANCE (B & I)
(Choice Based Credit System – University Elective)
SEMESTER-III

I. Number of Credits : 2

II. Course Objectives

The objectives of this course are:

- To equip the students with the knowledge of basic banking operation and Insurance industry.
- To recognize opportunities brought about by the dramatic changes that have occurred in the past decade in the banking and insurance industry.

III. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**Evolution of Banking** • *Brief Structure of Banks* • *Systems of Banking-Mixed,Branch,Unit,Group, Chain* • *RBI-Organization & Functions* • *Methods of Credit Control& Credit Creation.* • *Commercial Banking* • *Universal Banking*	**5**
2	**Bank Management** • *Sources and Uses of Funds in Banks* • *Balance sheet of a Bank* • *Value Chain Analysis in Banking Industry* • *Emergingtrends in Banking* • *Credit Cards* • *E-Banking*	**5**
3	**Retail Banking – An Introduction** • *Open Market Conditions and Role of Banks and Financial Institutions* • *Retail Banking –Concept and Importance.* • *Retail Banking Products- Housing Loan, Conveyance Loan, Personal Loan, Educational Loan, Loan for Retail Traders* • *Plastic Money*	**5**
4	**Marketing of Banking Services – Banking Products and Services** • *Distribution, Pricing and Promotion Strategy for Banking Services* • *Attracting and Retaining Bank Customers*	**5**

Module No.	Title/Topic	Classroom Contact Sessions
	• *Marketing Strategy of Credit Cards, Debit Cards, Saving Accounts and Different Types of Loans*	
5	**Introduction to Insurance Sector** • *Concept of Risk: Types of Risk, Risk Appraisal, Transfer and Pooling of Risks, Concept of Insurable Risk.* • *Concept of Insurance: Relevance of Insurance* • *Types of Insurance Organisations* • *Intermediaries in Insurance Business* • *Formation of Insurance Contract: Life, Fire, Marine and Motor Insurance Contracts* • *Principles of Insurance: Utmost Good Faith, Indemnity, Insurable Interest.*	5
6	**Practice of Life Insurance** • *Insurance Products, a Hedge Against Personal Risk (s)* • *Insurance Products, Alternative to Investment Products* • *Insurance Products, Collateral Security in the Rising Hire-Purchase Market Scenario* • *Marketing of Insurance Products- Life and Non Life Products* • *I.T in Insurance Business: Internet Based Delivery of Insurance Products, Servicing of Policies*	5
	Total	**30**

IV. Pedagogy

The course will emphasise self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will be as follows:

•	Classroom Contact Sessions	…	About 24 Sessions
•	Assignments	…	About 04 Sessions
•	Feedback	…	About 02 Sessions

The exact division among the above components will be announced by the instructor at the beginning of the semester as a part of detailed session-wise schedule.

V. Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks	Percentage of total internal evaluation
1	Quizzes	3	10	30	10
2	Presentations/Case Discussion	2	30	60	20
3	Synthesis Reports	2	60	120	40
4	Viva-voce	1	60	60	20
5	Attendance and Participation			30	10
			Total	**300**	**100**

The total marks will be divided by 10 and declared as Institute-level evaluation marks for the course. The Institute-level evaluation will constitute 30% of the total marks for the course.

VI. External Evaluation

The University examination will be for 70 marks and will be based on practical and a viva-voce.

VII. Learning Outcomes

At the end of the course, the student should have learnt:

- Knowledge about various functions associated with banking.
- Practice and procedures relating to deposit and credit, documentation, monitoring and control.
- An insight into banking services and banking technology.
- Understanding about the insurance sector and its products.

VIII. Reference Material

Text-Book

1. BhartiPathak, Indian Financial System, Pearson Education, Latest Edition
2. Gupta, P K. , Fundamentals of Insurance, Himalaya Publishing House, Latest Edition

Reference-Books

1. Latest Publications of Insurance Regulatory Authority of India
2. Khan M A, Introduction to Insurance, Educational Publication House, Aligarh, Latest Edition

Journals / Magazines / Newspapers

1. Journal on Banking Financial Services and Insurance Research
2. The Indian Banker

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICAL SCIENCE

PD261 : Astrophysics, Space and Cosmos-1 (ASC–1)

University Elective
Semester 3 Undergraduate Programme

Prerequisite:

Student having exposure of 10+2 level of Physics, and studying in bachelor and master program from any institute can join this course.

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	**-**	**2**	**2**	**2**
Marks	**-**	**100**	**100**	

C. Objective of the Course:

There have been frontier developments in recent months and past couple of years which involve major coming together of cosmology, physics and engineering sciences, such as the gravitational waves detected by LIGO and the fascinating images of shadows of the ultra-compact objects (e.g. black hole) at the center of our neighbouring galaxy M87. This development have spurred major interest in young students, citizens as well as scientific community as a whole and major research activities have started throughout the world & internationally in the academic centres. There is a huge interest in CHARUSAT students also. To meet up this demand & to create frontier research activity in terms of projects and proposals this course Astrophysics, Space and Cosmos has been devised. These will create important dividends in terms of frontier research emerging from CHARUSAT.

D. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Basic Astronomical Techniques	10
2.	The Frontier of Space	06
3.	The Life and Death of Stars	08
4.	The Universe	06

Total Hours (Theory): 0

Total Hours (Lab): 30

Total Hours: 30

E. Detailed Syllabus:

1.	**Basic Astronomical Techniques**	**10 Hrs**
	Description of Telescopes, Methods of observation, electromagnetic spectral window, resolution, sensitivity, noise, signal to noise ratio, background, aberrations, Telescopes at different wavelengths, Detectors at different wavelengths, Imaging techniques, spectroscopy, calibration, Atmospheric effects at different wavelengths, Astronomical data analysis, H-R diagram, sun and solar system, Sky Gazing.	6 (L) + 4 (P)
2.	**The Frontier of Space**	**6 Hrs**
	Outer reaches of earth's Atmosphere, Earth's Atmospheric Layers and Orbital Mechanism: Types of Orbits Launching of Satellites, Basics of Satellite subsystems, channel and link. Satellite Applications: Earth Observation, Scientific Study, Weather Forecast, Military Applications, GPS.	
3.	**The Life and Death of Stars**	**8 Hrs**
	Stars and Galaxies, The story of collapsing stars: Star formation and evolution, Interstellar Nebula, Red Giant, Planetary Nebulae, Planetary Dynamics, White Dwarfs, Red super Giant, Supernovae- Neutron star, Black hole, naked singularity. Gravitational Lensing, Accretion disks Around Compact Objects, Binary pulsar, Gravitational Waves.	
4.	**The Universe**	**6 Hrs**

	Universe in different scale: solar scale, Galactic scale, Cosmological scale, vastness of our universe, Expanding universe (Big bang, inflation), some interesting facts of our universe.	

F. Instructional Methods and Pedagogy

The topics will be discussed in interactive class room sessions using classical black-board teaching, ICT, hands-on-experiments and demonstration of experiments, whichever is relevant. Assignments, small projects and lab-exercise will be to the students. Student's needs to submit solution/report/results of above mentioned work, which will be eventually used for the evaluation purpose. Occasionally seminar/viva presentation will also be used as one of the evaluation tools.

G. Student Learning Outcomes:

Students will get preliminary knowledge about Astrophysics, Space and Cosmology. This is a level one course, which provides basis for the second level course. After completion of this course, students can start taking small project in this or allied fields.

H. Recommended Study Materials

❖ **Reference Book & Text Book:**

1. The Story of Collapsing Stars: Black Holes, Naked Singularities, and the Cosmic Play of Quantum Gravity, Prof. Pankaj S. Joshi
2. An Introduction to Cosmology 3ed, J.V. Narlikar
3. An introduction to modern cosmology, Andrew R. Liddle
4. Astronomy: Astronomy for Beginners: The Magical Science of Stars, Galaxies, Planets, Black Holes, Wormholes and much, much more! (Astronomy, Astronomy Textbook, Astronomy for Beginners), Miles Clarke
5. Satellite Communication, Dennis Roddy, Mc-Graw Hill publication
6. Satellite Communication, Timothy Pratt, Charles Bostian, Jeremy Allnut, John Wiley and Sons publication

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

List of University Elective Courses

For Second Year Fourth Semester Undergraduate Programme

(Academic year 2019-20)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	EC282.01: Prototyping Electronics with Arduino	EC / FTE
2	CE282.01: Web Designing	CE / FTE
3	CL282.01: Basics of Environmental Impact Assessment	CL / FTE
4	EE286: Computer Programming for Electrical Engineering	EE/FTE
5	IT282.01: Internet Technology and Web Design	IT/FTE
6	ME282.01: Material Science	ME/FTE
7	PH238.01: Cosmetics in daily life	RPCP/FPH
8	NR261.01: Life Style Diseases & Management	NURSING / FMD
9	PT192.01: Occupational Health & Ergonomics	ARIP / FMD
10	CA225: Programming the Internet	CMPICA / FCA
11	BM241: Health Care Management	I2IM / FMS
12	PD262: Astrophysics, Space and Cosmos-2 (ASC-2)	PDPIAS/FAS

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
EC282.01: PROTOTYPING ELECTRONICS WITH ARDUINO
B. TECH SEMESTER-IV (UNIVERSITY ELECTIVE)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	**0**	**2**	**2**	**2**
Marks	**0**	**100**	**100**	

A. Objective of the Course:

Objectives of this course are

- To aware students regarding Arduino Architecture
- To aware students regarding peripherals, sensors, wireless devices interfacing & Programming

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Getting Started with Arduino	02
2.	Arduino C	06
3.	GPIO Programming	08
4.	Sensors & Actuators	06
5.	Wireless Devices	08

Total Hours (Theory):30

Total Hours (Lab): 00

Total Hours: 30

C. Detailed Syllabus:

1.	**Getting Started with Arduino 02 Hours**	**10%**
1.1	Introduction to Arduino	1 Hr
1.2	Arduino Variants	
1.3	The Arduino UNO Board	1 Hr
1.4	Electronics Components	
2.	**Arduino C 06 Hours**	**20%**
2.1	C Data types	2 Hrs
2.2	Control Statements	
2.3	Functions & Arrays	
2.4	Using Modifying & Creating Arduino Libraries	2 Hrs
2.5	Advanced Coding & Memory Handling in Arduino	2 Hrs
3.	**GPIO Programming 08 Hours**	**25%**
3.1	LED Interfacing with Arduino	1 Hr
3.2	Driving 7 Segment Display using Arduino	1 Hr
3.3	Serial Communications & Soft serial Utility	2 Hrs
3.4	LCD Interfacing using Arduino	1 Hr
3.5	Matrix Keypad Interfacing using Arduino	1 Hr
3.6	Controlling Speed of DC Motor using PWM Techniques	1 Hr
3.7	Driving Stepper Motor using Arduino	1 Hr
4.	**Sensors & Acuators 06 Hours**	**20%**
4.1	Introduction to Sensors & Actuators	2 Hrs
4.2	Digital On/Off Sensors	
4.3	Analog Sensors	2 Hrs
4.4	Pulse Sensors	2 Hrs
5.	**Wireless Devices 08 Hours**	**25%**
5.1	Introduction to Wireless Devices	2 Hrs

5.2	Interfacing GSM & GPS Module with Arduino	
5.3	Interfacing Bluetooth Module with Arduino	2 Hrs
5.4	Interfacing Zigbee Module with Arduino	2 Hrs
5.5	Interfacing IR Module with Arduino	2 Hrs

D. Instructional Methods and Pedagogy

- Quiz
- Audio Visual Presentations
- White Board
- Online Demo

E. Student Learning Outcomes

- Upon completion of this course, students will understand the Arduno Architecture.
- Students can understand the programming GPIOs, Sensors, Actuators & Wireless Devices.

F. Recommended Study Materials

❖ **Reference Book & Text Book:**

6. Beginning of Android ADK with Arduino by Mario
7. Arduino CookBook, Michael Margolis, O'Reilly

❖ **Web Materials / Reading Materials:**

1. Lecture notes
2. Handouts
3. Chapter wise Assignment

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING
U & P U. PATEL DEPARTMENT OF COMPUTER ENGINEERING
CE282.01: WEB DESIGNING
B. TECH SEMESTER-IV (UNIVERSITY ELECTIVE)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Objective of the Course:

Students will learn

• To Learn how to link pages so that they create a Web site.

• To Design and develop a Web site using text, images, links, lists, and tables for navigation and layout.

• To Style web page using CSS, internal style sheets, and external style sheets.

• To Learn how to use graphics in Web design.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Web Designing Principles	02
2	Basic of Web Designing	03
3	Hyper Text Markup Language	06
4	Cascading Style Sheet Language	06
5	Extensible Mark-up Language	04
6	Adobe Dreamweaver	03
7	Adobe Photoshop	06

Total Hours (Theory):30

Total Hours (Lab): 00

Total Hours: 30

C. Detailed Syllabus:

1.	**Web Designing Principles02 Hours**	**7%**
	Basic principles involved in developing a web site, Planning process, Five Golden rules of web designing, Designing navigation bar, Page Design, Home Page Layout,Design Concept	
2.	**Basic of Web Designing 03 Hours**	**10%**
	Brief History of Internet, What is World Wide Web, Why create a web site, Web Standards and Protocols	
3.	**Hyper Text Markup Language06 Hours**	**20%**
	Basic structure of an HTML document, Mark up Tags, Heading-Paragraphs, Line Breaks, HTML Tags and Attributes , Creating an HTML layout	
4	**Cascading Style Sheets06 Hours**	**20%**
	Concept of CSS, Creating Style Sheet, CSS Properties, CSS Styling(Background, Text Format, Controlling Fonts), CSS Id and Class, CSS Selectors, Box Model(Introduction, Border properties, Padding, Properties, Margin properties)	
5	**Extensible Mark-up Language 04 Hours**	**13%**
	Basics of XML, XML Attribute, Elements, XML Validator	
6	**Adobe Dreamweaver 03 Hours**	**10%**
	Setting Up a Site with Dreamweaver, FTP - Site Upload Feature of Dreamweaver, Create various types of Links, Insert multimedia including text, image, animation & video	
7	**ADOBE Photoshop06 Hours**	**20%**
	Use of selection tools, Drawing and Painting tool, Transform B/W image to color, Layers, moving layer content using move tool Manipulating Images	

D. Instructional Methods and Pedagogy

- Project
- Audio Visual Presentations
- White Board
- Online Demo

E. Student Learning Outcomes

Upon successful completion of this course, the student will:

- Create an Information Architecture document for a web site.
- Select and apply mark-up languages for processing, identifying, and presenting of information in web pages.
- Combine multiple web technologies to create web components.
- Incorporate best practices in navigation, usability and written content to design websites that give users easy access to the information they seek.

F. Recommended Study Materials

❖ **Reference Book & Text Book:**

8. Html And CSS: Design And Build Websites, By Jon Duckett.
9. Learning Web Design: A Beginner's Guide To Html, Css, Javascript, And Web Graphics, By Jennifer Niederst Robbins
10. Beaird, Jason. 2010. The Principles Of Beautiful Web Design. 2nd Edition. Isbn-10: 098057689x
11. Adobe Dreamweaver Cs6: Classroom In A Book

❖ **Web Materials / Reading Materials:**

7. Lecture notes
8. URLs such as

https://www.w3.org/

http://www.land-of-web.com/category/tutorials

http://www.bypeople.com/design-trends-for-2011/

https://www.w3schools.com/

https://www.html5rocks.com/en/

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
MANUBHAI S. PATEL DEPARTMENT OF CIVIL ENGINEERING

CL282.01: BASICS OF ENVIRONMENTAL IMPACT ASSESSMENT

SEMESTER-IV (UNIVERSITY ELECTIVE)

Credits and Hours:

Teaching Scheme	Theory	Tutorial	Practical	Total	Credit
Hours/week	-	-	2	2	2
Marks	-	-	100	100	

A. Objectives of the Course:

The main objectives of the course are:

- To provide a basic understanding of the need, objective and the parameters considered for Environmental Impact Assessment (EIA) studies
- To provide an awareness on impact on resources and environment from development projects
- To introduce students to the legal, economic, administrative and technical process of preparing and/or evaluating environmental impact documents
- To learn laws related to EIA and auditing in India

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to EIA	05
2	Considerations for Environmental Assessment	07
3	Process of Impact Assessment	07
4	Tools for Assessing Environmental Impact	07
5	EIA in Indian Scenario	04

Total Hours (Theory): 30

Total Hours (Lab): 00

Total Hours: 30

C. Detailed Syllabus:

1	**Introduction to Environmental Impact Assessment**	**05 Hours**	**17%**
1.1	What is EIA?		
1.2	Objectives of EIA		
1.3	Brief History of Environmental Impact Analysis,		
1.4	EIA as research,		
1.5	EIA as decision making process,		
1.6	EIA in Global Affairs		
2	**Considerations for Environmental Assessment**	**07 Hours**	**23%**
2.1	Assessment Methodology		
2.2	Socioeconomic Impact Assessment		
2.3	Air Quality Impact Analysis		
2.4	Noise Impact Analysis		
2.5	Energy Impact Analysis		
2.6	Water Quality Impact Analysis		
2.7	Vegetation And Wild Life Impact Analysis		
2.8	Cumulative Impact Assessment		
2.9	Ecological Impact Assessment		
2.10	Risk Assessment		
3	**Process of Impact Assessment**	**07 Hours**	**23%**
3.1	Process for Environmental Impact Study		
3.2	Terms of Reference		
3.3	Stages in EIS Production: Screening, Scoping, Prediction, Evaluation, Reducing Impact, Monitoring, Conclusions		
3.4	Components of EIA Reports		
4	**Tools for Assessing Environmental Impact**	**07 Hours**	**23%**
4.1	Impact Assessment Methodologies - various Methods - Their Applicability		
4.2	Rapid EIA		
4.3	Strategic Impact Assessment		
4.4	Cumulative Impact Assessment		
5	**EIA in Indian Scenario**	**04 Hours**	**14%**
5.1	Provisions in the EIA Notification by MoEFCC		
5.2	Categorization of Industries for Seeking Environmental Clearance		
5.3	Procedure for Environmental Clearance		
5.4	Environmental Management Plan		
5.5	Case Study: Sardar Sarovar Dam, Narmada Project		

D. Instructional Method and Pedagogy:

At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams will be conducted as per pedagogy as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar will be conducted as per pedagogy as a part of internal theory evaluation.

E. Students Learning Outcomes:

On the completion of the course the students will be able to:

- Decide on the typical parameters to be considered in a EIA study of a developmental project
- Fully participate in interdisciplinary environmental report preparation teams
- Review and understand an EIA document for completeness and adequacy
- Review a typical development project plan and identify possible environmental effects and prepare appropriate initial studies
- Utilize EIA documents for policy development, project planning or for legal or political action planning

F. Recommended Study Materials:

Text Books:

4. Environmental Impact Assessment, Larry W Canter (2nd Edition), McGraw Inc., Singapore, 1996.
5. Environmental Impact Assessment – Handbook: John G Rau and D C Wooren, Mc-GrawHill.
6. Environmental Impact Assessment, A. K. Shrivastava, APH Publishing, New Delhi, 2003.

Reference Books:

1. Eccleston, H.C. Environmental Impact Statements. John Wiley & Sons, Inc. Canada, 2000.

2. World Bank, Environmental Assessment Sourcebook. Volume 1. World Bank Technical Paper No. 139, Washington, D.C, 1991.
3. Environmental Impact Analysis - a Decision Making Tool: By R K Jain, L. V. Urban and G.S. Stacey Publishers: Van Nostrand Reinhold New York.

Web Materials:

9. http://eia.unu.edu/course/index.html%3Fpage_id=173.html
10. http://envfor.nic.in/legis/eia/eia-2006.htm
11. http://envfor.nic.in/
12. http://nptel.ac.in/Clarify_doubts.php?subjectId=120108004&lectureId=5
13. http://cpcb.nic.in/
14. http://gpcb.gov.in/

EE286: COMPUTER PROGRAMMING FOR ELECTRICAL ENGINEERING
Semester-IV (Level II)
B. Tech (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	**Theory**	**Practical**	**Total**	**Credit**
Hours/week	---	2	2	2
Marks	---	50	50	

B. Examination Scheme:

Theory Marks		**Practical Marks**		**Total Marks**
Internal	External	Internal	External	
---	---	30	70	100

C. Course Objectives:

In this world of digitization, as an electrical engineer it is very important to opt for a precise computational alternative. The objectives of the course are:

i. To introduce the students with the fundamentals and detail knowledge of different computational techniques. (1, 4, 5)

ii. To learn how to develop the program using basic commands of MATLAB. (2, 4,6)

iii. To develop programs and user defined algorithms to provide optimized output. (1,5,6,7)

iv. Ability to perform and develop different computational skills using software recognized worldwide. (3,6)

v. To learn how to implement this computational technique in solving problems of power system. (9)

vi. To be able to participate in MATLAB coding competition. (11)

D. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	System of Linear Equations	08
2	Interpolation	04
3	Nonlinear Equations	06
4	Ordinary Differential Equations	08
5	Matrices and Eigenvalues	04

Total hours (Theory) :00Hrs
Total hours (Lab) :30Hrs
Total hours : 30Hrs

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	**Understanding**	**Application**	**Analyze**	**Evaluate**	**Create**
05	15	20	20	25	15

F. Course Outcomes (Learning Outcomes):

Upon successful completion of this course, a student will be able to

i. Students will be aware about the use of MATLAB for numerical method.
ii. To strengthen the fundamentals of mathematics
iii. To enhance the programming skills in MATLAB for electrical engineering.

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Laboratories will be conducted with the aid of multi-media projector.
- A student has to prepare a laboratory term work as per instruction given by lab instructor.
- A student has to prepare a laboratory term work as per instruction given by lab instructor.
- Attendance is compulsory in laboratory, which carries five marks of the overall evaluation.
- Two viva voce will be conducted during the semester and average of two will be considered as a part of overall evaluation.

H. Recommended Study Material:

Text Book:

1. Applied Numerical Methods Using Matlab, Won Young Yang, Wenwu Cao Tae-Sang Chung, John Morris, JOHN WILEY & SONS, INC.

Reference Book:

1. Computer Oriented Numerical Methods, V. Rajaraman, PHI Publication
2. Engineering Optimization: Theory and Practice, S S Rao, JOHN WILEY & SONS, INC.

Web Material:

- http://www.nptel.ac.in/courses/122104018/node109.html
- http://www.geos.ed.ac.uk/~yliu23/docs/lect_spline.pdf
- https://mat.iitm.ac.in/home/sryedida/public_html/caimna/interpolation/lagrange.html

I. List of experiments:

1. Introduction To Matlab Software and Work Environment. Introduction To Matlab basic Mathematical, trigonometric, Array and Matrix operations.
2. To Program various logical and Iterative loops in Matlab scripts
3. To Program Gauss elimination& Gauss-Jordan elimination method for solving linear equations and its validation with Matlab functions
4. To Program Jacobi Method & Gauss Seidel method for solving linear equations and its validation with Matlab functions
5. To Program Diagonalization methods for solving Matrices and Eigen values
6. To Program Power method for solving Matrices and Eigen values and its validation with Matlab functions
7. To Program Lagrange Polynomial interpolation method in Matlab scripts
8. To Program Bisection method & Regula Falsi method of solving nonlinear equation and its validation with Matlab functions
9. To Program Secant method and Newton-Raphson method of solving non linear equation and its validation with Matlab functions
10. To Program Euler's method of solving ordinary differential equations and its validation with Matlab functions
11. To Program Runge-Kutta method of solving ordinary differential equations and its validation with Matlab functions.

J. Course Assignments:

Final exams	**50%**
Lab Performance	**50%**

K. Course Assessments & Course Outcomes matrix (LOs or CLOs)

Assessment Methods	COs or LOs
Lab performance	CO No: 1,2,3,4,5,6,7,9 & 11
Final exam	CO No: 1,4,5,6,11

L. Teaching Methods & Learning Activities: (tick the applicable activities)

Telling/Explaining	√	Collaborating	√
Questioning	√	Experiments	√
Reading	√	Oral Presentations/Reports	
Demonstrating	√	Web Searching	√
Problem Solving	√		

M. Assessment Methods (Formal & Informal)(tick the applicable activities)

Test/Exam	
Lab Reports	√
Quiz	

N. Student Work load : (Total 50 Hours)

Course Readings	05 hrs	Lectures	00 hrs
Problem Solving	10 hrs	Experiments	30 hrs
Exams/Quizzes	00 hrs	Lab Pre work/Report	05 hrs

FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF INFORMATION TECHNOLOGY
IT282.01: INTERNET TECHNOLOGY AND WEB DESIGN
SEMESTER-IV (UNIVERSITY ELECTIVE)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	02	02	2
Marks	-	100	100	

A. Objective of the Course:

The main objectives for offering the course software project are:

- To describe the basic concepts for web development.
- To provide exposure in the field of Internet Technologies and Web Development.
- Learn the basic working scheme of the Internet and World Wide Web.
- To provide an intuition in web designing and development using Hyper Text Markup Language (HTML).
- Specify design rules in constructing web pages and sites.

B. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to WWW	03
2.	Current Trends on Internet	04
3.	HTML Programming Basics	06
4.	Web Publishing and Browsing	09
5.	Services on Internet	04
6.	Electronic Mail	04

Total hours (Theory): 30

Total hours (Lab): 00

Total hours: 30

C. Detailed Syllabus:

1. **Introduction to WWW** **03 Hours** **10%**
   Web Related browser functions, Browser Configuration, Browser Security Issues, Cookies, Spider, Intelligent Agents and special purpose browsers.
2. **Current Trends on Internet** **04 Hours** **13.33%**
   Introduction, Languages, Internet Phone, Internet Video, collaborative computing, e-commerce.
3. **HTML Programming Basics** **06 Hours** **20%**
   HTML page structure, HTML Text, HTML links, HTML document tables, HTML Frames, HTML Images, multimedia
4. **Web Publishing and Browsing** **09 Hours** **30%**
   Overview, SGML, Web hosting, Documents Interchange Standards, Components of Web Publishing, Document management, Web Page Design Consideration and Principles, Search and Meta Search Engines, Publishing Tools.
5. **Services on Internet** **04 Hours** **13.33%**
   E-mail, WWW, Telnet, FTP, IRC and Search Engine
6. **Electronic Mail** **04 Hours** **13.33%**
   Email Networks and Servers, Email protocols –SMTP, POP3, IMAp4, MIME6, Structure of an Email – Email Address, Email Header, Body and Attachments, Email Clients: Netscape mail Clients, Outlook Express, Web based E-mail. Email encryption- Address Book, Signature File.

D. Instructional Method and Pedagogy:

- In the very beginning, the course delivery pattern, prerequisites of the subject will be discussed.
- Multimedia and overhead Projectors, Chalk - Board and White - Board will be used for Class room and laboratory teaching.
- Quiz / Q-A session will be conducted by the concerned faculty / for the theory.
- Audio Visual Presentations through electronic means and related software, and on-line demonstrations from the authentic web sites of the other premium institutes.
- Internal tests (*as per the directions from the head and dean*) will be conducted as a part of the regular curriculum.
- Seminars on advanced topics related to this subject will be key features.
- Academic counselling will reduce the formal distance between / amongst the students and faculty every fortnight.
- Students will be provided the latest updates such as technical articles, e-resources, printed materials and projects from magazines and journals.
- Attendance is compulsory in lectures which is part of overall evaluation.

E. Student Learning Outcome:

At the end of the course the students will be able to:

- Review the current topics in Web & Internet technologies.
- Learn the basic working scheme of the Internet and World Wide Web.
- Understand fundamental tools and technologies for web design.
- Comprehend the technologies for Hypertext Mark-up Language (HTML).
- Specify design rules in constructing web pages and sites.

F. Recommended Study Material:

Text Books:

1. HTML 4 , Bryan Pfaffenberge, Bible.
2. Greenlaw R and Hepp E "Fundamentals of Internet and www" 2nd EL, Tata McGrawHill,2007.
3. Comer, "The Internet Book", Pearson Education, 2009

Reference Books:

1. M. L. Young,"The Complete reference to Internet", Tata McGraw Hill, 2007.
2. Godbole AS & Kahate A, "Web Technologies", Tata McGrawHill,2008.
3. Jackson, "Web Technologies", Pearson Education, 2008.
4. Leon and Leon, "Internet for Everyone", Vikas Publishing House.

Web Materials:

1. www.w3schools.com
2. www.tutorialspoint.com

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING
ME282.01: MATERIAL SCIENCE
B. TECH SEMESTER-IV (UNIVERSITY ELECTIVE)

Credits Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Objective of the Course:

- To give an overview on engineering requirements of materials and the importance of structure property relations of engineering materials on their selections and use.
- To understand the testing/ evaluation of strength and other properties of engineering materials.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1	Introduction to Material Science	05
2.	Metals and Alloys	09
3.	Non-Metallic Materials	08
4.	Materials Testing and Inspection	08

Total hours (Theory): 30

Total hours (Lab): 00

Total: 30

C. Detailed Syllabus:

1 Introduction to Material Science — 05 Hours — 16%

1.1 Classification of engineering materials

1.2 Engineering requirements of materials

1.3 Properties of engineering materials

1.4 Criteria for selection of materials for engineering applications

1.5 Concept of electronic materials, classification, properties and application of electronics materials

2	**Metals & Alloys**	**09 Hours**	**34%**

2.1 Classification of Metals

2.2 Ferrous Metals: Classification, Composition, Properties, Application (for plain carbon steel, alloy steel including stainless steel and cast iron)

2.3 Non-Ferrous Metals: Classification, Composition, Properties, Application (for copper, copper alloys, aluminum and aluminum alloys)

3	**Non-Metallic Materials**	**08 Hours**	**25%**

3.1 Introduction and classification of non-metallic materials

3.2 Classification of polymers on basis of thermal behavior (thermoplastics & thermosetting), properties and applications of polymers

3.3 Composites: Introduction, Characteristics of composites, Types and applications of composites

3.4 Other non-metallic materials-types, properties and applications. (like plastic, rubber, ceramics, refectories, insulators, wood etc.)

4.	**Materials testing and inspection**	**08 Hours**	**25%**

4.1 Introduction to Non-destructive testing

4.2 Radiography Testing

4.3 Dye Penetration Testing, Magnetic Particle Testing

4.4 Ultrasonic Testing

4.5 Macro-examination, Spark Test, Macro-etching, Fracture, Microscopic examinations

D. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi–media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

E. Student Learning Outcomes / objectives:

Upon successful completion of this course, the student will be able to:

1. Understand basic engineering properties of materials and its Application.
2. Understand fundamentals of various engineering steels and alloys and should have ability to select suitable materials based on their properties.
3. Classify non-metallic materials.
4. Examine various engineering materials.

F. Recommended Study Material:

Text Books

1. Narang G. B. S. and Manchanedy K., “Materials and Metallurgy”, Khanna Pub New Delhi, India
2. Dharmendra K umar and Jain S. K., “Material science and manufacturing process”, Vikas Pub House, New Delhi, India
3. R.S.Khurmi and R.S.Sedha, “Material Science”, S.Chand
4. D.S.Nutt, “Materials science and metallurgy”, S.K.Katariya and sons, Delhi

Reference Book:

1. Raghvan V., “Metallurgy for engineers”, Prentice Hall of India
2. Khanna O. P., “Material Science - A Text Book of Material Science & Metallurgy”, DhanpatRai Pub

Reading Materials, web materials with full citations:

1. http://ocw.mit.edu/OcwWeb/web/courses/courses/index.htm#MaterialsScienceandEngineering
2. http://nptel.iitm.ac.in/courses/Webcourse-contents/IIScBANG/Material%20Science/New_index1.html

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF PHARMACY
Ramanbhai Patel College of Pharmacy
University Level Elective for Undergraduate Students of CHARUSAT
SEMESTER IV
Cosmetics in Daily Life [PH238.01]

Credit and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hr/Week	Theory		Practical		Total
					Internal	External	Internal	External	
IV	**PH**	**Cosmetics in Daily Life**	**02**	**02**	**-**	**-**	**30**	**70**	**100**

Course Objectives

The course is structured to introduce the students, with diversified background, for preparation and evaluation of commonly used cosmetic formulations. The course will also impart preliminary information about the mechanism through which these products would act.

Pre requisite: None

Methodology and Pedagogy:

During the sessions the students will be exposed to the concept with suitable examples, and are expected to learn through active learning, by preparing, e.g case studies, seminars, theoretical projects etc. The exercises will be given in groups of 2-3 students. Audio – Visual and IT resources will be used to transact the components of studies.

Learning Outcome:

Up on completion of the course, students would be able to,

1. Describe the current market scenario of cosmetics
2. Describe the various type of cosmetics for skin and hair care
3. Suggest the type of dosage forms likely to be adopted for particular product
4. Describe the evaluation, packaging and labeling of various skin and hair care products.

Outline of the Course

Sr No.	Unit
1	Introduction to Cosmetics and Market Scenario

2	Skin care products
3	Hair care products

Detailed Syllabus

Sr.No	Units
1	Introduction to Cosmetics and Market Scenario
	Definition, Classification, Market Scenario in India and World Wide, Introduction to Solution, Suspension, Emulsion etc
2	Skin care products
	Anatomy and physiology of skin, classification of various skin care products. Formulation, evaluation, packaging and labeling of various skin care products like skin creams and lotions, suntan and anti sunburn, skin bleaching, skin tonics, anti aging cream.
3	Hair care products
	Anatomy and physiology of hair, classification of various hair care products. Formulation, evaluation, packaging and labeling of various hair care products like shampoo, conditioner, hair tonics, hair wave sets,

Core Books

1. Butler, H. ed., 2013. Poucher's perfumes, cosmetics and soaps. Springer Science & Business Media.
2. Cosmetics Formulation Manufacturing & Quality Control, P.P.Sharma, 4th Ed., Vandana Publications.
3. Handbook of Cosmetic Science and Technology, Andre Barel, Marc Paye, Howard I. Maibach, CRC Press.
4. Cosmetic technology, Nanda S, Nanda A, Khar RK., Birla Publications Pvt. Ltd.
5. Cosmetics: Science and Technology, Balsam S.M. and Sagarin Edward, 2nd Ed, Wiley Interscience.

Reference Books

1. Encyclopedia of Pharmaceutical Technology Vol.1-3, Swarbric, J and Bolyln, J. C., Marcel Dekker, Inc., New York.
2. Draelos, Z.D. ed., 2015. Cosmetic dermatology: products and procedures. John Wiley & Sons.
3. The Theory and Practice of Industrial pharmacy, Lachmann, L., Lieberman, H.A. &Kanig, J.I., Lea and Fibiger, CBS Publishers and Distributers, New Delhi.

Web References

1. www.cir-safety.org/
2. https://www.fda.gov/Cosmetics/
3. https://www.cosmeticseurope.eu/cosmetics
4. www.cdsco.nic.in/

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ManikakaTopawala Institute of Nursing

University Level Elective for Undergraduate students:
NR 261.01- Life Style Diseases & Management (LSDM)

I. Credits & Scheme:

Semester	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
IV	NR 261	Life Style Diseases & Management (LSDM)	02	02	-	-	30	70	100

II. Course Objectives: Upon completing the course, students will be able to

- Describe the National Health Programmes (NPDCS and NCCP) related to non-communicable diseases
- Understand the risk factors associated with non-communicable diseases and lifestyle disorders
- Understand the concept of health promotion
- Plan, implement, monitor and evaluate a health promotion programme
- Management of life style disorders

Course Outline:

Unit No.	Title of Unit	Prescribed Hours
I.	Introduction of NCDs/life-style disorders: • Overview of life-style disorders and epidemiology • Burden of life-style disorders in India and World • Risk factors for life-style disorders	4
II.	Management of Non-communicable diseases: • Obesity • Hypertension • Diabetic mellitus • Cardiovascular diseases • Chronic Kidney Diseases	20

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
Semester IV (University Elective)

PT192.01 OCCUPATIONAL HEALTH & ERGONOMICS

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
-	2	2	-	2	2		100-	100

CREDIT HOURS:

A. OBJECTIVES OF THE COURSE:

This course will introduce students to the basic concept of occupational health & Ergonomics Assessment and management for prevent the Occupational Hazards.

B. OUTLINE OF THE COURSE:

Sr. No.	Title of the unit	Minimum number of hours
1.	OCCUPATIONAL HEALTH	06
2.	ERGONOMICS	12
3.	ENVIRONMENTAL FACTORS & ERGONOMICS	06
4.	OCCUPATIONAL ERGONOMICS	12
5.	OCCUPATIONAL ERGONOMICS IN SPECIAL AREAS	12

Total hours (Theory) : 48

Total hours (Practical): 00

Total hours:48

C. DETAILED SYLLABUS:

1	**OCCUPATIONAL HEALTH**	**06hrs**
1.1 1.2 1.3 1.4	Introduction. Objective & basic concepts of Occupational health. Reorganization of health hazards. Early detection of Occupational diseases.	
2	**ERGONOMICS**	**12hrs**
2.1 2.2 2.3 2.4	Definitions of terms: Ergonomics, Ergonomists. Social significance of ergonomics. Biomechanical, Physiological & Anthropometric factors related to ergonomics. Ergonomical aspects of Postures like Sitting, Standing, Hand & arm postures. Ergonomical aspects of movements like Pushing, Pulling, Lifting & carrying.	
3	**ENVIRONMENTAL FACTORS & ERGONOMICS**	**06 hrs**
3.1	Noise, Light, Vibration, Climate & Chemical substances.	
4	**OCCUPATIONAL ERGONOMICS**	**12 hrs**
4.1 4.2 4.3 4.4	Work environment and control measures in work place. Prevention of occupational diseases and accidents. Effects of lifestyle and behavior on health. Health education in the workplace.	
5	**OCCUPATIONAL ERGONOMICS IN SPECIAL AREAS**	**12 hrs**
5.1 5.2 5.3 5.4	Industrial worker. Agriculture and rural areas. Working women. IT – BPO industries.	

D. INSTRUCTIONAL METHOD AND PEDAGOGY:

- Interactive class room sessions using black-board and audio-visual aids.
- Using the available technology and resources for E- learning.
- Students will be focused on self-learning, practical learning which will be guided and facilitated by the faculty.
- Students will be enabled for continuous evaluation.
- Case study, group discussions, role-plays and simulation exercises.

E. STUDENT LEARNING OUTCOMES/OBJECTIVES:

At the end of the semester the student will be able:

- ❖ To review the basics of Occupational health & Ergonomics
- ❖ To understand the problem associated with workplace.
- ❖ To clinically correlate the Occupational hazards and Ergonomics.

F. RECOMMENDED STUDY MATERIAL:

TEXTBOOKS:

6. Benjamin O. ALLI, Fundamental Principles of Occupational Health and Safety, Second edition, ILOInternational Labour Organization 2008
7. M. Rabiul&Ahasan: Occupational Health, Safety and Ergonomic issues in small and medium- sized Enterprises in a Developing Country. OULU 2002
8. WaldemarKarwowski& William S. Marras: Occupational Ergonomics – Design and management of Work System; CRC Press, Boca Raton, FL, 1999
9. Occupational Health : A manual for primary health care workers; WHO-EM/OCH/85/E/L

CA 225 Programming the Internet
Semester-IV

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. **Objective:** The objective of this course is to provide working knowledge of design effective and attractive web pages using HTML5 and CSS.

Pre-requisite: Basic knowledge of HTML

Methodology & Pedagogy: During the sessions, topics related to design the pages of a website will be covered with suitable examples and students will be required to design and develop entire web sites using HTML5 and CSS.

Learning Outcome: Upon Successful completion of the course, Students will learn the concepts of CSS from basic to advanced. Students will be able to design attractive web pages using HTML5 and CSS

B. **Outline of the Course:**

Week No.	Content
1	HTML5: Introduction, New Elements, Semantics, Media: Audio and Video
2-3	Forms: Input Types, Form Elements, Form Attributes, Graphics: Canvas
4	Introduction to Cascading Style Sheet (CSS), CSS layout, CSS essentials,
5-6	CSS selectors, CSS Box Model and Wrapper class
7	CSS Backgrounds and CSS Borders
8	CSS Text and Fonts and CSS List
9-10	CSS Tables, CSS Display, CSS Navigation and CSS Button
11-12	CSS Image Gallery and CSS 3:- Transforms, Transitions, Animation

Total Hours (Practical): 24

❖ **Core Books:**

1. Matthew MacDonald: HTML5: The Missing Manual, O'Reilly Media, August 2011.
2. Peter Gasston: The Book of CSS3: A Developer's Guide to the Future of Web Design, No Starch Press, April 2011.
3. Richard York: Beginning CSS: Cascading Style sheets for Web Design, Wrox Press (Wiley Publishing), 2014.

❖ **Reference Books**:

1. Ivan Bayross: Web Enabled Commercial Application Development using HTML, JavaScript, DHTML and PHP, 4^{th} revised edition, BPB Publication.
2. David Mc Farland: CSS: The Missing Manual, O'Reilly, 2006.

3. Thomas Powell: HTML & CSS: The Complete Reference, Fifth Edition Paperback, 2010

❖ **Web References:**

1. http://www.w3schools.com/html/html5_intro.asp[HTML5 notes]
2. http://www.w3schools.com/css/[CSS notes]
3. http://www.tutorialspoint.com/internet_technologies/html.htm [HTML notes]
4. http://james-star.com/answers/en/css3-hover-effect-transitions-transformations-and-animations/[Animation in CSS]

BM241: HEALTHCARE MANAGEMENT(HCM)
(Choice Based Credit System – University Elective)
YEAR 2, SEMESTER-IV

I. **Number of Credits : 2**

II. **Course Objectives**

The objectives of this course are:

- To help students understand, gain knowledge, and develop an appreciation of health care management by exploring all aspects of the field.
- To provide a foundation of applying managerial knowledge within health care sector.

IX. **Course Outline**

Module No.	Title/Topic	Classroom Contact Sessions
1	**Hospital Planning and Medical Care** • *Steps Involved in Hospital Planning* • *Managing Patient Care Service* • *Managing Support Services* • *Administration and Engineering Service*	**5**
2	**Health Sector Financing and Reforms** • *Health Care Financing in India and Developed Nations* • *National Health Spending* • *Health Sector Reforms* • *Resource Generation for Hospitals* • *Hospital Financing Alternatives*	**5**
3	**Role of Private Sector in Health Care** • *Private Health Sector* • *Expanding the Private Sector Health Care Financing* • *Public- Private Mix in Health Care Sector* • *Legal Issues in Health Care Sector*	**5**
4	**Health Care Economics and Finance and Marketing** • *Economic Appraisal of Health Services* • *Needs Vs Demand Vs Supply Model* • *Health Service Financing and Expenditure Survey, Budgeting, Control, Pricing, and Efficiency* • *Production and Cost of Health Care* • *Hospital Costs and Efficiency Marketing Health and Family Welfare* • *Social Marketing for Health and Family Welfare* • *Marketing for Hospitals* • *Health Services Marketing; Pharmaceutical Marketing* • *Patient as a Customer*	**5**
5	**Legal Aspects of Healthcare Management**	**5**

Module No.	Title/Topic	Classroom Contact Sessions
	• *Mitigate Liability Through Risk Management Principles* • *Develop Relationship Management Skills* • *Apply an Ethical Decision-Making Framework* • *Incorporate Employment Law Procedures* • *Manage Communication*	
6	**Health Care and Social Pol icy** • *Social Welfare* • *Approaches to Analysis* • *Distribution of Health Services in India*	**5**
	Total	**30**

X. Pedagogy

The course will emphasise self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will be as follows:

- Classroom Contact Sessions … About 24 Sessions
- Assignments … About 04 Sessions
- Feedback … About 02 Sessions

The exact division among the above components will be announced by the instructor at the beginning of the semester as a part of detailed session-wise schedule.

XI. Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks	Percentage of total internal evaluation
1	Quizzes	3	10	30	10
2	Presentations/Case Discussion	2	30	60	20
3	Synthesis Reports	2	60	120	40
4	Viva-voce	1	60	60	20
5	Attendance and Participation			30	10
			Total	**300**	**100**

The total marks will be divided by 10 and declared as Institute-level evaluation marks for the course. The Institute-level evaluation will constitute 30% of the total marks for the course.

XII. External Evaluation

The University examination will be for 70 marks and will be based on practical and a viva-voce.

XIII. Learning Outcomes

At the end of the course, the student should have learnt:

- The relationship of social need for health care services and contemporary issues prevailing to it.

XIV. Reference Material

Text-Book

1. S.L.Goel, Health Care Policies and Programmes, Deep and Deep Publications Pvt Ltd, Latest Edition

Reference-Books

1. G.D. Kunders, Designing for Total Quality in Health Care, Prism Books Pvt. Ltd., Bangalore, Latest Edition
2. B.M. Sakharkar, Principles of Hospital Administration and Planning, Jaypee Brothers Medical Publishers Pvt. Ltd., Latest Edition

Journals / Magazines / Newspapers

1. International Journal of Health Planning and Management
2. Health Policy and Planning Journal

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICAL SCIENCE

PD262 : Astrophysics, Space and Cosmos-2 (ASC-2)

University Elective

Semester-IV Undergraduate Programme

Prerequisite:

Student having exposure of 10+2 level of Physics, and studying in bachelor and master program from any institute can join this course.

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

C. Objective of the Course:

There have been frontier developments in recent months and past couple of years which involve major coming together of cosmology, physics and engineering sciences, such as the gravitational waves detected by LIGO and the fascinating images of shadows of the ultra-compact objects (e.g. black hole) at the center of our neighbouring galaxy M87. This development have spurred major interest in young students, citizens as well as scientific community as a whole and major research activities have started throughout the world & internationally in the academic centres. There is a huge interest in CHARUSAT students also. To meet up this demand & to create frontier research activity in terms of projects and proposals this course Astrophysics, Space and Cosmos has been devised. These will create important dividends in terms of frontier research emerging from CHARUSAT.

D. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Black holes and Shadows	10
2.	Gravitational Waves	06
3.	Sensors and Detectors for Satellite Imaging	08
4.	Cosmology	06

Total Hours (Theory): 0

Total Hours (Lab): 30

Total Hours: 30

C. Detailed Syllabus:

1.	**Black holes and Shadows**	**10 Hrs**
	Spacetime, Schwarzschild metric, Null Geodesics in Schwarzschild metric, Photon sphere, Einstein ring, Shadow of Black hole, shadow of Naked singularity, recent observation of black hole image (M87 Galaxy), experiments using imaging technique	8 (L) + 2 (P)
2.	**Gravitational Waves**	**6 Hrs**
	Introduction to Gravitational Waves; recent development for the detection of Gravitational waves; Overview of Gravitational Wave detectors (i.e. LIGO, KAGRA, etc.) and its scientific and technologic importance; Outline of Interferometers & its applications; Construction and working of Michelson Interferometer & its application potentiality in Advance- LIGO; Demonstration experiment of Michelson Interferometer.	4 (L) + 2 (P)
3.	**Sensors and Detectors for Satellite Imaging**	**8 Hrs**
	Introduction to sensors, Signal, Parameters, Types of Sensors: CCD and CMOS, CMOS sensors: Architecture Configuration and Working, CCD sensors: Architecture Configuration and Working.	
4.	**Cosmology**	**6 Hrs**
	Galaxies: General Description, The Milky-Way, M87 Galaxy, Rotation Curves of Spiral Galaxies, Galaxy Formation, Expanding Universe, Hubble's Law, Dark matter and Dark Energy.	

D. Instructional Methods and Pedagogy

The topics will be discussed in interactive class room sessions using classical black-board teaching, ICT, hands-on-experiments and demonstration of experiments, whichever is relevant. Assignments, small projects and lab-exercise will be to the students. Students needs to submit solution/report/results of above mentioned work, which will be eventually used for the evaluation purpose. Occasionally seminar/viva presentation will also be used as one of the evaluation tools.

H. Student Learning Outcomes:

After completion of Astrophysics, Space and Cosmology level one and two courses, students will explore themselves for taking projects, which may eventually fulfill their curiosity in these and allied fields. It may open up new avenues in their career.

F. Recommended Study Materials

- **Reference Book & Text Book:**

1. The Story of Collapsing Stars: Black Holes, Naked Singularities, and the Cosmic Play of Quantum Gravity, Prof. Pankaj S. Joshi
2. An Introduction to Cosmology 3ed, J.V. Narlikar
3. An introduction to modern cosmology, Andrew R. Liddle
4. Astronomy: Astronomy for Beginners: The Magical Science of Stars, Galaxies, Planets, Black Holes, Wormholes and much, much more! (Astronomy, Astronomy Textbook, Astronomy for Beginners), Miles Clarke
5. Satellite Communication, Dennis Roddy, Mc-Graw Hill publication
6. Satellite Communication, Timothy Pratt, Charles Bostian, Jeremy Allnut, John Wiley and Sons publication.
7. CMOS/CCD Sensors and Camera Systems, Second Edition, Gerald Holst, Terrence S Lomheim, SPIE publication.
8. Review article: Advanced LIGO, The LIGO Scientific Collaboration, Classical and Quantum Gravity, Volume 32, Number 7 (2015)